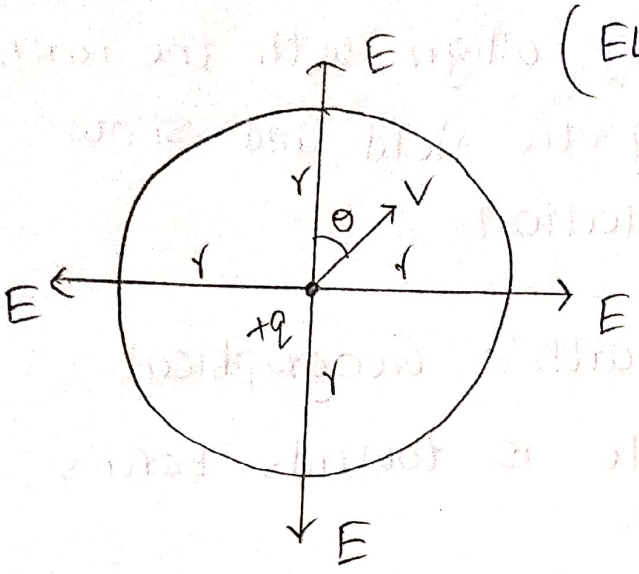
MOVING CHARGES AND MAGNETISM

(ELECTRO MAGNETISM)

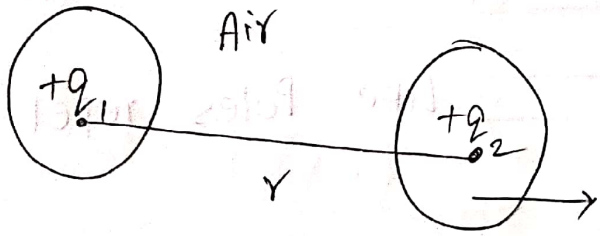


$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$

* 'E' depends on magnitud of charge and presence of "surrounding medium"

But doesnot depend on "speed of charge" & "direction of motion".

* Mono-charges "do exist".



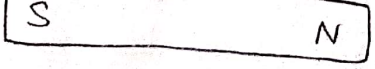
$$E = \frac{1}{4\pi\epsilon_0}\frac{q_1}{r^2}$$

Force acting on charge q_2, $F = Eq_2$

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$$

Here $E = \frac{F}{q}$

* A moving charge can produce both "electric & magnetic fields."



Bar magnet

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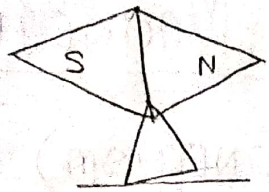
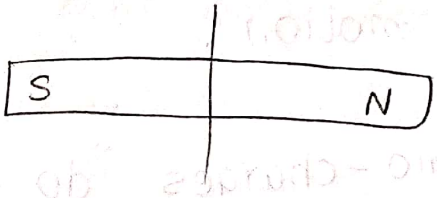


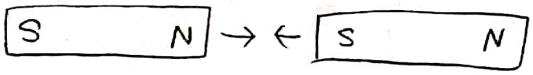
Magnetic compass needle

* They allign with the Earth Magnetic field and show deflection.

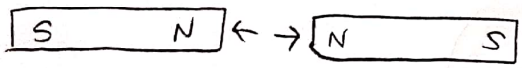
* South Pole is towards Earth's Geographical South Pole & North pole is towards Earth's Geographical North Pole.



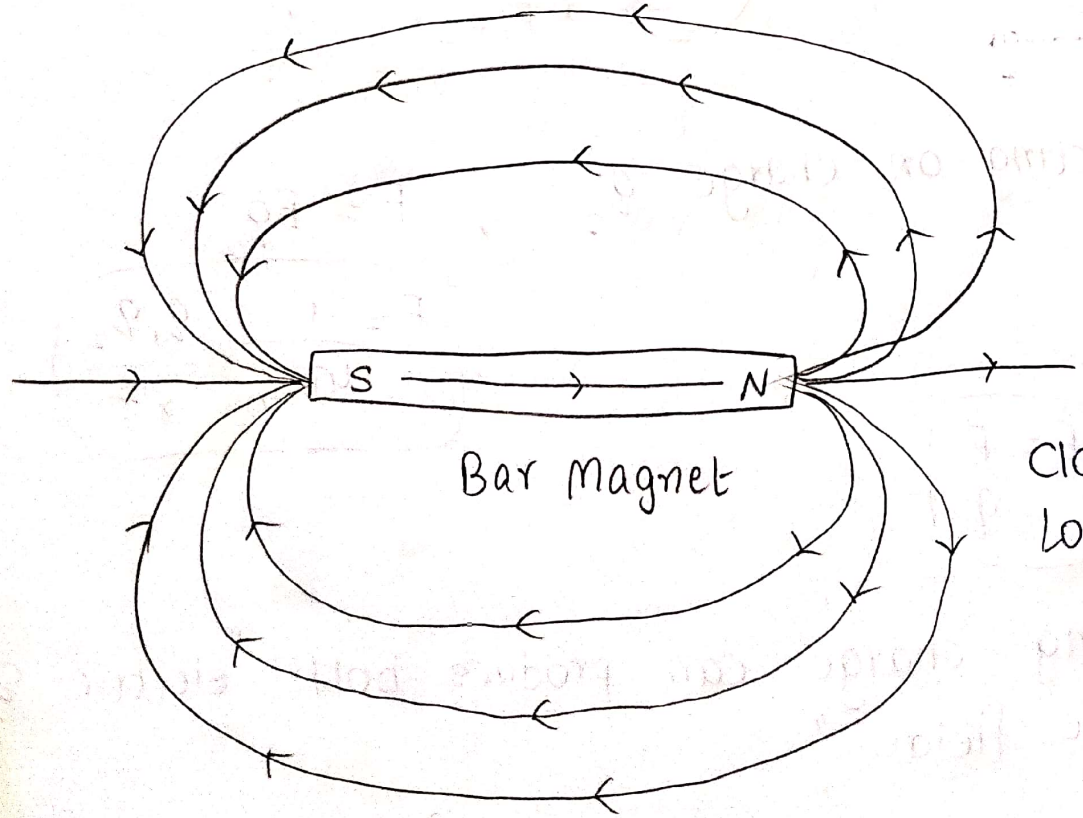
After cutting, each piece behaves as a new magnet.



Unlike Poles attract



Like Poles repel



* Outside the magnet, direction of field curves is from N-Pole to S-Pole.
* Inside the magnet, direction of field curves is from S-Pole to N-Pole.

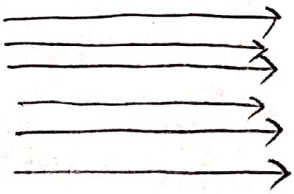
Direction of $\vec{B}$:-

⊗ ⊗ ⊗ ⊗ ⊗ ⊗

↳ Represents direction of magnetic field is into the plane of board

⊙ ⊙ ⊙ ⊙ ⊙

↳ Represents direction of magnetic field is out of plane of board.



In the plane of board & in the direction of arrow.

Force acting on a moving charge in a magnetic Field:-

Ampere observed that,

① Stationary charges do not effect by magnetic field.

② Magnetic force acting on a moving charge,

a) $F_m \propto B$ (Strength of magnetic field)

b) $F_m \propto q$ (magnitude of charge)

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c) $F_m \propto V$ (Speed of charge)

d) $F_m \propto \sin\theta$ | angle b/w $\bar{B}$ & $\bar{V}$ is θ)

By Combining,

$$F_m \propto BqV\sin\theta$$

$$\boxed{F_m = kBqV\sin\theta}$$

* If all quantities are measured in S.I unit then $k=1$

$$\Rightarrow \boxed{F_m = BqV\sin\theta}$$

In Vector form, $\boxed{\bar{F}_m = q\cdot(\bar{V}\times\bar{B})}$

then, $\boxed{B = \dfrac{F_m}{qV\sin\theta}}$

Direction of Magnetic field :-

If $q = 1C$; $V = 1m/s$ and $\theta = 90°$ then

$$\boxed{B = F_m}$$

* B is the magnetic field force acting on a unit +ve charge particle moving with a unit velocity $\perp^{ar}$ to $\bar{B}$

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* B is a vector.

Units:- Tesla (T) → S.I. Unit

Guass (G) → C.G.S. Unit

$1\,T = 10^4$ Guass

$T = \frac{Wb}{m^2}$

Methods to find F_m :- (Direction)

① Fleming's Left Hand Rule.

Thumb - Force

Fore finger - Magnetic field

Middle finger - Velocity.

② Right hand Palm Rule

Palm - Force

Fore Finger - B

Thumb - Velocity

③ Maxwell Right hand Thumb Rule.

Points:-

① $\bar{F}_m = q(\bar{v} \times \bar{B})$; $\bar{a} = \frac{\bar{F}_m}{m}$

② $\bar{F}_m \perp \bar{v} \Rightarrow \bar{F}_m \cdot \bar{v} = 0 \Rightarrow \bar{a} \cdot \bar{v} = 0$

③ $\bar{F}_m \perp \bar{B} \Rightarrow \bar{F}_m \cdot \bar{B} = 0 \Rightarrow \bar{a} \cdot \bar{B} = 0$

④ $\bar{F} \parallel \bar{v} \times \bar{B}$ if q is +ve

$\bar{F} \parallel -(\bar{v} \times \bar{B})$ if q is -ve

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(5) $\bar{V} \cdot (\bar{V} \times \bar{B}) = 0$

$\bar{B} \cdot (\bar{V} \times \bar{B}) = 0$

(6) Power given by magnetic field,

$P = \bar{F}_m \cdot \bar{V} = 0$

$\frac{W}{t} = 0$

$\boxed{W = 0}$ i.e, work done by magnetic field

$\Rightarrow \boxed{\Delta KE = 0}$ is always zero.

∴ Magnetic field will never accelerate or deccelera-te the charged particle but it can change the direction of motion.

Path of the Charged Particle in Magnetic field:-

a) When charged particle is projected at an angle $\theta = 0°$ (or) $180°$ with field:-

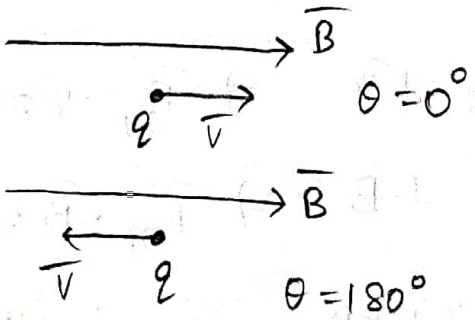
$F_m = BqV \sin\theta$

Here $\boxed{\sin\theta = 0}$

$\Rightarrow F_m = 0$

$\Rightarrow \bar{a} = 0$

$\Rightarrow \boxed{\bar{V} = \text{constant}}$



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i.e, It moves along "straight line path"

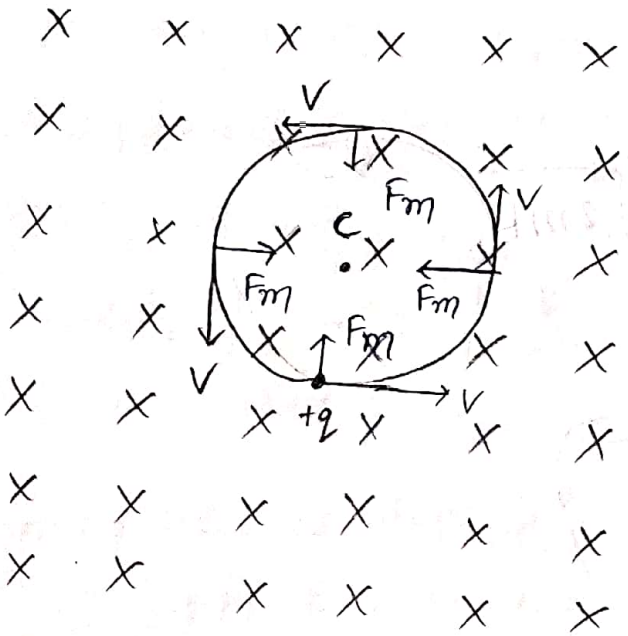
i.e, $\boxed{\bar{V} = \bar{u}}$

$\Rightarrow u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$

$\Rightarrow \boxed{u_x^2 + u_y^2 + u_z^2 = V_x^2 + V_y^2 + V_z^2}$

b) When charged particle is projected at an $\theta = 90^\circ$ with field :-

i) Projected from a point inside the field :-



$\bar{B}$ (Uniform Magnetic Field)

Here $\theta = 90^\circ$

$\Rightarrow \sin\theta = 1$ (max)

$\boxed{F_m = BqV \text{ (max)}}$

Here charged particle follows a "circular path"

In this case,

① $F_m = BqV$ & $F_m = \frac{mv^2}{r}$ (centripetal force)

$BqV = \frac{mv^2}{r}$

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$$r = \frac{mv}{Bq} \quad (or) \quad v = \frac{Bqr}{m}$$

Momentum of Charged Particle,

$$\bar{P} = m\bar{v}$$

K.E of moving charge,

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{B^2q^2r^2}{m^2}\right)$$

$$K = \frac{B^2q^2r^2}{2m} \text{ Joules}$$

$$1\,eV = 1.6 \times 10^{-19}\ J$$

(or) $K = \frac{p^2}{2m} \Rightarrow P = \sqrt{2mK}$

Radius of Circle,

$$r = \frac{P}{Bq} = \frac{\sqrt{2mK}}{Bq}$$

* If the Charged particle is accelereated through a P.d of V-volt before entering into magnetic field, then $W = \Delta K$

$$\Rightarrow qV = \frac{1}{2}m(v^2 - u^2)$$

$$qV + \frac{1}{2}mu^2 = \frac{1}{2}mv^2$$

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If $u = 0$ then $\frac{1}{2}mv^2 = qV$

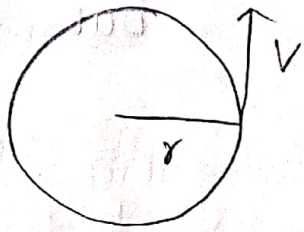
$$K = qV$$

Radius of Circular Path,

$$r = \frac{p}{Bq} = \frac{\sqrt{2mK}}{Bq} = \frac{\sqrt{2mVq}}{Bq}$$

Angular Velocity of Charged Particle,

$$\omega = \frac{v}{r} = \frac{Bq}{m}$$



Time Period of revolution,

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{Bq}$$

Frequency of revolution, $f = \frac{\omega}{2\pi} = \frac{Bq}{2\pi m}$

$\omega, f, T \longrightarrow$ depends on B, q & m

But doesn't depends on V & r.

Specific Charge, $\alpha = \frac{q}{m}$

In terms of specific charge,

$\omega = B\alpha$; $T = \frac{2\pi}{B\alpha}$; $f = \frac{B\alpha}{2\pi}$

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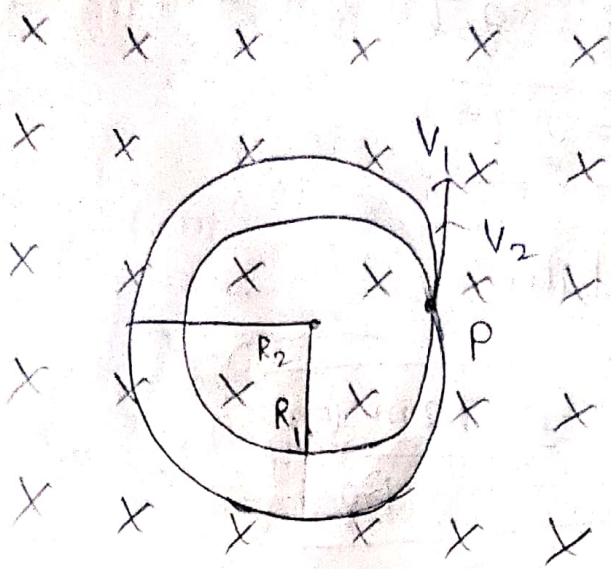
Ex:-



$V_1 < V_2$

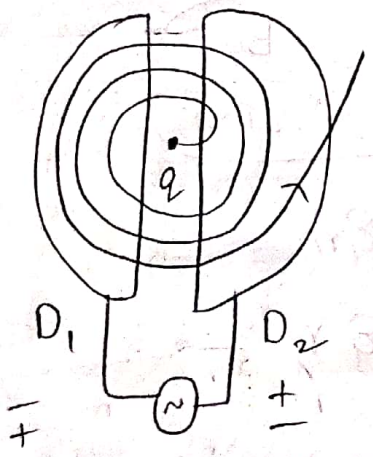
Here, $q_1 = q_2$

& $m_1 = m_2$

Sol:- then $R_1 < R_2$

but $T_1 = T_2$ ⟶ They return to P at same time.

Cyclotron:- A device used to accelerate charged particle.



high frequency oscillator

* D_1 & P_2 are hollow metal boxes called Dees
* Oscillator can produce electric field blw gap only.
* D_1 & D_2 can change their polarities alternatively at equal time intervals.

Principal of Cyclotron :- Resonance

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* A Strong Magnetic field is applied to $\perp^{lar}$ to the plane of dees.

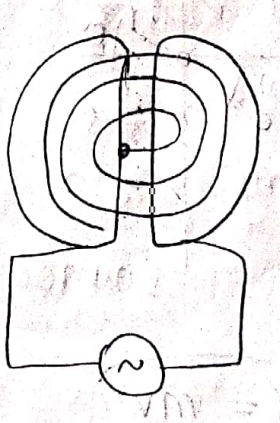
> Frequency of Charged Particle = Frequency of Oscillator

Ex:- Singly charged heavy ions of mass 'm' each are accelerated in a Cyclotron so that their maximum orbital radius is r. The frequency of cyclotrons oscillator is equal to f and the effective accelerating voltage across the dees is V. Neglecting the gap blw dees & ignoring the relativistic effect. The Total time of acceleration of each ion inside the cyclotron is

A) $\frac{\pi m f r^2}{2eV}$ B) $\frac{\pi^2 m f r^2}{eV}$ C) $\frac{\pi^2 m f r^2}{2eV}$ D) $\frac{\pi m f r^2}{eV}$

Sol:- $W = K.E$

$$n(eV) = \frac{B^2 q^2 r^2}{2m} \rightarrow (1)$$



Time taken for one complete revolution $= T = \frac{1}{f}$

Time spent in each dee $= \frac{T}{2} = \frac{1}{2f}$

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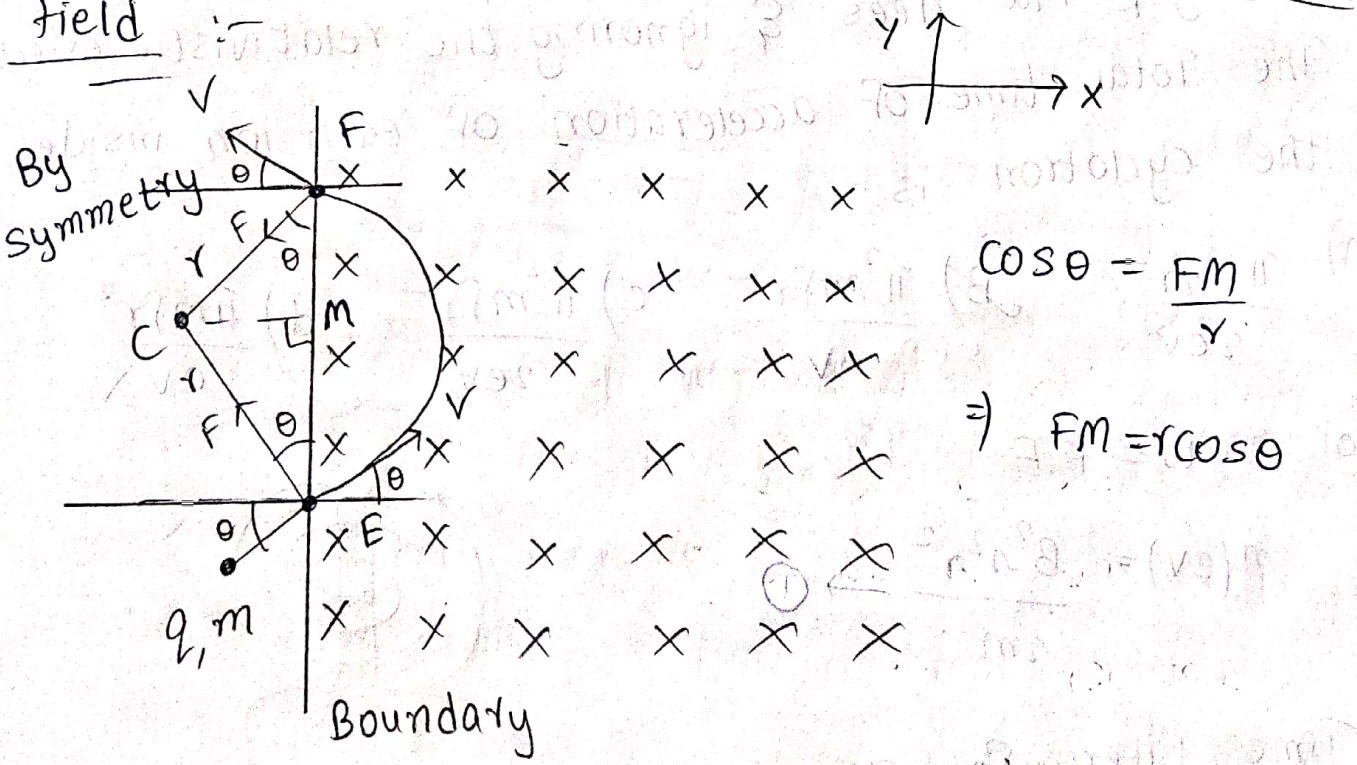
Total time spent, $t = nT_1 = \frac{n}{2f}$

From ①, $n = \frac{B^2q^2r^2}{2mev}$ & $f = \frac{Bq}{2\pi m}$

$\Rightarrow t = \frac{B^2q^2r^2}{4mevf} \times \frac{fm}{fm}$

$$\boxed{t = \frac{\pi^2 m f r^2}{ev}}$$

Charged Particle is projected from outside the field :-



$\cos\theta = \frac{FM}{r}$

$\Rightarrow FM = r\cos\theta$

Intercept, $EF = 2FM$

$= 2r\cos\theta$

$$\boxed{r = \frac{mv}{Bq}} \rightarrow ①$$

Angle turned by particle in magnetic field,

$$\phi = \pi - 2\theta$$

Time spent by the charged particle in magnetic field, $t = \frac{T}{2\pi}\phi$

$$t = \frac{2\pi m}{Bq\,2\pi}(\pi - 2\theta)$$

2π rad $\rightarrow T$

1 rad $\rightarrow \frac{T}{2\pi}$

$\phi \rightarrow \frac{T}{2\pi}\phi$

$$\boxed{t = \frac{m}{Bq}(\pi - 2\theta) = \frac{(\pi - 2\theta)}{B\alpha}}$$

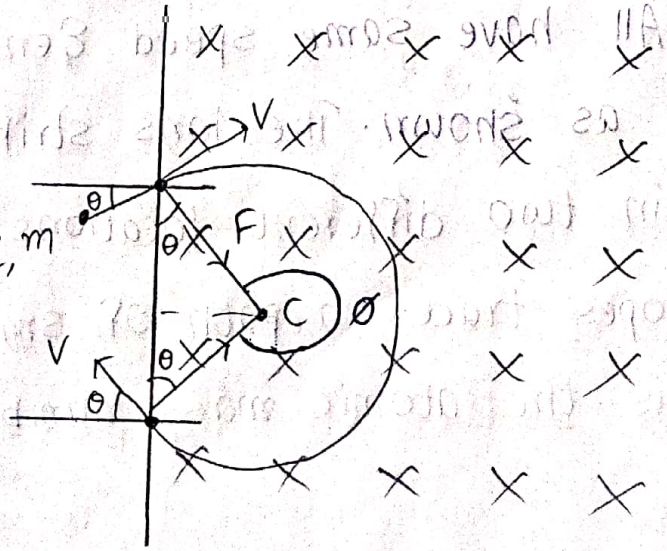
Impulse acting on the charged particle due to magnetic field

$$\vec{J} = \vec{P_f} - \vec{P_i} = m\left(v\cos\theta(-\hat{i}) + v\sin\theta\,\hat{j}\right) - m\left(v\cos\theta\,\hat{i} + v\sin\theta\,\hat{j}\right)$$

$$= -2mv\cos\theta\,\hat{i}$$

$$\boxed{\vec{J} = 2mv\cos\theta(-\hat{i})}$$

If the charged particle is +ve,



$$\phi = \pi + 2\theta$$

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Angle turned by particle in magnetic field

$$\phi = \pi + 2\theta$$

Time spent by charged particle in field,

$$t = \frac{T}{2\pi}\phi$$

$$t = \frac{2\pi m}{Bq\,2\pi}(\pi + 2\theta)$$

$$\boxed{t = \frac{m(\pi+2\theta)}{Bq} = \frac{\pi+2\theta}{B\alpha}}$$

Impulse acting on the charged particle due to magnetic field,

$$\bar{J} = \bar{P}_f - \bar{P}_i$$

$$\boxed{\bar{J} = 2mv\cos\theta\,(-\hat{i})}$$

Ex:- A Beam of single ionised atom of carbon, each charge +ve. All have same speed Enter a mass spectometer as shown. The ions strike the photographic plate in two different locations 5cm apart. The C_{12}^{6} isotopes traces a path of smaller radius 15cm. What is the atomic mass number of second isotope.

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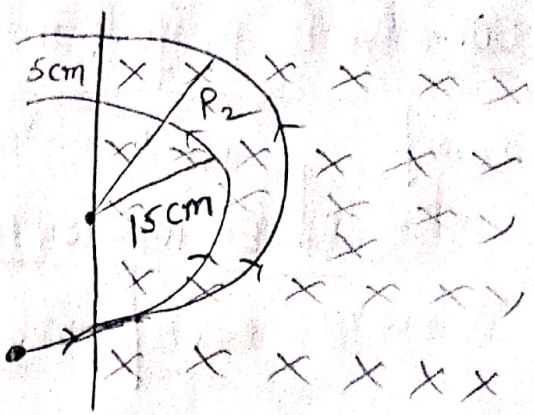
Sol:- From fig,



$$2R_2 - 2R_1 = 5\,cm$$

$$R_2 - R_1 = 2.5\,cm \rightarrow (1)$$

$$\Rightarrow R_2 = 17.5\,cm$$

$$R = \frac{mV}{Bq}$$

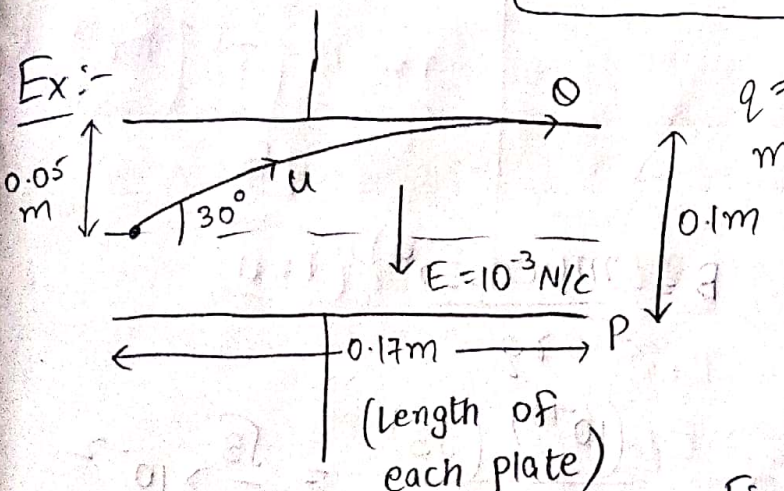
$$\Rightarrow R \propto m \Rightarrow \frac{R_1}{R_2} = \frac{m_1}{m_2}$$

$$\Rightarrow \frac{15}{17.5} = \frac{12}{x}$$

$$\Rightarrow x = 4\left(\frac{17.5}{5}\right)$$

$$\boxed{x = 14\ amu}$$

Ex:-



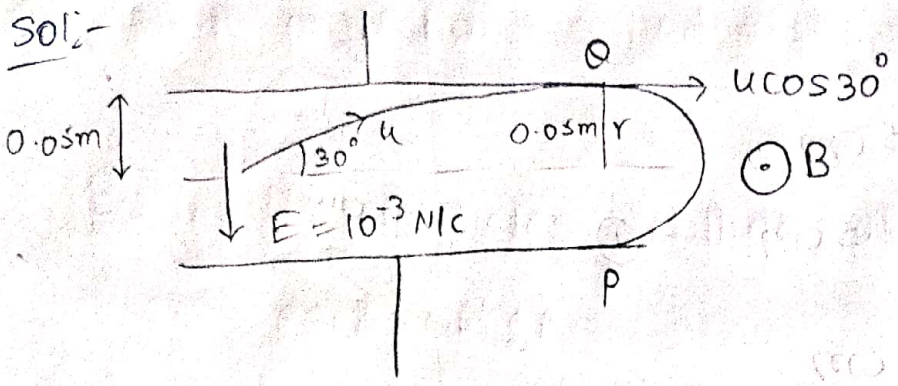
$q = 10^{-6}\,C$

$m = 10^{-10}\,kg$

[Neglect gravity]

Just outside the plate magnetic field is present. Find the velocity of projection of the charged particle and magnitude of the magnetic field $\perp^{ar}$ to the plane of figure, if it has to graze the plate at P and Q $\parallel^{el}$ to surface of the plate.

Sol:-



$$a = \frac{F}{m} = \frac{Eq}{m} = \frac{(10^{-3})(10^{-6})}{10^{-10}} = 10\ m/s^2$$

$$t = \frac{l}{u\cos 30^\circ} \rightarrow (1)$$

$$v_y^2 - u_y^2 = 2a_y y$$

$$0 - u_y^2 = 2(-10)(0.05)$$

$$u_y = 1\ m/s$$

$$\Rightarrow u\sin\theta = 1$$

$$u\sin 30^\circ = 1$$

$$\boxed{u = 2\ m/s}$$

$$\& \quad r = \frac{mu\cos 30^\circ}{Bq} \Rightarrow B = \frac{mu\cos 30^\circ}{rq}$$

$$= \frac{(10^{-10})(\sqrt{3})}{(0.05)(10^{-6})} = \frac{\sqrt{3}}{5} \times 10^{-2}$$

Ex:- A charged particle $+q$, mass m is placed at a distance d from an another charged particle $(-2q, 2m)$ in a uniform magnetic field as shown

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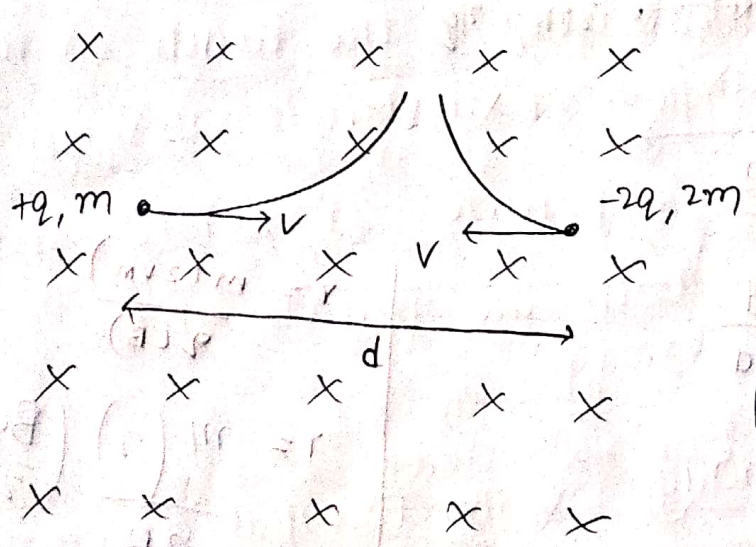
· if the particles are projected towards each other with equal speed v.

① Find the max value of projection speed v_m so that the two particles don't collide

② Find the time after which collision occurs b/w the particles if the projection speed is equal to v_m ③ Assuming the collision to be perfectly inelastic. Find the radius of path subtended.



$T = \frac{2\pi m}{Bq}$ which is same for both

Sol:-

① Condition for no collision,

$$r_1 + r_2 < d$$

$$\frac{mv_m}{Bq} + \frac{2mv_m}{B(2q)} < d$$

$$\frac{2mv_m}{Bq} < d$$

$$v_m < \frac{Bqd}{2m}$$

$$(v_m)_{max} = \frac{Bqd}{2m}$$

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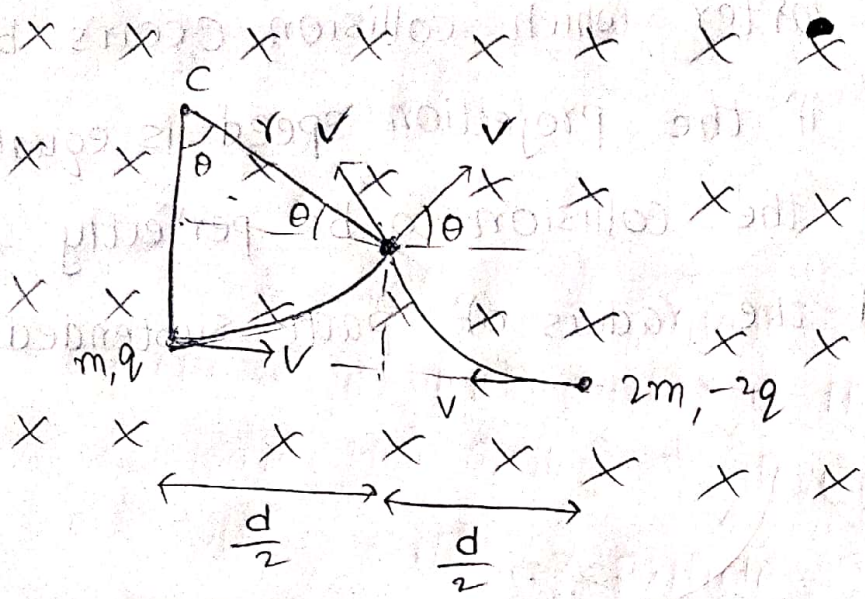
② Given projection speed $= 2v_m = \frac{Bqd}{m}$

(V)

Both the particles collide at a distance $\left(\frac{d}{2}\right)$ from each other.



Sol:- $\sin\theta = \frac{d}{2r} = \frac{d}{2d}$

$\Rightarrow \theta = \frac{\pi}{6}$

$\Rightarrow \omega t = \frac{\pi}{6}$

$\Rightarrow \boxed{t = \frac{\pi}{6\omega} = \frac{\pi m}{6Bq}}$

$r = \frac{m(2v_m)}{q(B)}$

$r = \frac{m(2)}{qB}\left(\frac{Bqd}{2m}\right)$

$\Rightarrow \boxed{r = d}$

③ L.C.L.M, $\vec{P_i} = \vec{P_f}$

$\Rightarrow mv\cos\theta\,\hat{i} + mv\sin\theta\,\hat{j} + 2mv\cos\theta\,(-\hat{i})$

$+2mv\sin\theta\,\hat{j} = 3m\vec{v_c}$

$\left[\theta = \frac{\pi}{6}\right]$

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$$3\bar{V}_C = \frac{\sqrt{3}}{2}V\hat{i} + \frac{V}{2}\hat{j} - \sqrt{3}V\hat{i} + V\hat{j}$$

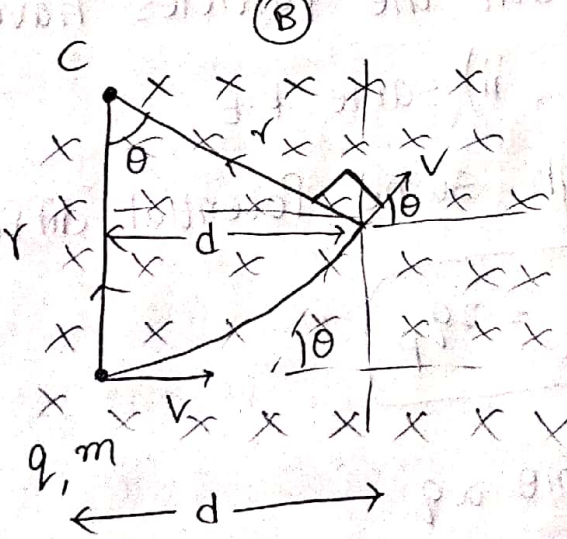
$$3\bar{V}_C = \frac{-\sqrt{3}}{2}\hat{i} + \frac{3V}{2}\hat{j}$$

$$3|\bar{V}_C| = \sqrt{\frac{3}{4} + \frac{9}{4}}V = \sqrt{3}V$$

$$\boxed{|\bar{V}_C| = \frac{V}{\sqrt{3}}}$$

$$R = \frac{(3m)V_C}{Bq} = \frac{(3m)V}{\sqrt{3}Bq} = \sqrt{3}d$$

Angle of Deviation of a Charged Particle :-



Angle of deviation

$$\sin\theta = \frac{d}{r} = \frac{Bqd}{mv}$$

$$\boxed{\theta = \sin^{-1}\left(\frac{Bqd}{mv}\right)}$$

Points:- If $r > d \Rightarrow \sin\theta < 1$

$$\Rightarrow \boxed{\theta < \frac{\pi}{2}\ \text{rad}}$$

If $r = d \Rightarrow \sin\theta = 1$

$$\Rightarrow \boxed{\theta = \frac{\pi}{2}}$$

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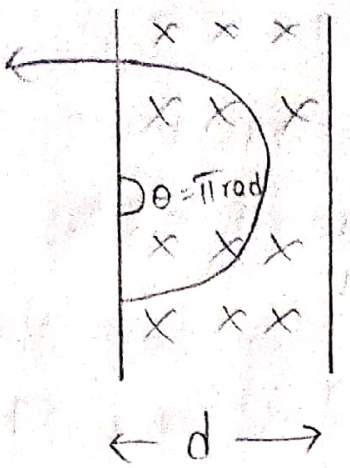
If $r < d \Rightarrow \sin\theta > 1$; doesn't exist



Ex:- A proton & an α-particle are projected in a magnetic field which exists in the width of region d. Compare the angles of deviation suffered by the proton & the α-particle. If before entering the magnetic field, both the particles have

i) same momentum ii) same K.E

iii) are accelerated through same Potential difference

Sol:- $\boxed{m_\alpha = 4m_p \; ; \; q_\alpha = 2q_p}$

i) $\sin\theta = \dfrac{Bqd}{mv} \Rightarrow \sin\theta \propto q$

$$\frac{\sin\theta_p}{\sin\theta_\alpha} = \frac{q_p}{q_\alpha} = \frac{1}{2}$$

ii) $\sin\theta = \dfrac{Bdq}{\sqrt{2mK}} \Rightarrow \sin\theta \propto \dfrac{q}{\sqrt{m}}$

$$\frac{\sin\theta_p}{\sin\theta_\alpha} = \frac{q_p}{q_\alpha}\sqrt{\frac{m_\alpha}{m_p}} = \left(\frac{1}{2}\right)\left(\sqrt{\frac{4}{1}}\right) = 1$$

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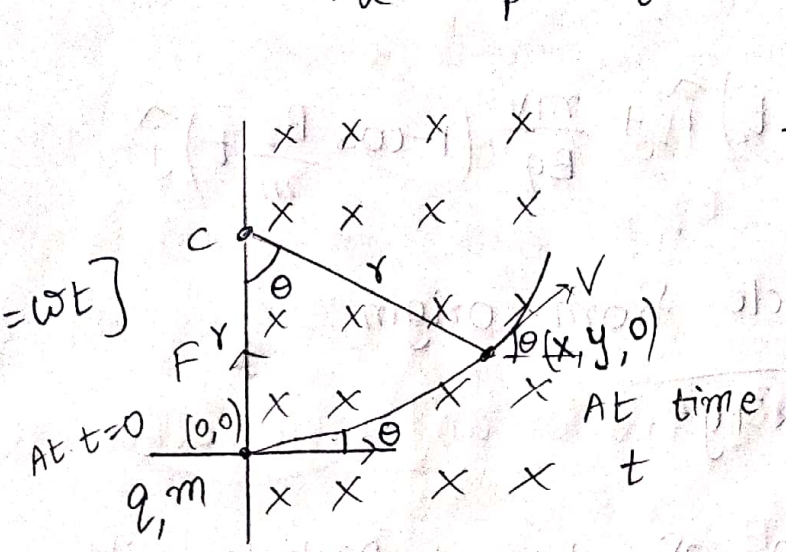
iii) $\sin\theta = \frac{dqB}{\sqrt{2mVq}} \Rightarrow \sin\theta \propto \sqrt{\frac{q}{m}}$

$$\frac{\sin\theta_p}{\sin\theta_m} = \sqrt{\frac{q_p}{q_\alpha} \cdot \frac{m_\alpha}{m_p}} = \sqrt{\frac{1}{2}\left(\frac{4}{1}\right)} = \sqrt{2}$$

*



$[\theta = \omega t]$

$\bar{B} = B(-\hat{k})$

$\bar{V} = V\hat{i}$

$\bar{F} = q(\bar{V} \times \bar{B})$

$= +\left(\hat{i} \times -\hat{k}\right)$

$= +\hat{j}$

i.e, particle motion is confined in x-y plane

radius $r = \frac{mV}{Bq} = \frac{P}{Bq} = \frac{\sqrt{2mk}}{Bq}$

At any time t,

Velocity of a particle,

$\bar{V} = V\cos\theta\,\hat{i} + V\sin\theta\,\hat{j}$

$\bar{V} = V\cos(\omega t)\hat{i} + V\sin(\omega t)\hat{j}$

$$\bar{V} = V\cos\left(\frac{Bq}{m}t\right)\hat{i} + V\sin\left(\frac{Bq}{m}t\right)\hat{j}$$

Clearly, $V^2 = V_x^2 + V_y^2 = \text{constant}$; $\left[\frac{q}{m} = \alpha\right]$

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Position coordinates of particle at time 't'.

$$\vec{r}_0 = x\hat{i} + y\hat{j}$$

Where $x = r\sin\theta$

$$y = r - r\cos\theta = r(1-\cos\theta)$$

$$\vec{r}_0 = \frac{mv}{Bq}\sin\left(\frac{Bq}{m}t\right)\hat{i} + \frac{mv}{Bq}\left(1-\cos\frac{Bq}{m}t\right)\hat{j}$$

Distance of particle from origin,

$$|\vec{r}_0| = \sqrt{x^2+y^2}$$

Direction of motion of charged particle with x-axis is,

$$\tan\theta = \frac{v_y}{v_x}$$

$$\boxed{\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right)}$$

Points:-

① At $t = \frac{T}{4} = \frac{2\pi}{4\omega} = \frac{\pi}{2\omega}$

$$\Rightarrow \vec{v} = v\hat{j}$$

$$\vec{r}_0 = r\hat{i} + r\hat{j}$$

② At $t = \frac{T}{2} = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$

$$\Rightarrow \vec{v} = -v\hat{i}$$

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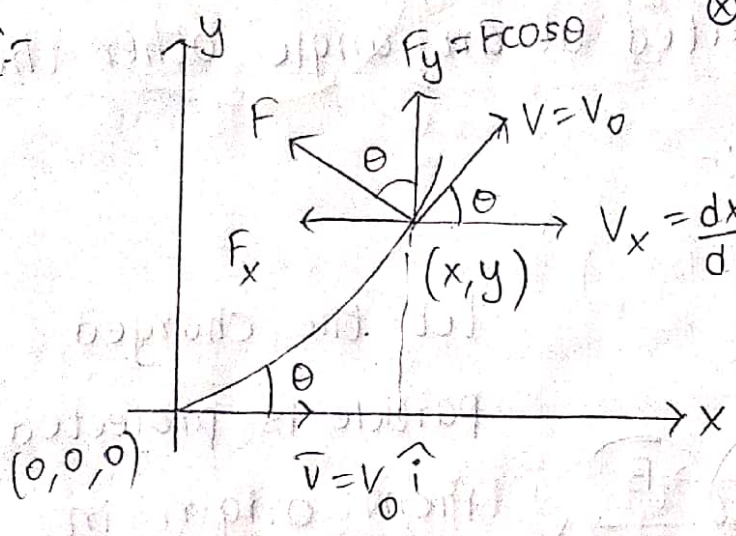
& $\overline{r_0} = 2r\hat{j}$

Ex:- A particle of charge q and mass m is projected from origin with a velocity $\overline{v} = v_0\hat{i}$ in a non-uniform magnetic field $\overline{B} = -B_0 x \hat{k}$ where B_0 & v_0 are +ve constants of proper directions. Find max +ve x-coordinate of particle during its motion

Sol:-



$\otimes\ \overline{B} = -B_0 x \hat{k}$

Force on the particle, $F = Bqv\sin 90°$

$F = B_0 x q V_0$

Now, $F_y = F\cos\theta$

$ma_y = B_0 x q V_0 \cos\theta$

$a_y = \frac{B_0 q}{m} x V_0 \cos\theta$

$\frac{dV_y}{dx}\left(\frac{dx}{dt}\right) = \frac{B_0 q}{m} x V_0 \cos\theta$

We have,

$V_0^2 = V_x^2 + V_y^2$

At max x-coordinate,

$V_x = 0$

$\Rightarrow \boxed{V_y = V_0}$

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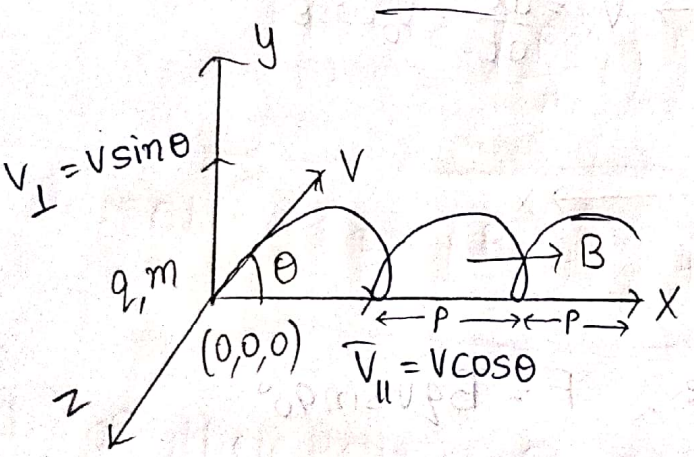
$$\int_0^{V_0} dV_y = \int_0^{x_{max}} \frac{B_0 q}{m} x \cdot dx$$

$$V_0 = \frac{B_0 q}{m} \frac{x_{max}^2}{2}$$

$$\boxed{x_{max} = \sqrt{\frac{2mV_0}{Bq}}}$$

When particle is projected at an angle other than $0°, 90°$ & $180°$ with $\bar{B}$:-



Let the charged particle is projected from origin in xy-plane at an angle θ with x-axis,

We have, $\bar{F} = q(\bar{V} \times \bar{B})$

$$\bar{F} = q\left[(V\cos\theta\,\hat{i} + V\sin\theta\,\hat{j}) \times B\hat{i}\right]$$

$$\boxed{\bar{F} = Bqv\sin\theta\,(-\hat{k})}$$

i.e, motion of charged particle lies in yz-plane

(Here z is -ve always)

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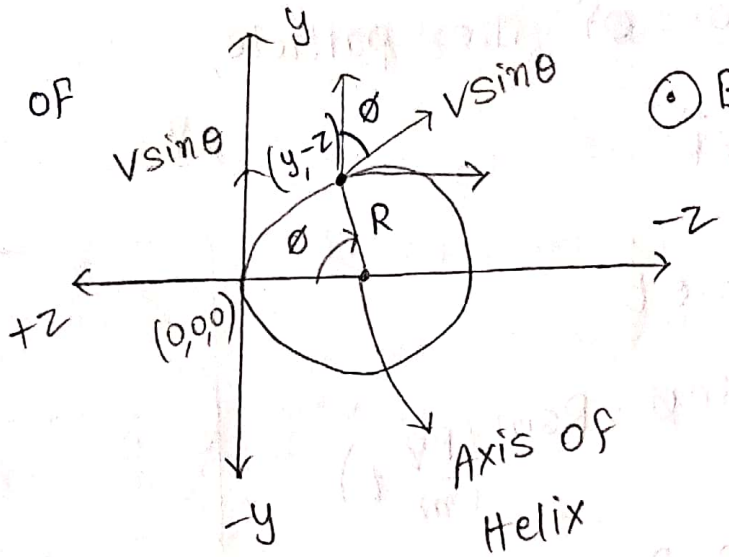
Path of Charged particle is "Helix (or) Helical path"

* Axis of Helix is parallel to x-axis

Projection of the path



$\phi = \omega t$

$y = R\sin\phi$

$-z = R(1-\cos\phi)$

Radius of Helix

$$R = \frac{mv\sin\theta}{Bq}$$

Time Period of Revolution,

$$T = \frac{2\pi m}{Bq}$$

[T doesn't depend on R (or) V]

* At any time 't',

Velocity vector of a particle is,

$$\vec{V} = V_x\hat{i} + V_y\hat{j} + V_z\hat{k}$$

$$\vec{V} = V\cos\theta\,\hat{i} + V\sin\theta\cos\phi\,\hat{j} + V\sin\theta\sin\phi\,(-\hat{k})$$

$$\vec{V} = V\cos\theta\,\hat{i} + V\sin\theta\cos(\omega t)\,\hat{j} + V\sin\theta\sin(\omega t)\,(-\hat{k})$$

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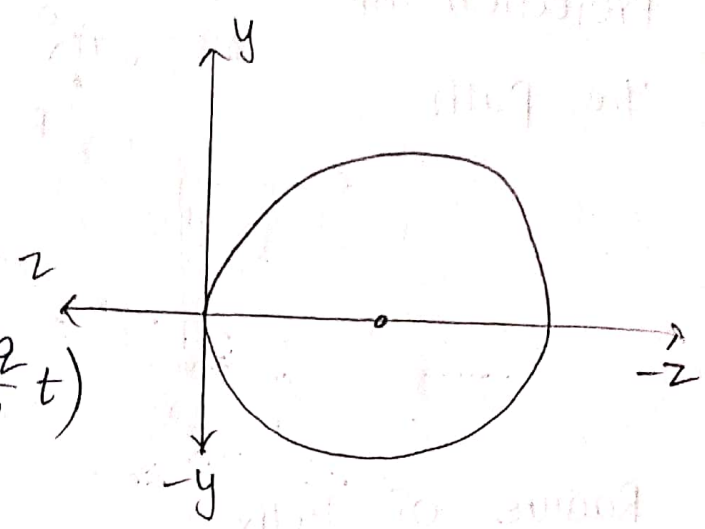
$$\bar{V} = V\cos\theta\,\hat{i} + V\sin\theta\cos\left(\frac{Bq}{m}t\right)\hat{j} + V\sin\theta\sin\left(\frac{Bq}{m}t\right)(-\hat{k})$$

Position vector of the particle,

$$\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$$



Here $x = V\cos\theta t$

$$y = R\sin\phi = R\sin\left(\frac{Bq}{m}t\right)$$

$$|z| = R - R\cos\phi$$

$$= R\left(1 - \cos\left(\frac{Bq}{m}t\right)\right)$$

$$\bar{r} = V\cos\theta t\,\hat{i} + R\sin\left(\frac{Bq}{m}t\right)\hat{j} + R\left(1-\cos\left(\frac{Bq}{m}t\right)\right)(-\hat{k})$$

Pitch of the Helix:- It is the distance travelled by the particle along B-direction during one complete revolution

i.e, $x = (V\cos\theta)T$

$$x = (V\cos\theta)\left(\frac{2\pi m}{Bq}\right)$$

$$x = \frac{2\pi m V\cos\theta}{Bq}$$

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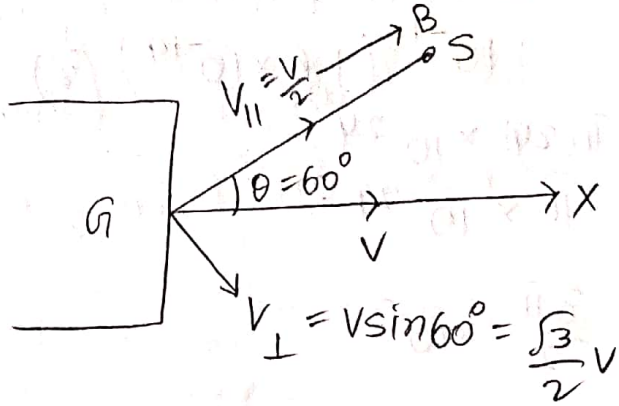
Ex:- An electric gun G emits electrons of energy 2×10^3 ev travelling in +ve x-direction. The e^- are required to hit the spot S where GS=0.1m and the line GS make 60° with x-axis. A Uniform magnetic field $\vec{B}$ parallel to GS exists in the region outside the electron gun. Find the min value of B needed to make the e^- hit S.

Sol:-



$K = 2\times10^3$ eV

$\frac{1}{2}mv^2 = 2\times10^3 \times 1.6\times10^{-19}$ J

$\frac{1}{2}(9\times10^{-31})v^2 = 2\times16\times10^{-17}$

$v^2 = \frac{4\times16}{9}\times10^{14}$

$v = \frac{2\times4}{3}\times(10^{14})^{1/2} = 2.66\times10^7$ m/s

$v = 2.66\times10^7$ m/s

Let the particle hit the spot 'S' after n-revolutions then,

$GS = np$

$$0.1 = n\left(\frac{2\pi m}{Bq}\right)\left(\frac{V}{2}\right)$$

$$B = \frac{n(2\pi)(mv)}{(0.1)\,q\,(2)}$$

For $B = B_{min}$; $n = 1$

$$B_{min} = \frac{2\pi\left(9\times10^{-31}\right)\left(\frac{8}{3}\times10^{7}\right)}{10^{-1}\left(1.6\times10^{-19}\right)(2)}$$

$$= \frac{\pi(24)\times10^{-24}}{16\times10^{-21}}$$

$$= \frac{3\pi}{2}\times10^{-3}$$

$$= \frac{3}{2}\left(\frac{22}{7}\right)\times10^{-3}$$

$$\boxed{B_{min} = 4.7\times10^{-3}\ \text{Tesla}}$$

Ex:- A particle of charge per mass 'α' is released from origin with velocity $\bar{v} = v_0\hat{i}$ in a uniform magnetic field $\bar{B} = -B_0\hat{k}$. If the particle passes through $(0, y, 0)$ then y is

A) $\frac{-2V_0}{B_0\alpha}$ B) $\frac{V_0}{B_0\alpha}$ C) $\frac{2V_0}{B_0\alpha}$ D) $\frac{-V_0}{B_0\alpha}$

Sol:- $\bar{F} = q(\bar{v}\times\bar{B}) = +ve\left(\hat{i}\times(-\hat{k})\right) = +\hat{j}$

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& $\bar{v} \perp \bar{B}$ $\Rightarrow$ Particle is in Circular Path of radius $r = \frac{mv_0}{B_0 q} = \frac{v_0}{B_0 \alpha}$

It passes through $(0, 2r, 0)$

$\Rightarrow y = 2r = \frac{2v_0}{B_0 \alpha}$

Ex:- A Charged Particle of Specific charge 'α' moves with a velocity $\bar{v} = v_0 \hat{i}$ in a magnetic field $\bar{B} = \frac{B_0}{\sqrt{2}}(\hat{j} + \hat{k})$ then

A) Path of particle is helix

~~B)~~ Path of particle is circle

~~C)~~ Distance moved by the particle in time $t = \frac{\pi}{B_0 \alpha}$ is $\frac{\pi v_0}{B_0 \alpha}$

D) Velocity of particle after time $t = \frac{\pi}{B_0 \alpha}$ is $\frac{v_0}{2}\hat{i} + \frac{v_0}{2}\hat{j}$

Sol:- $\bar{v} = v_0 \hat{i}$ & $\bar{B} = \frac{B_0}{\sqrt{2}}(\hat{j} + \hat{k})$ $\Rightarrow \bar{v} \perp \bar{B}$

$\Rightarrow$ Circular Path.

Distance = Speed x time $= v_0\left(\frac{\pi}{B_0 \alpha}\right) = \frac{\pi v_0}{B_0 \alpha}$

At any time t, $|\bar{v}| =$ constant

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$$\Rightarrow \left|\frac{V_0}{2}\hat{i} + \frac{V_0}{2}\hat{j}\right| = \sqrt{\left(\frac{V_0}{2}\right)^2 + \left(\frac{V_0}{2}\right)^2}$$

$$= \frac{V_0}{\sqrt{2}} \quad (\neq V_0)$$

$\Rightarrow$ Option D is wrong.

Ex:- Taking the radius of Earth as 6400 km and the horizontal component of Earth magnetic field at the Equator is 0.4×10^{-4} wb/m^2. With what min momentum could a proton circle the Earth at equator?

A) 6.4×10^{-17} kg m/s

~~B) 4.1×10^{-17} kg m/s~~

C) 7.1×10^{-19} kg m/s

D) 4.1×10^{-19} kg m/s

Sol:- $F_m = F_c \Rightarrow Bqv\sin\theta = \frac{mv^2}{r}$ $[\theta = 90^\circ]$

$$\Rightarrow Bqr = mv = p$$

$$\Rightarrow p = (4\times10^{-5})(1.6\times10^{-19})(64\times10^{5})$$

$$= 4096 \times 10^{-20}$$

$$= \boxed{4.096 \times 10^{-17} \text{ kg m/s}}$$

Ex:- A Particle of charge per mass 'α' is released from origin with velocity $\vec{v} = v_0\hat{i}$ in a magnetic

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field

$$\bar{B} = -B_0 \hat{k} \quad \text{for} \quad x \leq \frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha} \quad \&$$

$$\bar{B} = 0 \quad \text{for} \quad x > \frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha}$$

The x-coordinate of particle at time $t > \frac{\pi}{3B_0\alpha}$ would be

A) $\frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha} + \frac{\sqrt{3}}{2} V_0 \left(t - \frac{\pi}{3B_0\alpha}\right)$

B) $\frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha} + V_0 \left(t - \frac{\pi}{3B_0\alpha}\right)$

C) $\frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha} + \frac{V_0}{2} \left(t - \frac{\pi}{3B_0\alpha}\right)$

D) $\frac{\sqrt{3}}{2} \frac{V_0}{B_0 \alpha} + \frac{V_0 t}{2}$

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Motion of a charged particle in Combined Electric & magnetic fields:-

Lorentz Force:-

Net Force $\bar{F} = \bar{F}_e + \bar{F}_m$

$\Rightarrow$ acceleration of particle,

$$\bar{a} = \frac{\bar{F}}{m}$$

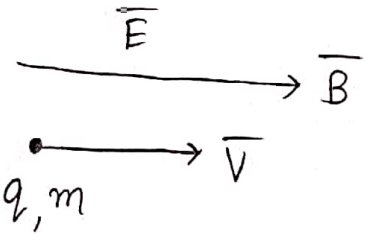
Cases:-

① All the three vectors $\bar{V}, \bar{E}$ & $\bar{B}$ are collinear / Parallel:-

Here $\bar{V} \parallel \bar{E} \parallel \bar{B}$

$$\bar{F}_m = q(\bar{V} \times \bar{B}) = 0$$

$$\bar{F}_e = q\bar{E}$$



Net Force (Lorentz Force),

$$\bar{F} = \bar{F}_e + \bar{F}_m$$

$$\bar{F} = q\bar{E} + \bar{0}$$

$$\boxed{\bar{F} = q\bar{E}}$$

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$$\bar{a} = \frac{\bar{F}}{m} = \frac{q\bar{E}}{m} = \text{constant}$$

Velocity of particle at any time 't'

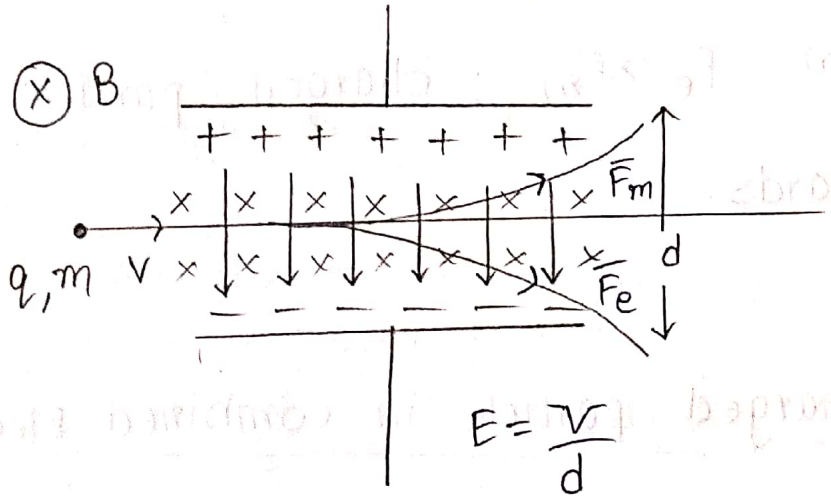
$$\bar{V}_f = \bar{u} + \bar{a}t$$

$$\boxed{\bar{V}_f = \bar{V} + \frac{q\bar{E}}{m}t}$$

i.e, path of the particle is straight line.

(2) All the three vectors are $\perp^{lar}$ to each other :-

i.e, $\bar{V} \perp \bar{E} \perp \bar{B}$



If both the fields are acting then

$$\bar{F} = \bar{F}_e + \bar{F}_m$$

$$\bar{F} = q\bar{E} + q(\bar{V} \times \bar{B})$$

For one particular velocity (V_0),

$$\boxed{\bar{F} = 0}$$

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$$q\bar{E} + q(\bar{V}_0 \times \bar{B}) = 0$$

$$\bar{E} = -(\bar{V}_0 \times \bar{B})$$

$$|\bar{E}| = |\bar{V}_0 \times \bar{B}|$$

$$E = V_0 B \sin 90^\circ$$

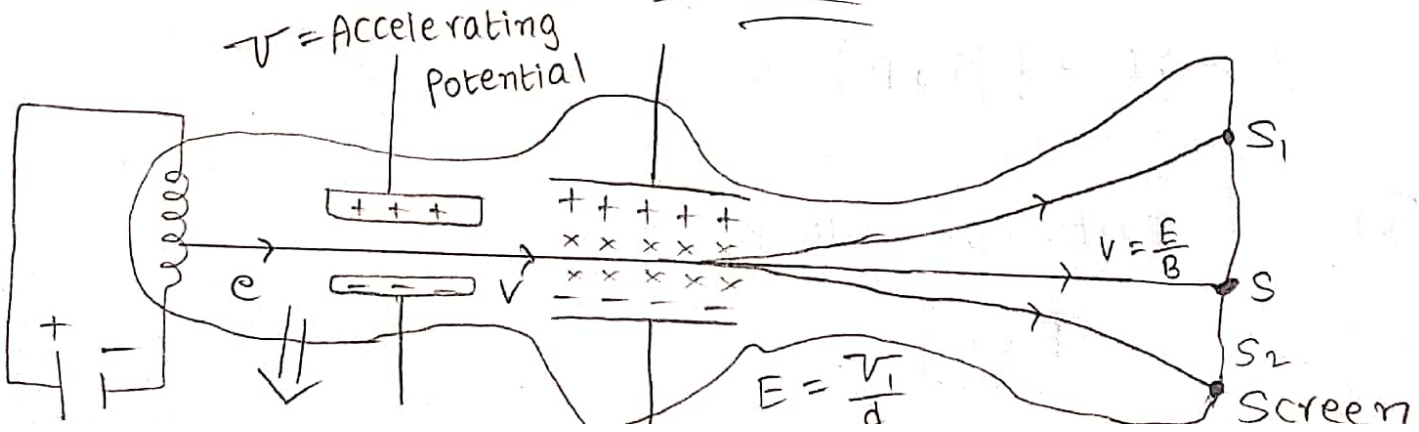
$$\boxed{V_0 = \frac{E}{B}}$$ → It is called Velocity Sector.

* IF $V > V_0$ then $F_m > F_e$; charged particle deflect upwards.

* IF $V < V_0$ then $F_e > F_m$; charged particle deflect downwards.

Motion of a charged particle in combined Electric & magnetic fields :- (Application)

J-J Thomson's Experiment :-



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$$eV = \frac{1}{2}mv^2$$

$$ev = \frac{1}{2}m\left(\frac{E}{B}\right)^2$$

$$\boxed{\frac{e}{m} = \frac{E^2}{2vB^2} = 1.76\times10^{11}\ C/kg}$$

Specific Charge

③ When $\bar{V} \perp \bar{E}$ & $\bar{V} \perp \bar{B}$ but $\bar{E} \parallel \bar{B}$:-

Let $\bar{E} = E\hat{j}$

$\bar{B} = B\hat{j}$

$\boxed{\bar{V} = V_0\hat{i}}$

Only due to electric field,

$$\bar{F}_e = q\bar{E} = qE\hat{j}$$

$$\Rightarrow \bar{a}_y = \frac{\bar{F}_e}{m} = \frac{qE}{m}\hat{j}$$ (Tangential acceleration)

Velocity after time 't' is,

$$\bar{V}_y = \cancel{\bar{u}_y}^{\,0} + \bar{a}_y t$$

$$\bar{V}_y = \frac{qE}{m}t\,\hat{j}$$

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Displacement of charged particle,

$$\bar{y} = \cancel{\bar{u}_y t}^{0} + \frac{1}{2}\bar{a}_y t^2 = \frac{1}{2}\bar{a}_y t^2$$

$$\boxed{\bar{y} = \frac{1}{2}\frac{qE}{m}t^2\hat{j}}$$

Only due to magnetic field,

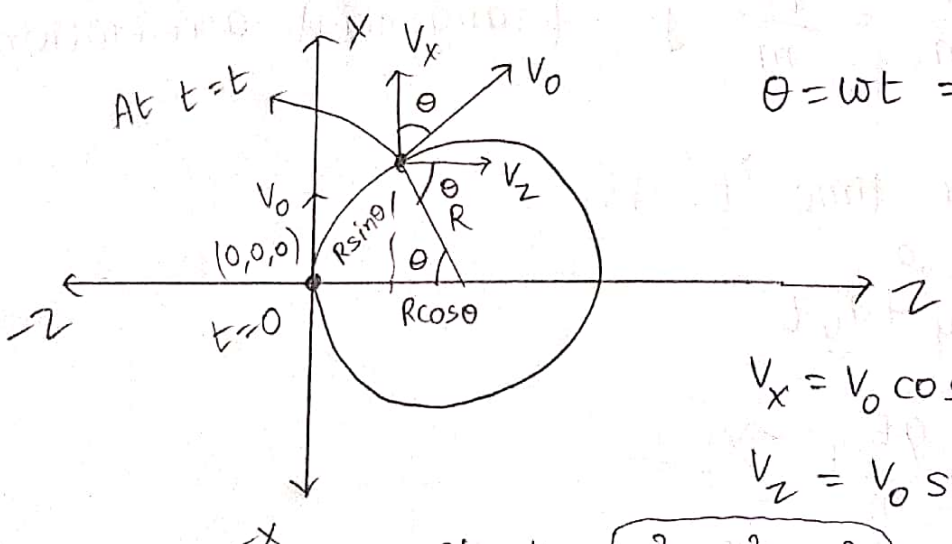
$$\bar{F}_m = q(\bar{V}_0 \times \bar{B})$$

$$\bar{F}_m = q(V_0\hat{i} \times B\hat{j})$$

$$\boxed{\bar{F}_m = qV_0B\hat{k}}$$

$$\bar{a}_m = \frac{\bar{F}_m}{m} = \frac{qV_0B}{m}\hat{k} \quad \text{(radial acceleration)}$$

Clearly, motion of charged particle is a circular path in xz-plane, which is $\perp^{lar}$ to $\bar{B}$



$$\theta = \omega t = \frac{Bq}{m}t$$

$$V_x = V_0\cos\theta$$

$$V_z = V_0\sin\theta$$

Clearly, $\boxed{V_0^2 = V_x^2 + V_y^2}$

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Velocity of the charged particle as a function of time,

$$\bar{V} = \bar{V_x} + \bar{V_y} + \bar{V_z}$$

$$\bar{V} = V_0 \cos\left(\frac{Bq}{m}t\right)\hat{i} + \frac{qE}{m}t\,\hat{j} + V_0 \sin\left(\frac{Bq}{m}t\right)\hat{k}$$

Position of the charged particle as a function of time,

$$\bar{r} = \bar{x} + \bar{y} + \bar{z}$$

From fig,

$$\sin\theta = \frac{x}{R} \Rightarrow x = R\sin\theta$$

where $R = \frac{mV_0}{Bq}$

& $z = R(1-\cos\theta)$

$$\bar{r} = \frac{mV_0}{Bq}\sin\left(\frac{Bq}{m}t\right)\hat{i} + \frac{1}{2}\frac{qE}{m}t^2\hat{j} + \frac{mV_0}{Bq}\left(1-\cos\frac{Bq}{m}t\right)\hat{k}$$

Clearly, Resultant path is the combined path of the straight line & circle called Helix with increasing pitch.

$$\left[\text{Pitch} = V_{\parallel} \times t\right] \Rightarrow \text{Pitch} = \frac{qEt}{m} \times \frac{2\pi m}{Bq}$$

M-II :- Given $\bar{V} = V_0\hat{i}$; $\bar{E} = E\hat{j}$; $\bar{B} = B\hat{j}$

Lorentz Force, $\bar{F} = \bar{F_e} + \bar{F_m}$

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$$m\frac{d\bar{v}}{dt} = q(\bar{E} + \bar{v}\times\bar{B})$$

Let $\bar{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ be the velocity of the charged particle at any time 't',

$$m\frac{d}{dt}(v_x\hat{i}+v_y\hat{j}+v_z\hat{k}) = q\left(E\hat{j} + (v_x\hat{i}+v_y\hat{j}+v_z\hat{k})\times B\hat{j}\right)$$

$$m\frac{dv_x}{dt}\hat{i} + m\frac{dv_y}{dt}\hat{j} + m\frac{dv_z}{dt}\hat{k} = q\left(E\hat{j} + (v_x B\hat{k} - v_z B\hat{i})\right)$$

Comparing the coefficients on both sides.

$$m\frac{dv_x}{dt} = -qBv_z$$

$$\frac{dv_x}{dt} = \frac{-qBv_z}{m} \rightarrow (1)$$

$$m\frac{dv_y}{dt} = qE$$

$$\frac{dv_y}{dt} = \frac{qE}{m} \rightarrow (2)$$

$$m\frac{dv_z}{dt} = qv_xB$$

$$\frac{dv_z}{dt} = \frac{qB}{m}v_x \rightarrow (3)$$

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From eq ②,

$$\frac{dV_y}{dt} = \frac{qE}{m}$$

$$dV_y = \frac{qE}{m}\,dt$$

$$\int_0^{V_y} dV_y = \int_0^t \frac{qE}{m}\,dt$$

$$\boxed{V_y = \frac{qE}{m}\,t}$$

$$\frac{dy}{dt} = \frac{qE}{m}\,t$$

$$\int_0^y dy = \int_0^t \frac{qE}{m}\,t\,dt$$

$$\boxed{y = \frac{1}{2}\frac{qE}{m}\,t^2}$$

Clearly eq ① & ③ are "Coupled equations"

These equations can be decoupled by differentiating again.

$$\frac{d^2V_x}{dt^2} = \frac{-qB}{m}\left(\frac{dV_z}{dt}\right)$$

$$\frac{d^2V_x}{dt^2} = \frac{-qB}{m}\left(\frac{qB}{m}V_x\right)$$

$$\boxed{\frac{d^2V_x}{dt^2} = \frac{-q^2B^2}{m^2}V_x} \longrightarrow ④$$

and $$\frac{d^2V_z}{dt^2} = \frac{qB}{m}\left(\frac{dV_x}{dt}\right)$$

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$$\frac{d^2 v_z}{dt^2} = \frac{-q^2 B^2}{m^2} v_z \quad \longrightarrow (5)$$

Clearly eq (4) & (5) represent S.H.M with

$$\omega = \frac{Bq}{m}$$

Let the solutions of eq (4) & (5) be

$$v_x = A\cos\omega t + B\sin\omega t$$

$$v_z = C\cos\omega t + D\sin\omega t$$

[A, B, C, D → constants]

Conditions:-

(1) At $t=0$; $v_z = 0$ & $v_x = v_0$

$\Rightarrow$ $C = 0$

and $v_0 = A\cos 0 + B\sin 0$

$A = v_0$

From (1), $\frac{dv_x}{dt} = 0$ (at $t=0$)

$$\frac{dv_z}{dt} = \omega v_0$$

$$\frac{dv_x}{dt} = -v_0 \omega \sin\omega t + B\omega\cos\omega t$$

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At $t=0$, $\frac{dV_x}{dt}=0$.

$$0 = -V_0\omega\sin 0 + B\omega\cos 0$$

$$\Rightarrow \boxed{B=0}$$

& $\frac{dV_z}{dt} = D\omega\cos\omega t$

At $t=0 \Rightarrow \omega V_0 = D\omega\cos 0$

$$\boxed{D=V_0}$$

$\therefore V_x = V_0\cos\omega t$ & $V_z = V_0\sin\omega t$

Finally Velocity, $\bar{V} = \bar{V_x} + \bar{V_y} + \bar{V_z}$

$$= V_0\cos\left(\frac{Bq}{m}t\right)\hat{i} + \frac{qE}{m}t\hat{j} + V_0\sin\left(\frac{Bq}{m}t\right)\hat{k}$$

Again, $V_x = V_0\cos\omega t$

$$\frac{dx}{dt} = V_0\cos\omega t$$

$$\int_0^x dx = \int_0^t V_0\cos\omega t$$

$$x = \frac{V_0}{\omega}(\sin\omega t) = \frac{mV_0}{Bq}\sin\omega t = R\sin\frac{Bq}{m}t$$

and $V_z = V_0\sin\omega t$

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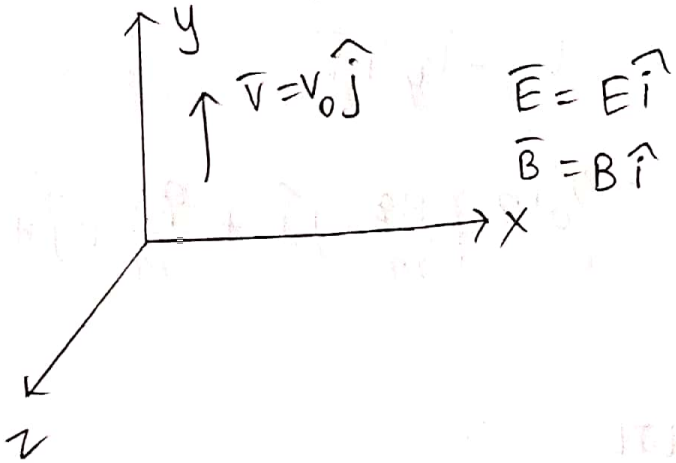
$$\frac{dz}{dt} = V_0 \sin\omega t$$

$$\int_0^z dz = \int_0^t V_0 \sin\omega t\, dt$$

$$\boxed{z = \frac{V_0}{\omega}(1-\cos\omega t)}$$

$$\boxed{z = \frac{mV_0}{Bq}\left(1-\cos\frac{Bq}{m}t\right)}$$

Ex:- Find Velocity as a function of time t,



Sol:- $$\bar{V} = \frac{qE}{m}t\,\hat{i} + V_0\cos\omega t\,\hat{j} + (V_0\sin\omega t)(-\hat{k})$$

Here $\hat{i} = \frac{\bar{E}}{|\bar{E}|}$; $\hat{j} = \frac{\bar{V}_0}{|\bar{V}_0|}$ $\left[\because \bar{V}\times\bar{B} = \hat{j}\times\hat{i} = -\hat{k}\right]$

$$\hat{k} = \frac{\bar{B}\times\bar{V}_0}{BV_0}$$

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$$\bar{v} = \frac{q\bar{E}}{m}t + \bar{v}_0 \cos\omega t + \frac{\sin\omega t}{B}(\bar{v}_0 \times \bar{B})$$

(4) $\bar{E} \perp \bar{B}$ and Charged particle is released from rest in combined fields:-

Take $\bar{E} = E\hat{j}$; $\bar{B} = B\hat{k}$

$$\bar{F}_m = q(\bar{v} \times \bar{B}) = +(\hat{j} \times \hat{k})$$

$$\bar{F}_m = \hat{i}$$

* Initially, Electric Force acts on the charged particle so, particle starts moving in $\bar{E}$ direction. Soon after this, magnetic force starts acting.
* Charged particle moves in circular path in xy-plane due to magnetic Force
* Now, resultant path is the combined motion of Straight line and circular path called "Cycloid Path".

Straight line ⟶ y direction

Circular path ⟶ x-y plane

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Lorentz Force,

$$\bar{F} = \bar{F}_e + \bar{F}_m$$

$$m\frac{d\bar{v}}{dt} = q\left(\bar{E} + \bar{v}\times\bar{B}\right) \qquad [v_z = 0]$$

Let $\bar{v} = v_x\hat{i} + v_y\hat{j}$ be the velocity at any time 't'

$$m\frac{dv_x}{dt}\hat{i} + m\frac{dv_y}{dt}\hat{j} = q\left(E\hat{j} + (v_x\hat{i} + v_y\hat{j}) \times B\hat{k}\right)$$

$$m\frac{dv_x}{dt}\hat{i} + m\frac{dv_y}{dt}\hat{j} = q\left(E\hat{j} + (v_x B)(-\hat{j}) + v_y B\hat{i}\right)$$

Comparing the coefficients,

$$m\frac{dv_x}{dt} = qBv_y \Rightarrow \frac{dv_x}{dt} = \frac{qBv_y}{m} \rightarrow (1)$$

& $$m\frac{dv_y}{dt} = q(E - v_x B) \Rightarrow \frac{dv_y}{dt} = \frac{q}{m}(E - v_x B) \rightarrow (2)$$

Eq (1) & (2) are Coupled equations,

$$\frac{d^2v_x}{dt^2} = \frac{qB}{m}\frac{d(v_y)}{dt} = \frac{qB}{m}\cdot\frac{q}{m}(E - v_x B)$$

$$\Rightarrow \frac{d^2v_x}{dt^2} = \frac{q^2B}{m^2}(E - v_x B) \rightarrow (3)$$

and $$\frac{d^2v_y}{dt^2} = \frac{-qB}{m}\left(\frac{dv_x}{dt}\right)$$

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$$\frac{d^2V_y}{dt^2} = \frac{-qB}{m}\left(\frac{qB}{m}V_y\right)$$

$$\frac{d^2V_y}{dt^2} = \frac{-q^2B^2}{m^2}V_y \quad \rightarrow (4)$$

This is S.H.M, $\boxed{\omega = \frac{Bq}{m}}$

Let the solution of this equation is,

$$V_y = A\sin(\omega t + \phi)$$

At $t=0$; $V_y = 0$

$$0 = A\sin(0+\phi)$$

$$\Rightarrow 0 = A\sin\phi \Rightarrow \boxed{\phi = 0}$$

At $t=0$; $V_x = 0$

$$\frac{dV_y}{dt} = \frac{qE}{m} \quad \rightarrow (5)$$

we have, $V_y = A\sin\omega t$

$$\frac{dV_y}{dt} = A\omega\cos\omega t$$

$$\frac{qE}{m} = A\omega\cos 0$$

$$A = \frac{qE}{m\omega} = \frac{qE}{m}\left(\frac{m}{Bq}\right) = \frac{E}{B}$$

$$\boxed{V_y = \frac{E}{B}\sin\left(\frac{Bq}{m}t\right)}$$

From (1),

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$$\frac{dv_x}{dt} = \frac{qB}{m}\left(\frac{E}{B}\right)\sin\omega t$$

$$\Rightarrow \int_0^{v_x} dv_x = \int_0^t \frac{qE}{m}\sin\omega t$$

$$\boxed{v_x = \frac{qE}{m\omega}(1-\cos\omega t)}$$

Finally,

$$\boxed{\vec{v} = \frac{E}{B}(1-\cos\omega t)\hat{i} + \frac{E}{B}\sin\omega t\,\hat{j}}$$

$$\left(\omega = \frac{Bq}{m}\right)$$

Position Vector,

$$\vec{r} = \vec{x} + \vec{y} \qquad \vec{x} = \int_0^x v_x\,dt \quad \& \quad \vec{y} = \int_0^y v_y\,dt$$

$$\boxed{\vec{r} = \frac{E}{B\omega}(\omega t - \sin\omega t)\hat{i} + \frac{E}{B\omega}(1-\cos\omega t)\hat{j}}$$

Here $\frac{E}{B\omega} = R =$ Radius of Circle

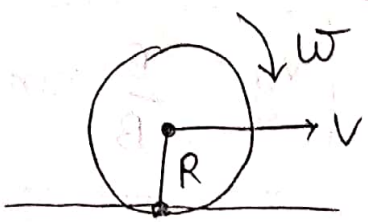
Speed,

$$|\vec{v}| = \frac{E}{B}\sqrt{(1-\cos\omega t)^2 + (\sin\omega t)^2}$$

$$= \frac{E}{B}\sqrt{2 - 2\cos\omega t} = \frac{2E}{B}\sin\left(\frac{\omega t}{2}\right)$$

$v = R\omega$

$$\frac{E}{B} = R\omega \Rightarrow \boxed{R = \frac{E}{B\omega}}$$



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$V=0$ when $\sin\left(\frac{\omega t}{2}\right)=0$

$\Rightarrow \frac{\omega t}{2} = 0, \pi, 2\pi, \ldots$

$t = 0, \frac{2\pi}{\omega}, \frac{4\pi}{\omega}, \ldots$

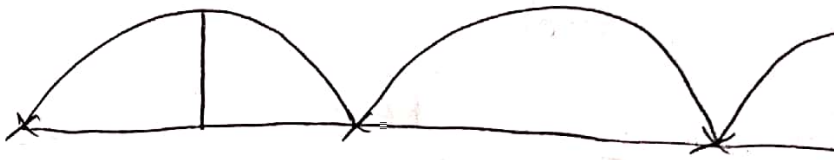
Distance travelled by charged particle before two instants that comes to rest momentarily,

$$d = \int_0^{2\pi/\omega} v\,dt = \int_0^{2\pi/\omega} \frac{2E}{B}\sin\left(\frac{\omega t}{2}\right)$$

$$= \frac{\frac{2E}{B}}{\frac{\omega}{2}}\left[\cos\left(\frac{\omega t}{2}\right)\right]_{2\pi/\omega}^{0}$$

$$= \frac{4E}{B\omega}\left[1+1\right] = \frac{8E}{B\omega} = 8R$$

$$\boxed{d = \frac{8E}{B\omega}}$$



Position co-ordinates of the particle:-

$$\frac{dx}{dt} = \frac{E}{B}(1-\cos\omega t)$$

$$\int_0^x dx = \int_0^t \frac{E}{B}(1-\cos\omega t)\,dt$$

$$\boxed{x = \frac{E}{B\omega}(\omega t - \sin\omega t)}$$

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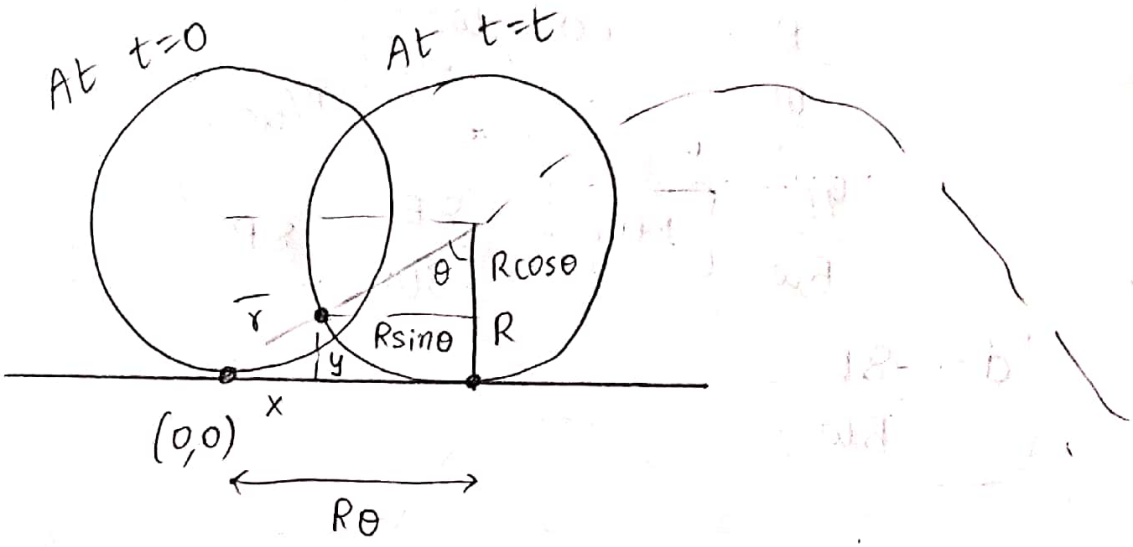
and ~~dVy~~ $\frac{dy}{dt} = \frac{E}{B}\sin\omega t$

$$\int_0^y dy = \int_0^t \frac{E}{B}\sin\omega t\, dt$$

$$\boxed{y = \frac{E}{B\omega}(1-\cos\omega t)}$$

$y_{max} = \frac{2E}{B\omega}$ when $\cos\omega t = -1$

$\Rightarrow \omega t = \pi$



$\vec{r} = x\hat{i} + y\hat{j}$

$\vec{r} = R(\theta - \sin\theta)\hat{i} + R(1-\cos\theta)\hat{j}$

where $\theta = \omega t$

$R = \frac{E}{B\omega}$

$V_x = \frac{dx}{dt} = R\omega(1-\cos\omega t)$ & $V_y = \frac{dy}{dt} = R\omega\sin\omega t$

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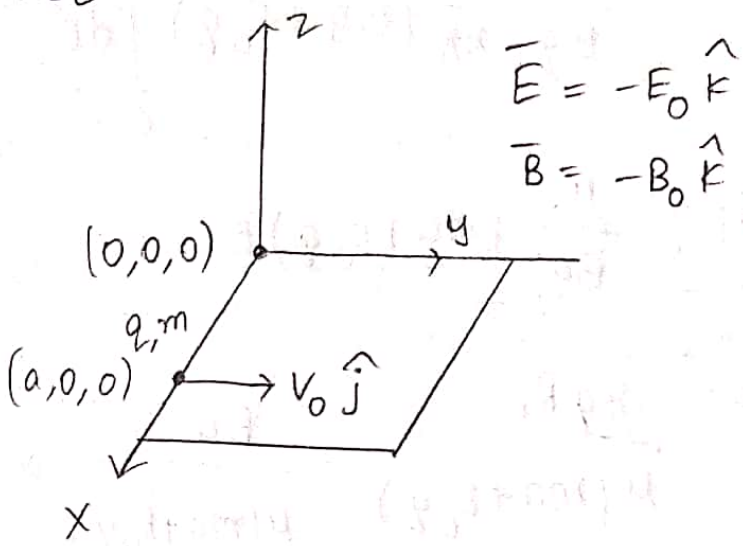
Ex:- The particle enters into the field at $(a,0,0)$ with velocity $= V_0 \hat{j}$. The particle starts moving into a circular path on a plane. If the coefficient of friction b/w the particle & plane is μ then calculate

① The time when the particle comes to rest

② the time when the particle hit the centre.

③ Distance travelled by the particle when it comes to rest.

Sol:-



$$\bar{E} = -E_0 \hat{k}$$

$$\bar{B} = -B_0 \hat{k}$$

① At any time t,

$$N = mg + E_0 q \longrightarrow ①$$

$$BqV = \frac{mv^2}{R} \longrightarrow ② \quad \text{[Radial Force]}$$

$$\Rightarrow \boxed{R = \frac{mv}{Bq}}$$

and

$$-m\frac{dv}{dt} = \mu N \longrightarrow ③ \quad \text{[Tangential Force]}$$

$$-m\frac{dv}{dt} = \mu(mg + E_0 q)$$

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$$\Rightarrow \int_{V_0}^{0} dv = \frac{-\mu}{m}(E_0 q + mg) \int_0^t dt$$

$$\boxed{t = \frac{mV_0}{\mu(E_0 q + mg)}}$$

(2) We have,

$$dR = \frac{m dv}{Bq}$$

$$\int_{R_i}^{0} dR = \frac{-m}{Bq} \frac{\mu}{m} (mg + E_0 q) \int_0^t dt$$

$$R_i = \frac{\mu}{Bq} (mg + E_0 q) t$$

$$t = \frac{Bq R_i}{\mu(mg + E_0 q)} = \frac{Bq}{\mu(mg + E_0 q)} \times \frac{mV_0}{Bq}$$

$$t = \frac{mV_0}{\mu(mg + E_0 q)}$$

(3) $$-m \frac{dv}{dl} \times \frac{dl}{dt} = \mu(mg + E_0 q)$$

$$-\int_{V_0}^{0} v dv = \frac{\mu}{m} (mg + E_0 q) \int_0^l dl$$

$$\boxed{l = \frac{V_0^2 m}{2\mu(mg + E_0 q)}}$$

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Ex:- A particle of specific charge α starst moving under the action of Electric field, $\bar{E} = E_0 \hat{i}$ & $\bar{B} = B_0 \hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} + 3\hat{j})$ Find the value of x_0.

Sol:- Total work done = Change in kinetic Energy.

$$W_E + W_m^{\nearrow 0} = \frac{1}{2} m (v^2 - u^2)^0$$

$$Eq(\Delta x) = \frac{1}{2} m v^2$$

$$(\Delta x) \frac{E_0 \alpha}{} = \frac{1}{2} (25)$$

$$\boxed{\Delta x = \frac{25}{2 E_0 \alpha}}$$

Ex:- A particle of charge 'q' & mass 'm' starts moving from origin under the action of electric field $\bar{E} = E_0 \hat{i}$, $\bar{B} = B_0 \hat{i}$ with velocity $\bar{v} = V_0 \hat{j}$. Find the time at which speed of particle becomes $2V_0$.

Sol:- Due to magnetic field,

$$\bar{V_0} = V_y \hat{j} + V_z \hat{k}$$

$$\Rightarrow V_0^2 = V_y^2 + V_z^2 \rightarrow (1)$$

Due to electric field,

$$\bar{V_x} = \frac{Eq}{m} t \, \hat{i}$$

Given speed $= 2V_0 \Rightarrow V_x^2 + V_y^2 + V_z^2 = 4V_0^2$

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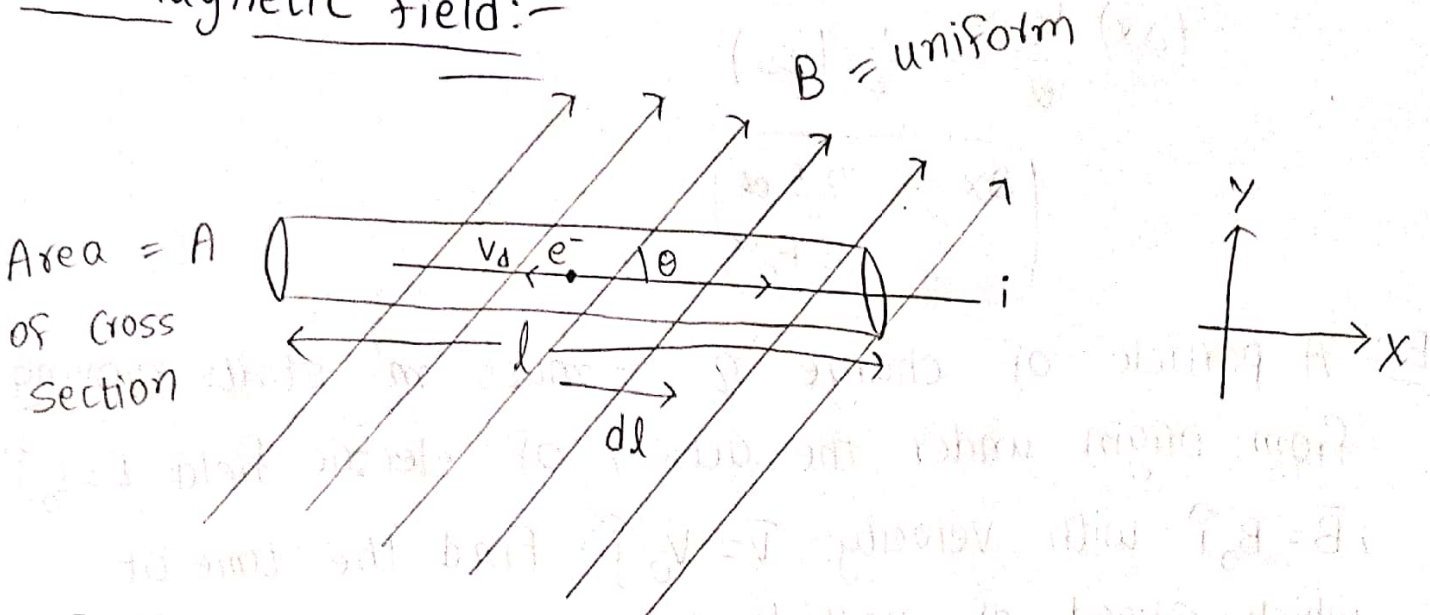
$$\Rightarrow V_x^2 + V_0^2 = 3V_0^2 + V_0^2$$

$$V_x = \sqrt{3}V_0$$

$$\Rightarrow \sqrt{3}V_0 = \frac{Eqt}{m}$$

$$\boxed{t = \frac{\sqrt{3}mV_0}{Eq}}$$

Force acting on a current carrying conductor in magnetic field:-



Force on each electron,

$$\bar{f} = -e(\bar{V}_d \times \bar{B}) \longrightarrow (1)$$

n = no. of charge carriers per unit volume

⇒ Charge contained in elementary volume

$$dq = (nA\,dl)e$$

Force on this element is,

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$$d\bar{f} = -neA\,d\ell\,(\bar{V}_d \times \bar{B})$$

If $d\bar{\ell}$ is the vector in the direction of current,

$$\bar{V}_d = V_d(-\hat{i})$$

$$d\bar{\ell} = d\ell(+\hat{i})$$

$$d\bar{f} = neAV_d(d\bar{\ell} \times \bar{B})$$

$$\boxed{d\bar{f} = i(d\bar{\ell} \times \bar{B})} \quad \text{where } i = neAV_d$$

$$\boxed{\bar{a}\cdot(\bar{b}\times\bar{c}) = [\bar{a}\,\bar{b}\,\bar{c}] = -\bar{b}\cdot(\bar{a}\times\bar{c})}$$

$$d\bar{f} = i\,d\bar{\ell} \times \bar{B}$$

$i\,d\bar{\ell}$ ↓ Current element, which is in the direction of current

Now, Total force acting on the wire,

$$\bar{F} = \int d\bar{f} = \int i\,d\bar{\ell} \times \bar{B}$$

$$= i\int d\bar{\ell} \times \bar{B}$$

$$\boxed{\bar{F} = i(\bar{l} \times \bar{B})}$$

Simply,

$F = qVB\sin\theta$

$= \left(\frac{q}{t}\right)(Vt)(B\sin\theta)$

$= i\ell B\sin\theta$

$= i(\bar{\ell} \times \bar{B})$

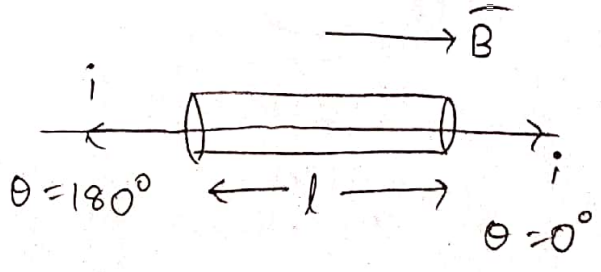
Points:-

$$|\bar{F}| = i\ell B\sin\theta$$

① If $\theta = 0^\circ$ or 180°

$\sin\theta = 0$

$\Rightarrow \boxed{F = 0}$



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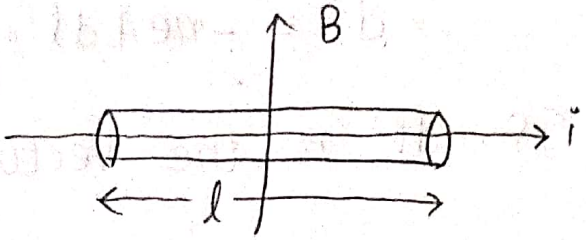
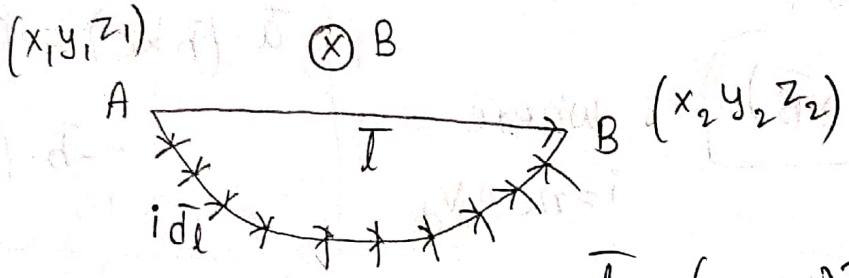
② If $\theta = 90^\circ$ then

$\sin\theta = 1$

$\Rightarrow \boxed{F = ilB}$ (max)



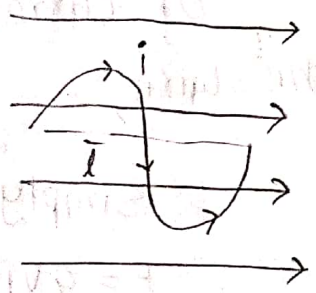
③ Conductor of any random shape,



$$\bar{l} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

$$\bar{F} = i(\bar{l} \times \bar{B})$$

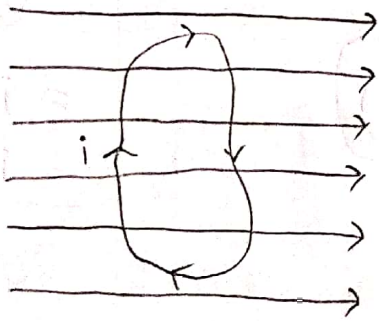
Ex:-



$\bar{l} \parallel \bar{B}$

$\Rightarrow \boxed{\bar{F}_{net} = 0}$

Ex:-



$\bar{B}$ = Uniform

Here $\bar{l} = 0$

$\Rightarrow \boxed{\bar{F}_{net} = 0}$

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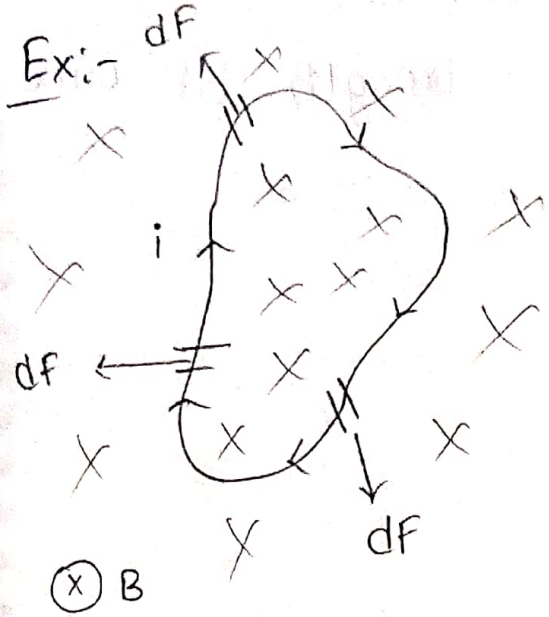
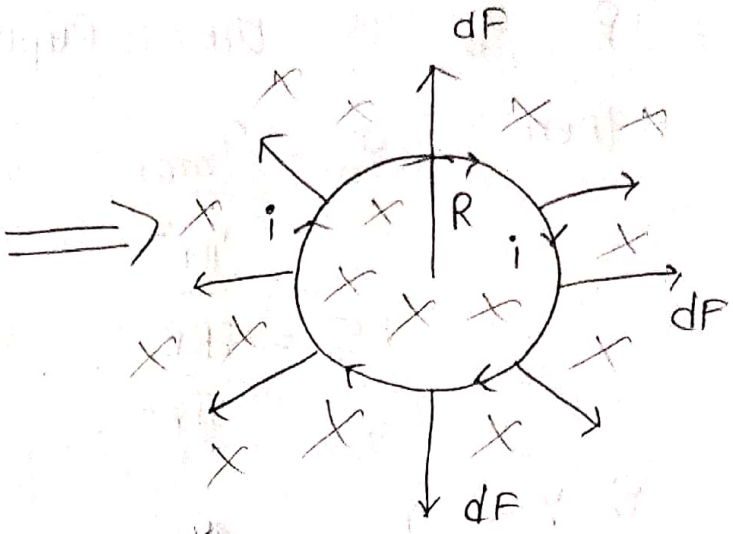
Ex:-



$F_{net} = 0$



$F_{net} = 0$

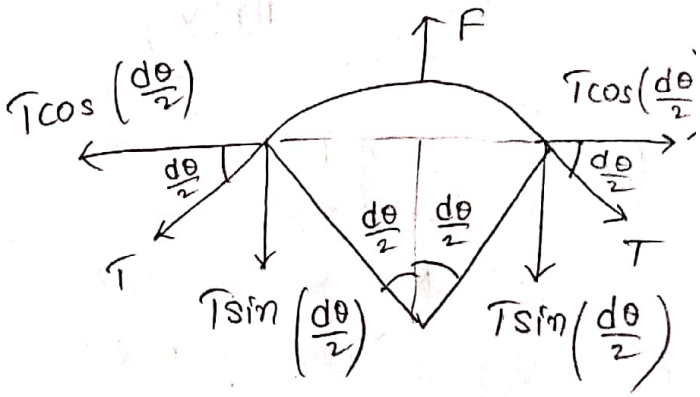
To find extra Tension developed in the wire,

In equillbrium,

$$F = 2T\sin\left(\frac{\theta}{2}\right)$$

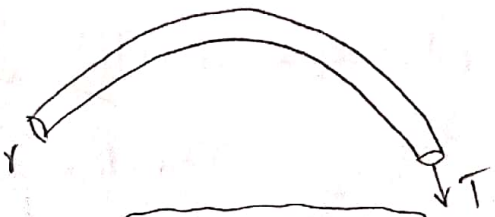
$$iR(d\theta)B = 2T\left(\frac{d\theta}{2}\right)$$

$$\boxed{T = iRB}$$



$l = R\,d\theta$

$$T = \frac{iIB}{2\pi R} \qquad [l = 2\pi R]$$



$$\text{Strain} = \frac{dr}{r}$$

Stress developed in the wire,

$$\sigma = \frac{T}{\pi r^2}$$

$$Y = \frac{\text{stress}}{\text{strain}}$$

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If σ_0 is the Ruptured strength of wire

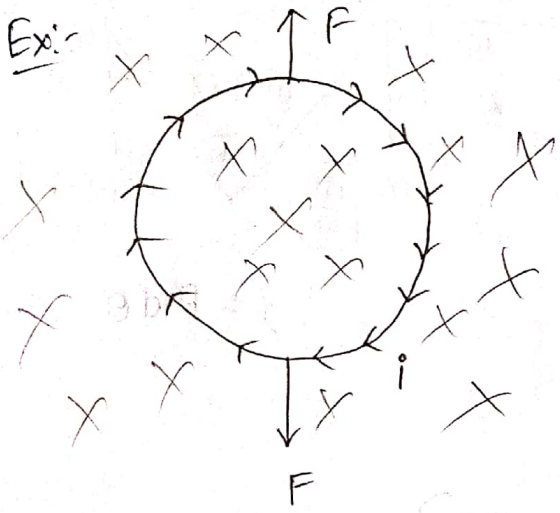
then, $\sigma_0 = \frac{\tau_{max}}{\pi r^2}$

$$\sigma_0 = \frac{iRB}{\pi r^2}$$

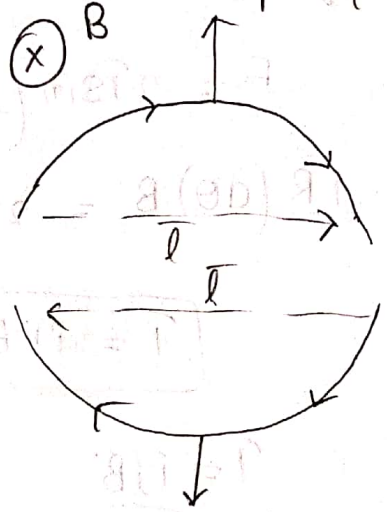
& $Y = \frac{\tau}{\pi r^2} \times \frac{\Delta R}{dR}$

$$\frac{dR}{R} = \frac{\tau}{\pi r^2 Y}$$

Ex:-



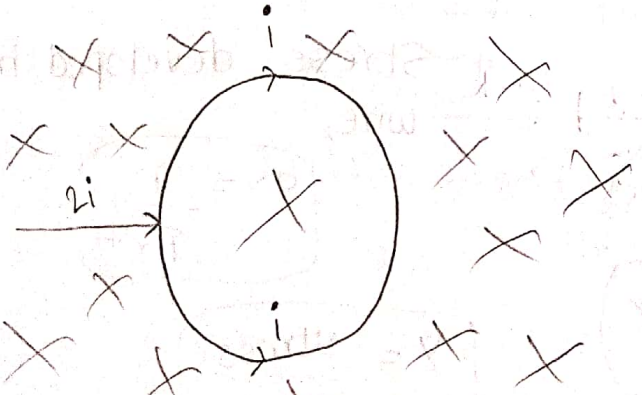
$\approx$



$F_1 = i(2R)B$

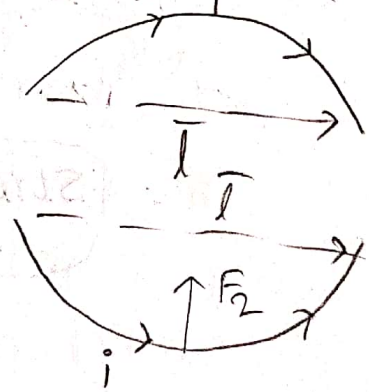
$F_2 = i(2R)B$

Ex:-



$\vec{F}_{net} \neq 0$

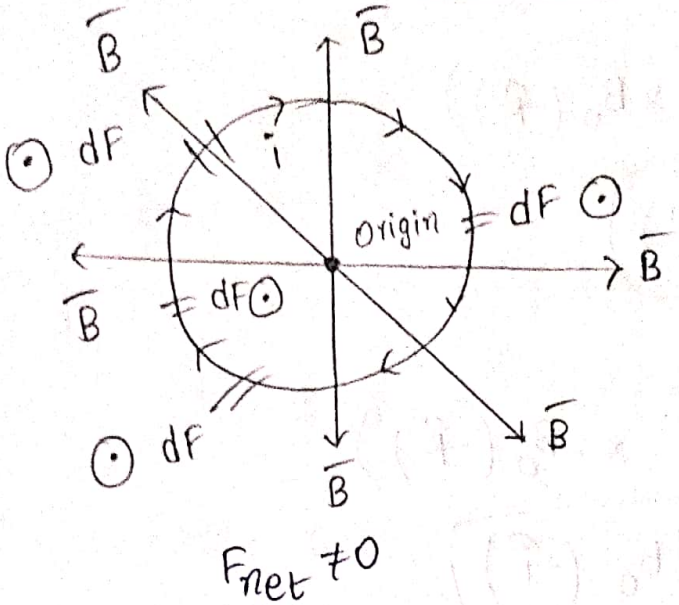
=



$F_1 = i(2R)B$

$$F_{net} = F_1 + F_2 = 4iRB$$

Radial Magnetic Field :-



$F_{net} \neq 0$

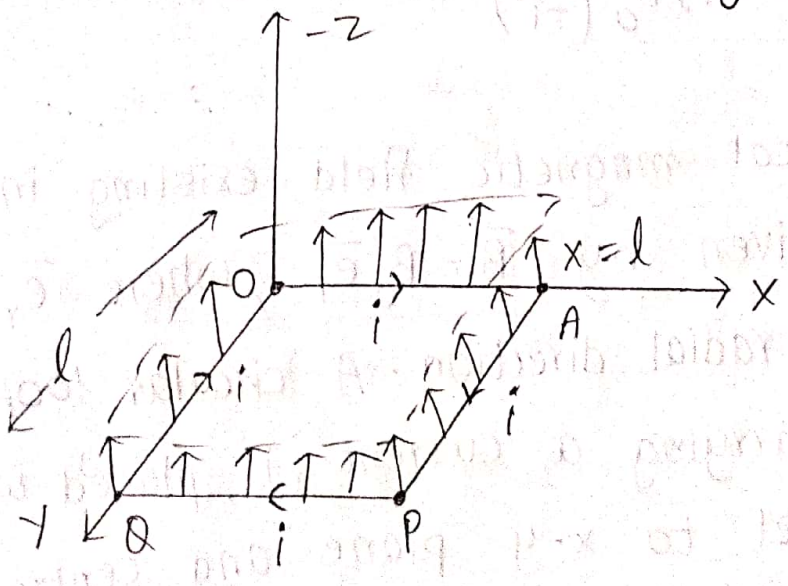
$$dF = i\,dl\,B \quad \odot$$

$$F_{net} = \int dF = \int i\,dl\,B$$

$$= iB \int dl$$

$$F_{net} = iB(2\pi R)$$

Ex:- A square loop of edge l and carrying current i with its edges $\parallel^{el}$ to x-y axes is placed in a magnetic field $\bar{B} = B_0\left(1+\frac{x}{l}\right)(-\hat{k})$. Find net magnetic force experienced by loop.



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Sol:- Force on QO,

$$\bar{F}_1 = i(\bar{l} \times \bar{B})$$

$$= i\left(l(-\hat{j}) \times B_0(-\hat{k})\right)$$

$$\boxed{\bar{F}_1 = i l B_0 (+\hat{i})}$$

Force on AP,

$$F_2 = i\left(l\hat{j} \times 2B_0(-\hat{k})\right)$$

$$\boxed{\bar{F}_2 = 2 i l B_0 (-\hat{i})}$$

Clearly forces on sides OA & PQ are equal & opposite to each other, so they cancel out.

$$\bar{F}_{net} = \bar{F}_1 + \bar{F}_2$$

$$= i l B_0 (\hat{i}) + 2 i l B_0 (-\hat{i})$$

$$= i l B_0 (-\hat{i})$$

Ex:- A hypothetical magnetic field existing in a region is given by $\bar{B} = B_0 \bar{e}_r$, where $\bar{e}_r$ = unit vector along radial direction. A circular loop of radius 'a' carrying a current 'i', placed with its plane parallel to x-y plane and centre at (0,0,d). Find the magnitude of radial force acting on the loop.

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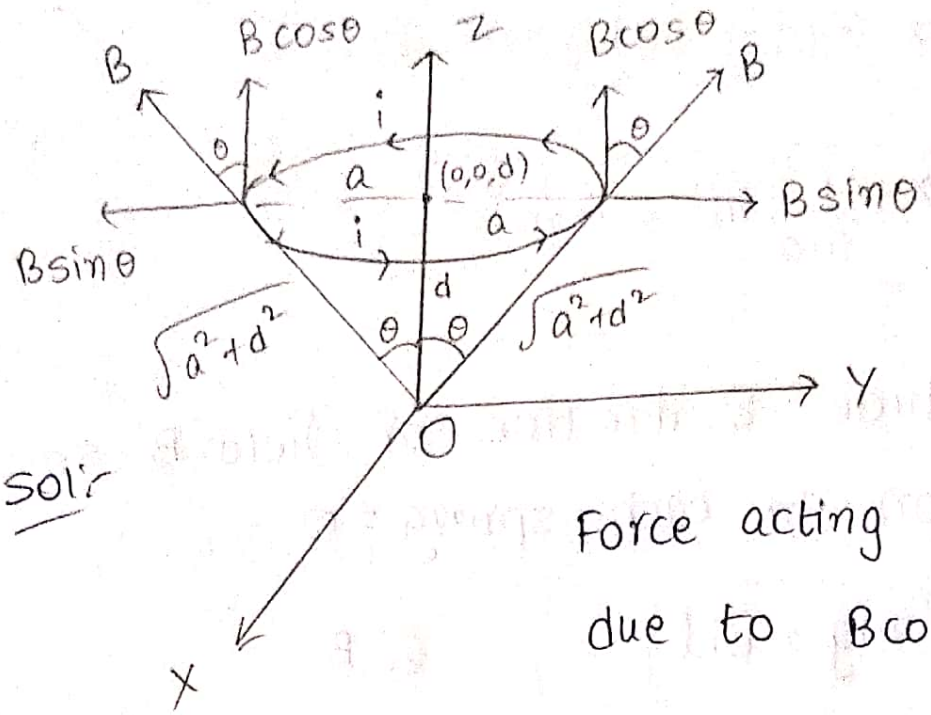
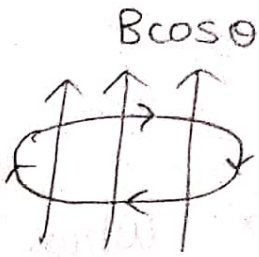


Sol:-

Force acting onthe wire due to $B\cos\theta = 0$

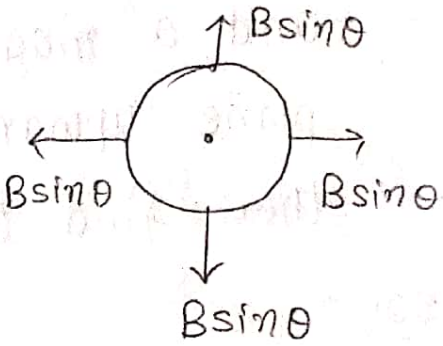


Force acting on the wire due to $B\sin\theta$

$$= i\,(B\sin\theta)\,2\pi a$$

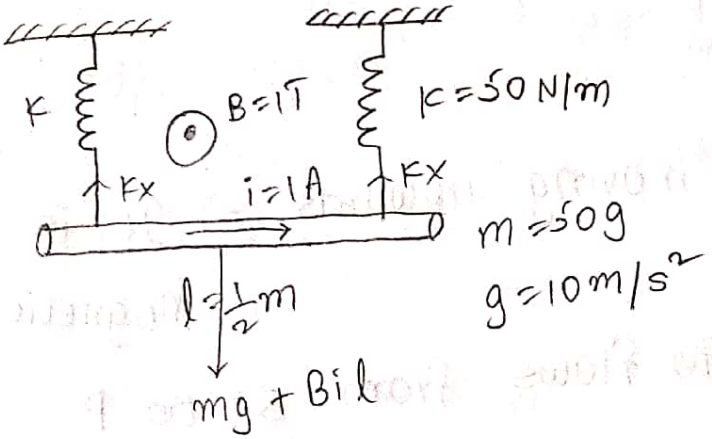
$$= iB(2\pi a)\left(\frac{a}{\sqrt{a^2+d^2}}\right)$$

$$= \frac{2\pi i B a^2}{\sqrt{a^2+d^2}}$$



Ex:-

For an outwards magnetic field B, Find the deformaton of the springs



In equilibrium, $2Kx = Bil + mg$

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$\Rightarrow 100x = \frac{1}{2} + \frac{1}{2}$

$$x = \frac{1}{100} m = 1cm$$

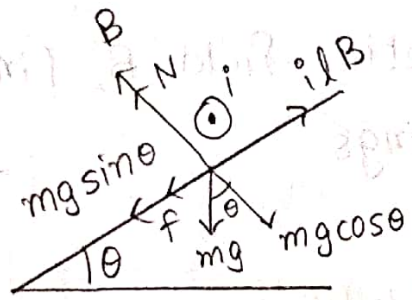
Find the magnitude & direction of field 'B' so that deformation in each spring = 0.

$$mg = Bil$$

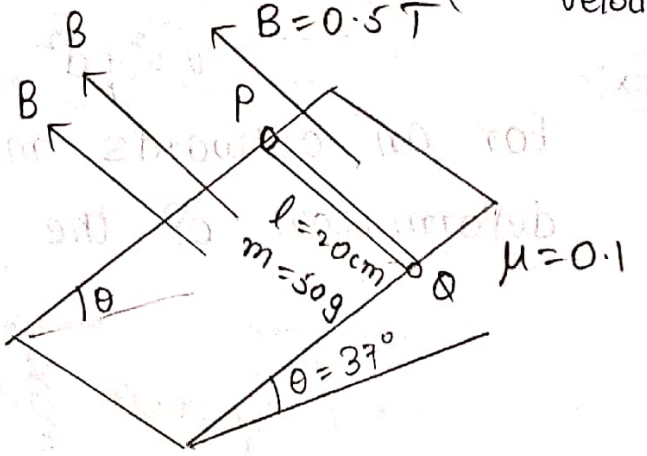
⊗ B

Ex:- When a current 'i' is passing through the wire and a magnetic field is applied normal to the plane upwards, the wire starts moving upwards then find the magnitude of current. (with const. velocity)

Sol:-



Only Sliding



Since wire is moving upwards ⇒ It is due to Magnetic Force

⇒ Current ~~flows~~ flows from Q to P

Since wire is moving with const. Velocity

$$ilB = mg\sin\theta + f$$

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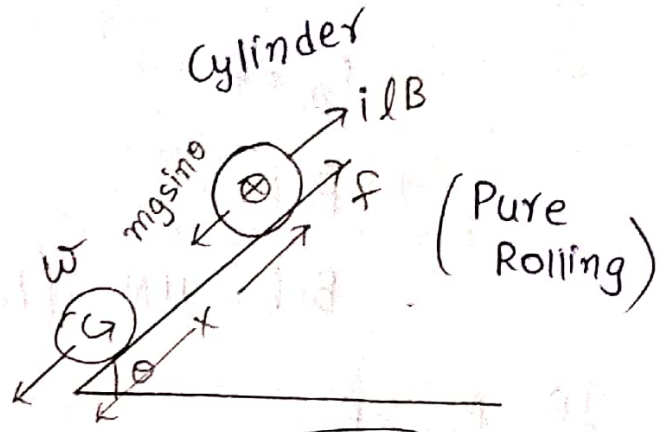
$$ilB = mg\sin\theta + \mu mg\cos\theta$$

$$i\left(\frac{1}{5}\right)\left(\frac{1}{2}\right) = \frac{1}{20}(10)\left(\frac{3}{5}\right) + \frac{1}{10}\left(\frac{1}{20}\right)(10)\left(\frac{4}{3}\right) \quad [\theta = 37^\circ]$$

$$\boxed{i = 3.4\,A}$$

*



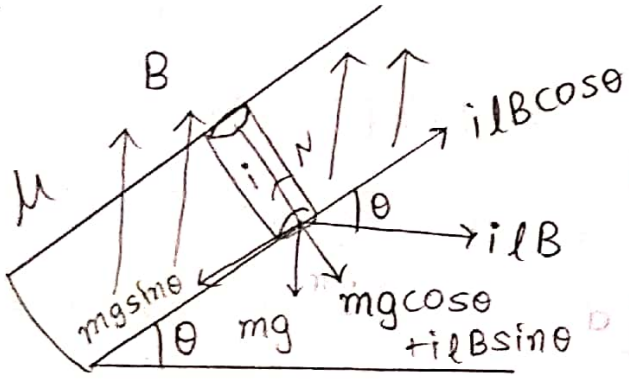
$$W_g + W_B + W_f = \Delta K$$

$$(mg\sin\theta)x - ilBx - \mu mg\cos\theta x = \frac{1}{2}mV^2 + \frac{1}{2}I\omega^2$$

$$\boxed{V = R\omega}$$

$$I = \frac{mR^2}{2}$$

Ex:-



[In Pure rolling, $W_f = 0$]

$$N = mg\cos\theta + ilB\sin\theta$$

Wire is in equilibrium,

$$\mu N + mg\sin\theta = ilB_{max}\cos\theta$$

and

$$mg\sin\theta = ilB_{min}\cos\theta + \mu N$$

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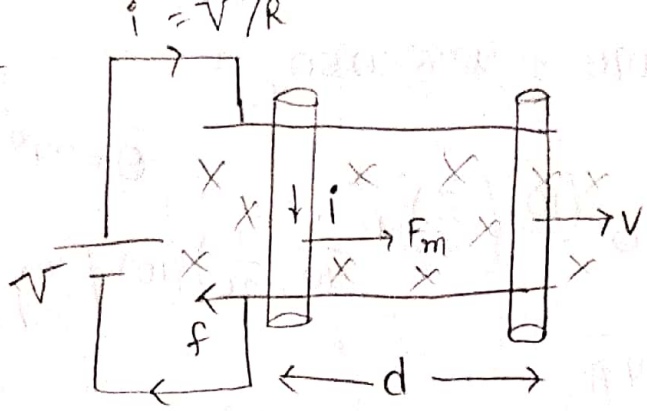
Ex:-



$$F_m = f$$

$$\Rightarrow Bil = f$$

$$Bil = \mu N \quad \text{(Limiting equilibrium)}$$

If $F_m > f_L$

$$a = \frac{F_m - f_L}{m} = \text{constant}$$

$$v^2 - u^2 = 2as$$

$$v^2 - u^2 = 2\left(\frac{F_m - f_L}{m}\right)d$$

$$\frac{1}{2}m(v^2 - u^2) = F_m d - f_L d$$

$$\Rightarrow \boxed{W_m + W_f = \Delta K}$$

In case of pure rolling,

$$W_m + \cancel{W_f}^{\,0} = \Delta K$$

$$Bild = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

[In pure rolling, W_f on Horizontal Surface $= 0$]

$$\omega = \frac{V}{r} \quad \& \quad I = \frac{mr^2}{2}$$

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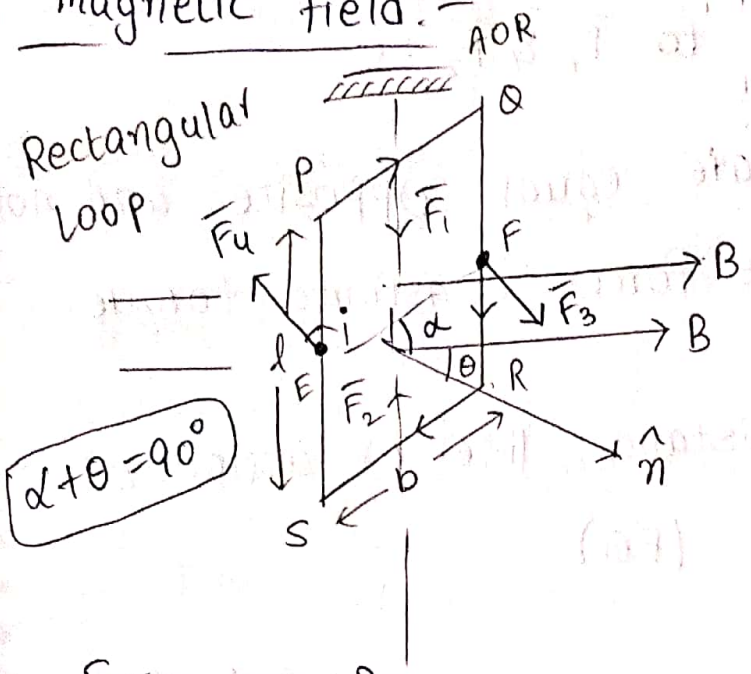
Torque acting on a current-carrying loop in Magnetic field:-

Rectangular Loop



Normal to the plane of coil makes an angle 'θ' with magnetic field.

Force on PQ,

$$\bar{F}_1 = i(\bar{b} \times \bar{B}) \quad (\downarrow)$$

Force on RS,

$$\bar{F}_2 = i(\bar{b} \times \bar{B}) \quad (\uparrow)$$

Net force of $\bar{F}_1$ & $\bar{F}_2$ is zero.

Force on QR,

$$\bar{F}_3 = i(\bar{l} \times \bar{B})$$

$$= ilB \ (\odot)$$

Force on SP,

$$\bar{F}_4 = i(\bar{l} \times \bar{B})$$

$$= ilB \ (\otimes)$$

$$\bar{F}_{net} = \bar{F}_1 + \bar{F}_2 + \bar{F}_3 + \bar{F}_4 = \bar{0}$$

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Clearly, $\bar{F}_1$ & $\bar{F}_2$ are collinear forces

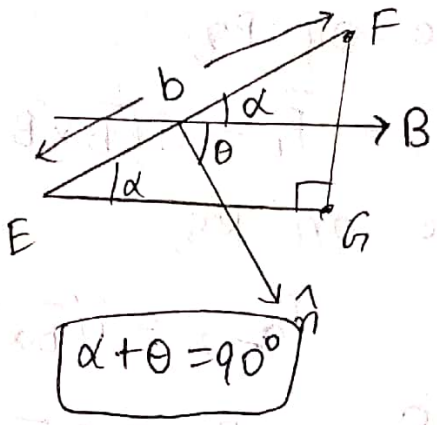
$\Rightarrow$ Torque due to $\bar{F}_1$ & $\bar{F}_2 = 0$

where as $\bar{F}_3$ & $\bar{F}_4$ are equal, opposite and non collinear forces. These forces constitute torque.

τ = Force $\times$ $\perp^{lar}$ distance line of action of forces (EG)

$= BiIb\cos(90-\theta)$

$$\boxed{\tau = BiIb\sin\theta}$$



$$\cos\alpha = \frac{EG}{EF}$$

$$EG = EF\cos\alpha = b\cos\alpha$$

$$\tau = BiA\sin\theta$$

If the coil contains N-turns,

$$\boxed{\tau = BiAN\sin\theta}$$

In Vector form,

$$\boxed{\bar{\tau} = \bar{m} \times \bar{B}}$$

→ Torque always acts along the axis of rotation

Here $\boxed{\bar{m} = NiA\hat{n} = \text{magnetic moment of loop}}$

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This formula is used for a coil of any geometrical shape.

Comparission blw Electrostatics & Magnetostatics

Electrostatics	Magnetostatics
① Electric dipolemoment, $\bar{P} = q \times 2\bar{l}$	① Magnetic moment, $\bar{m} = NiA\,\hat{n}$
② Torque $\bar{\tau} = \bar{P} \times \bar{E}$	② Torque $\bar{\tau} = \bar{m} \times \bar{B}$
③ P.E stored in the dipole, $U = -\bar{P}\cdot\bar{E} = -PE\cos\theta$	③ P.E stored in magnetic dipole, $U = -\bar{m}\cdot\bar{B} = -mB\cos\theta$
④ Work done by field, $W = U_i - U_f$ $W = -PE\cos\theta_1 + PE\cos\theta_2$ $W = PE(\cos\theta_2 - \cos\theta_1)$	④ Work done by the field, $W = U_i - U_f$ $W = mB(\cos\theta_2 - \cos\theta_1)$

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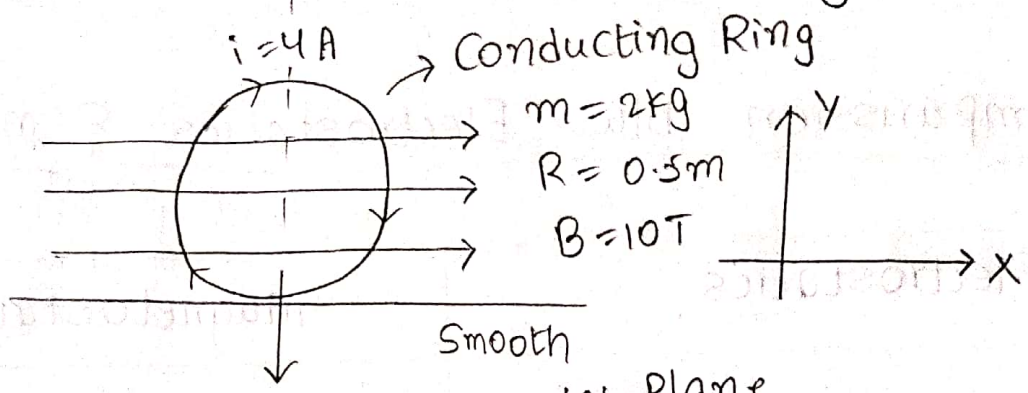
Ex:- A Horizontal Magnetic field B=10T is switched on. Find initial acceleration of ring.



Sol:-

$\bar{\tau} = \bar{m} \times \bar{B}$

$\bar{m} = iA\hat{n}$

$= 4(\pi R^2)(-\hat{k})$

$= \pi(-\hat{k})$

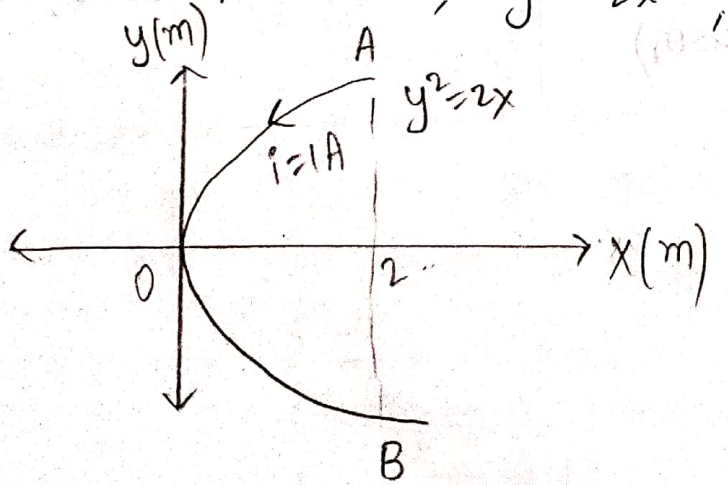
$\bar{\tau} = \pi(10)(-\hat{k} \times \hat{i})$

$\bar{\tau} = 10\pi(-\hat{j})$

$\bar{\tau} = I\bar{\alpha}$

$\bar{\alpha} = \frac{\bar{\tau}}{I} = \frac{10\pi}{\left(\frac{MR^2}{2}\right)}(-\hat{j}) = \frac{20\pi}{(2)\left(\frac{1}{4}\right)} = \boxed{40\pi(-\hat{j})}$ rad/s²

Ex:- Find Magnetic Force on the wire which is of the form parabola, $y^2 = 2x$, $i = 1A$



$\bar{B} = -4\hat{k}$

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Sol:- $\bar{F} = i(\bar{l} \times \bar{B})$

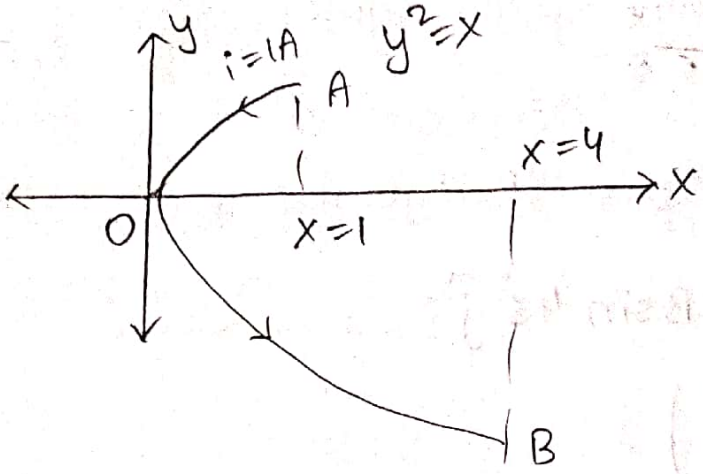
$\bar{l} = \bar{AB} = 4(-\hat{j})$

$\bar{F} = (1)\left(4(-\hat{j}) \times -4\hat{k}\right)$

$(A) = (2,2)$; $B = (2,-2)$

$= 16\hat{i}$

Ex:-



$\bar{B} = -2\hat{k}$

Find unit vector in the direction of force.

Sol:-

$A = (1,1)$ $\quad B = (4,-2)$

$\bar{AB} = \bar{OB} - \bar{OA} = (4\hat{i} - 2\hat{j}) - (\hat{i} + \hat{j}) = 3\hat{i} - 3\hat{j}$

$\bar{F} = i(\bar{l} \times \bar{B}) = 1\left(3\hat{i} - 3\hat{j} \times -2\hat{k}\right)$

$= 6\hat{j} + 6\hat{i} = 6(\hat{i} + \hat{j})$

$\frac{\bar{F}}{|F|} = \frac{6(\hat{i}+\hat{j})}{6\sqrt{2}} = \hat{i} + \hat{j}$

Ex:- At time $t=0$, frame is at rest as shown. With its side $\parallel^{el}$ to x & y axes. Each side of the frame is of mass m & length l. Find

i) Torque about O acting on the frame due to the magnetic field.

ii) Find angle by which the frame rotate in Δt

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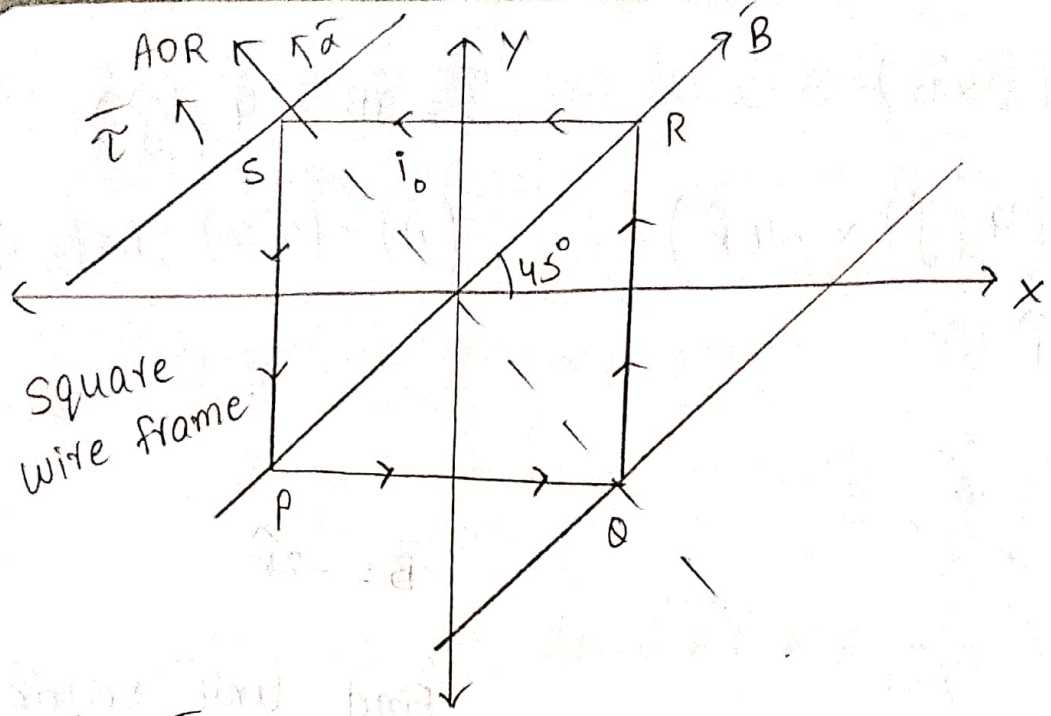


Sol: $\bar{B} = B\cos 45^\circ \hat{i} + B\sin 45^\circ \hat{j}$

$$\bar{B} = B\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$$

$$|\bar{\tau}| = |\bar{m} \times \bar{B}|$$

$$= \frac{i l^2 B}{\sqrt{2}} \left[\hat{k} \times (\hat{i}+\hat{j})\right]$$

$$|\bar{\tau}| = \frac{i l^2 B}{\sqrt{2}} (\hat{j}-\hat{i})$$

$\bar{m} = iNA\,\hat{n}$

$= i l^2 (1)\,\hat{k}$

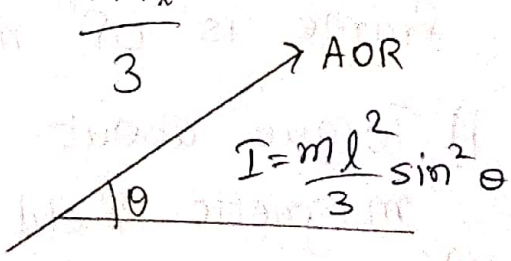
Immediate angular acceleration,

$$\bar{\alpha} = \frac{\bar{\tau}}{\bar{I}} = \frac{\frac{i l^2 B}{\sqrt{2}}(\hat{j}-\hat{i})}{\frac{2ml^2}{3}}$$

$$\bar{\alpha} = \frac{3Bi}{2\sqrt{2}m}(\hat{j}-\hat{i})$$

$$I = 4\left[\frac{ml^2}{3}\sin^2 45^\circ\right]$$

$$= \frac{2ml^2}{3}$$

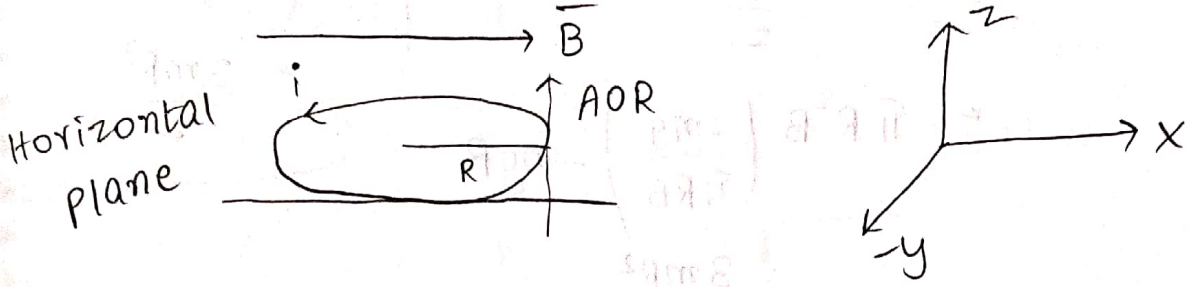


$$I = \frac{ml^2}{3}\sin^2\theta$$

$$\theta = \frac{1}{2}\alpha(\Delta t)^2$$

$$\theta = \frac{3iB}{4\sqrt{2}m}\Delta t^2$$

Ex:- A Circular wire loop lies on a horizontal plane, B is horizontal. How ~~long~~ large can the current I be before one edge will lift off surface



Sol:-

$$\tau_{mB} = \vec{m} \times \vec{B}$$

$$= i\pi R^2 \hat{k} \times B\hat{i}$$

$$= i\pi R^2 B\hat{j}$$

Torque due to gravity,

$$\tau_g = \vec{r} \times \vec{F}$$

$$= R(-\hat{i}) \times mg(-\hat{k})$$

$$= mgR(-\hat{j})$$

Just to lift off,

$$\tau_{mB} = \tau_{mg}$$

$$\pi R^2 B i_{max} = mgR \Rightarrow \boxed{i_{max} = \frac{mg}{\pi RB}}$$

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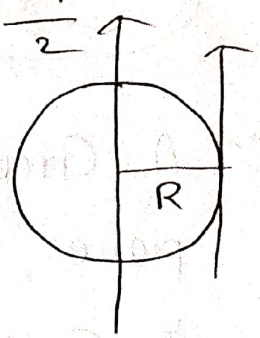
* If $i = \frac{2mg}{\pi RB}$ then find immediate angular acceleration

$$\alpha = \frac{\tau_{net}}{I} = \frac{\tau_{mB} - \tau_g}{\frac{3mR^2}{2}}$$

$$= \frac{\pi R^2 B i - mgR}{\frac{3mR^2}{2}}$$

$$= \frac{\pi R^2 B \left(\frac{2mg}{\pi RB}\right) - mgR}{\frac{3mR^2}{2}}$$

$$= \frac{mgR \times 2}{3mR^2} = \boxed{\frac{2g}{3R}}$$



$I = \frac{mR^2}{2} + mR^2$

$= \frac{3mR^2}{2}$

Linear acceleration of centre of ring ; $a = R\alpha$

$$\boxed{a = \frac{2g}{3}}$$

Ex:- A Circular coil of radius r and has n turns carries a current i. It prevents from slipping down by mean of rigid support along the line $\perp^{lar}$ to the line of greatest slope on smooth inclined plane. The Total mass of coil is m. In which direction and of what min magnitude

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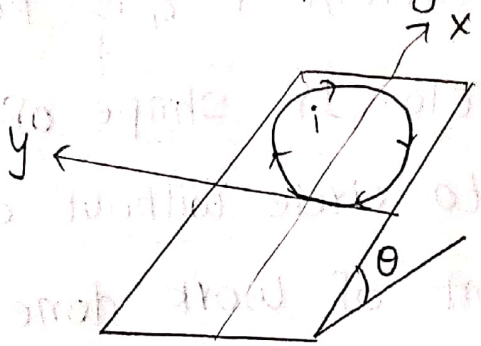
should "B" applied to that the coil starts tipping over the line of rigid support.



Sol:-

Ex:- A metallic wire is folded to form a square of side a and has current i. $\vec{\mathcal{E}}$ is kept ⊥lar to uniform magnetic field. If shape of loop is Change from square to circle without changing the length and current. Find amount of work done by external agent.

Sol:-

$$W_{ext} = U_f - U_i = -\vec{m}_2 \cdot \vec{B} + \vec{m}_1 \cdot \vec{B}$$

$$= \vec{B}(\vec{m}_1 - \vec{m}_2) \quad [\theta = 0^\circ]$$

$$= B(m_1 - m_2)$$

$$= B\left(ia^2 - i\pi\left(\frac{4a^2}{\pi^2}\right)\right)$$

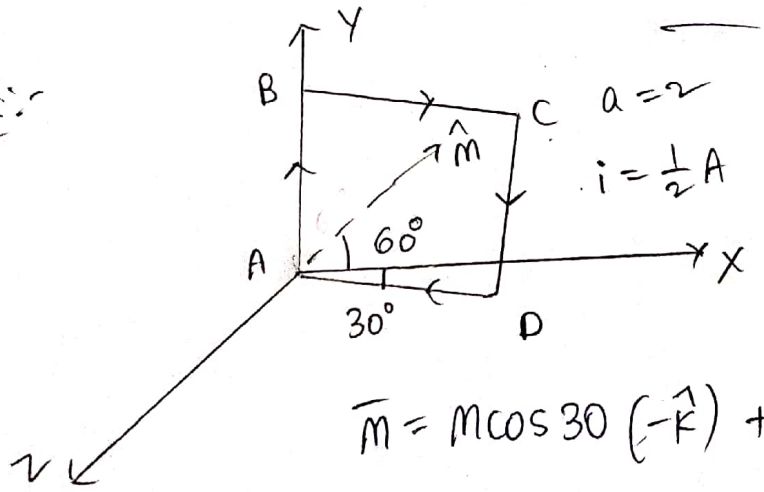
$$= Bi(a^2)\left(1 - \frac{4}{\pi}\right)$$

$4a = 2\pi r$

$r = \frac{2a}{\pi}$

Magnetic Moment Calculation:-

Ex:-



$a = 2$

$i = \frac{1}{2}A$

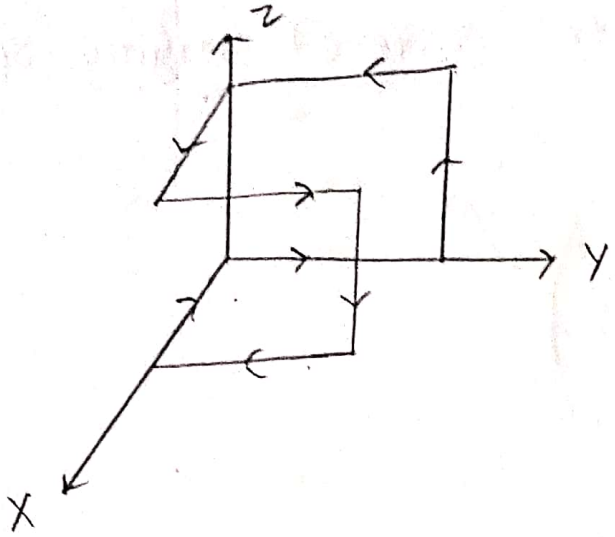
$m = iA = \frac{1}{2}(2)^2 = 2$

$$\vec{m} = m\cos 30(-\hat{k}) + m\cos 60(\hat{i})$$

$$= 2 \times \frac{\sqrt{3}}{2}(-\hat{k}) + 2\left(\frac{1}{2}\right)(\hat{i})$$

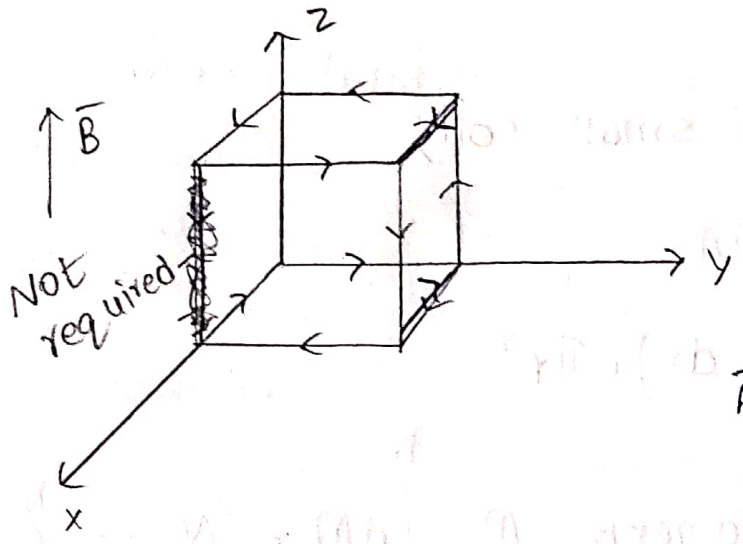
$$\vec{m} = \hat{i} - \sqrt{3}\hat{k}$$

Ex:-



Find Magnetic moment of loop

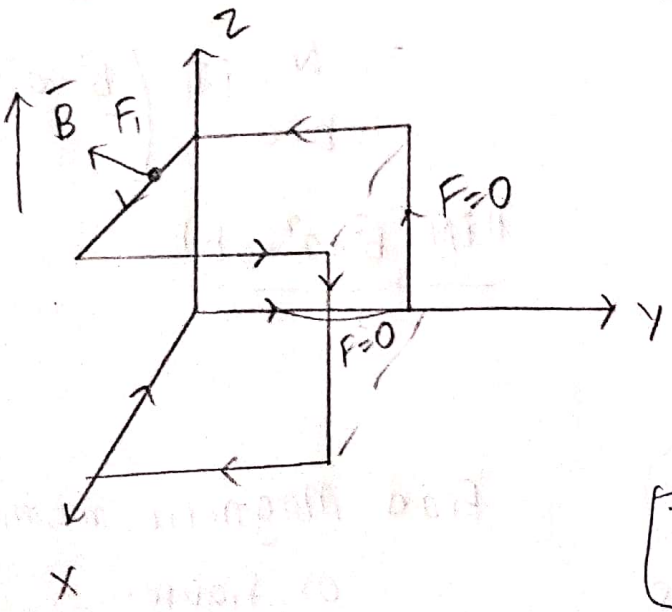
Sol:- Gap is filled with two wires that carry equal currents in opposite direction.



$$\bar{m} = \bar{m}_1 + \bar{m}_2 + \bar{m}_3$$

$$\bar{m} = m\hat{j} + m\hat{k} + m(-\hat{k}) = m\hat{j}$$

Torque on the wire $\bar{\tau} = \bar{m} \times \bar{B} = mB\hat{i}$



$\bar{\tau} = \bar{F} \times \perp^{ler}$ distance b/w two forces

$= iBl \times l\hat{i}$

$= il^2B\hat{i}$

$$\boxed{\bar{\tau} = mB\hat{i}}$$

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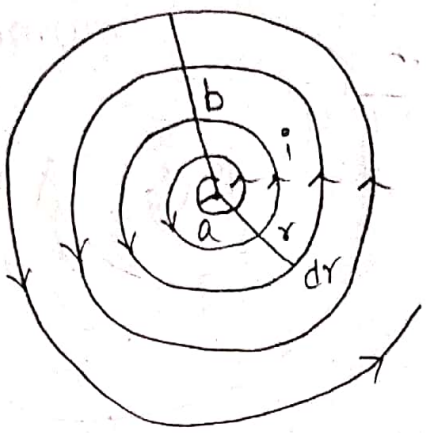
Magnetic moment due to current carrying spiral coil:-



Let total no. of turns in gap b-a, along radial direction = N

Magnetic moment of small coil,

$$dM = (dN)\,iA$$

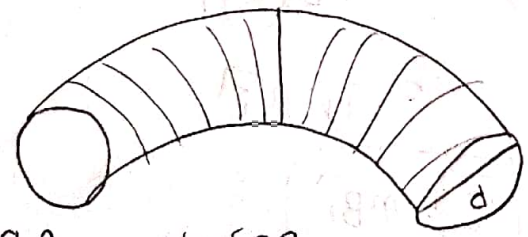
$$= \left(\frac{N}{b-a}\,dr\right) i\,\pi r^2$$

Total magnetic moment, $M = \int_a^b dM = \frac{N}{b-a}\,i\pi \int_a^b r^2$

$$= \frac{N}{b-a}\,i\pi\left(\frac{b^3-a^3}{3}\right)$$

$$= \frac{\pi i N\left(b^2+a^2+ab\right)}{3}$$

Ex:-



$i = 0.8\,A$ $N = 500$ Half Tone

$d = 5\,cm$

Find Magnetic moment of figure.

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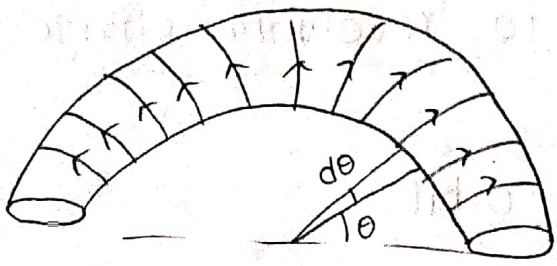
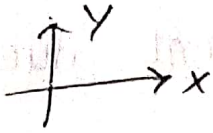
Sol:-





$N \rightarrow \pi$

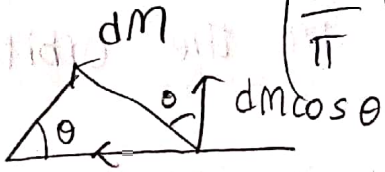
$1 \rightarrow \frac{N}{\pi}$

$d\theta \rightarrow \frac{N}{\pi} d\theta$

$d\vec{M} = dM\cos\theta\,\hat{j} + dM\sin\theta\,(-\hat{i})$

Here $dM = dNiA$

$= \left(\frac{N}{\pi} d\theta\right) i \frac{\pi d^2}{4}$



$$M = \int_0^{\pi} d\vec{M}$$

$$= \int_0^{\pi} \frac{N}{4} i d^2 \cos\theta\, d\theta\, \hat{j} - \int_0^{\pi} \frac{N}{4} i d^2 \sin\theta\, \hat{i}$$

($dM\sin\theta$)

$$= \frac{N}{4} i d^2 \left[\sin\theta\right]_0^{\pi} \hat{j} + \frac{N}{4} i d^2 \left[\cos\theta\right]_0^{\pi} \hat{i}$$

$$= \frac{N}{4} i d^2 (0) + \frac{N}{4} i d^2 (-2)\hat{i}$$

$$= -\frac{N}{2} i d^2 \hat{i} = -\frac{(500)}{2}(0.8)\left(\frac{25}{10^4}\right)$$

$$= -\frac{4}{2} \times 25 \times 10^{-2} = 50 \times 10^{-2}$$

$$= \boxed{0.5}$$

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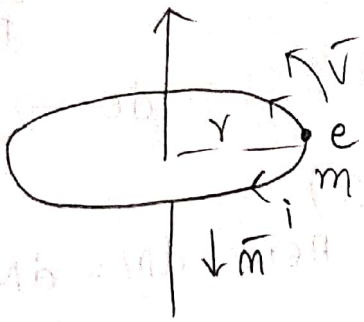
Magnetic moment due to revolving charge:-

① Due to electron in orbit:-



Current in the orbit, $i = fe = \frac{ve}{2\pi r}$

Magnetic moment due to orbital motion,

$$M = iA = \frac{ev}{2\pi r}(\pi r^2)$$

$$M = \frac{evr}{2\pi}(\pi)$$

$$\boxed{M = \frac{evr}{2}} \quad (\downarrow)$$

We have, $mvr = \frac{nh}{2\pi}$

$$vr = \frac{nh}{2\pi m}$$

$$\boxed{M = \frac{enh}{4\pi m}}$$

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For first orbit, $n=1$

$$\mu = \frac{eh}{4\pi m} = \text{Bohr magnetron}$$

$$\mu = \frac{(1.6\times10^{-19})(6.62\times10^{-34})}{4\times\pi\times9.1\times10^{-31}}$$

$$\mu = 9.27\times10^{-24}\ Am^2$$

Angular momentum of revolving electron,

$$\vec{L} = m(\vec{r}\times\vec{v})$$

$$= mvr \quad (\uparrow)$$

$$\frac{\mu}{L} = \frac{evr}{2\times mvr}$$

$$\frac{\mu}{L} = \frac{e}{2m}$$ ⟹ It is called gyrometric ratio.

In Vector form,

$$\vec{\mu} = \frac{-e}{2m}\vec{L}$$

Ex: A Uniform ring of radius R and charge q & mass m rotate about its axis with angular Velocity ω. Find ratio of Magnetic moment & Angular moment.

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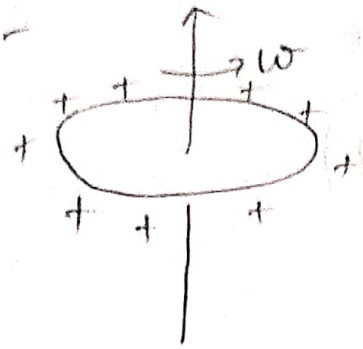
Sol:-



Current in the ring,

$$i = fq = \frac{\omega q}{2\pi} = \frac{Vq}{2\pi r}$$

Magnetic moment,

$$m = iA = \frac{qVR}{2}$$

Angular moment about AOR,

$$L = I\omega$$

$$L = mR^2\omega = \frac{mR^2V}{R} = mVR$$

$$\frac{m}{L} = \frac{qVR}{2mVR} = \frac{q}{2m}$$

$$\boxed{\frac{m}{L} = \frac{q}{2m}}$$

Ex:- If two charged particles each of charge q & mass m are connected to the ends of a rigid mass less rod and is rotated about an axis passing through the centre and $\perp^{ler}$ to the length. Find the ratio of magnetic moment & Angular moment.

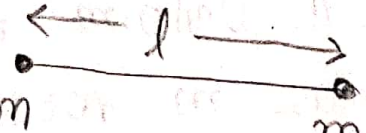
Sol:-

$$m = 2i\pi\left(\frac{l}{2}\right)^2 = \frac{i\pi l^2}{2} = \frac{qV\pi l^2}{2(2\pi r)}$$

$$= \frac{qV\pi l}{2\pi}$$

$$L = mvr \times 2$$

$$= 2mv\frac{l}{2}$$



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$$L = mvl$$

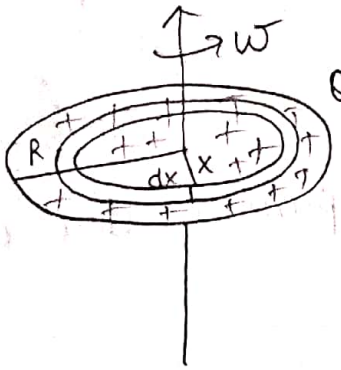
Now, $$\frac{m}{L} = \frac{qvl/2}{mvl}$$

$$\boxed{\frac{m}{L} = \frac{q}{2m}}$$

∴ This formula is valid

Ex:- A non conducting disc of mass M, radius R & has surface charge density σ is rotating with Angular velocity ω. Find the magnetic dipole moment of disc.

Sol:-



Q $\quad \sigma = \frac{Q}{\pi R^2}$ = constant

Charge on elementary ring,

$$dq = \sigma (2\pi x)\, dx$$

$$= \frac{Q}{\pi R^2} (2\pi x)\, dx$$

$$dq = \frac{2Q}{R^2} x\, dx$$

Current in the circular ring due to spinning,

$$di = (dq)\frac{\omega}{2\pi} = \frac{2Q}{R^2}\left(\frac{\omega}{2\pi}\right) x\, dx$$

Magnetic moment due to elementary ring,

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$$dM = (di)A$$

$$dM = \frac{2Q}{R^2}\left(\frac{\omega}{2\pi}\right) x\,dx \cdot (\pi x^2)$$

Total magnetic moment,

$$M = \int_0^R dM = \frac{Q\omega}{R^2}\int_0^R (x^3\,dx)$$

$$\boxed{M = \frac{Q\omega R^2}{4}} \Rightarrow M = \frac{\sigma(\pi R^2)\omega R^2}{4}$$

$$\boxed{M = \frac{\sigma\omega R^4\pi}{4}}$$

Angular momentum of spinning disc about its own axis,

$$L = I\omega = \frac{mR^2}{2}\omega$$

$$\frac{M}{L} = \frac{\frac{Q\omega R^2}{4}}{\frac{mR^2}{2}\omega} = \frac{Q}{2m} \Rightarrow \boxed{\frac{M}{L} = \frac{Q}{2m}}$$

This formula is applicable for any geometrical shape

Short-Cut:-

$$\frac{M}{L} = \frac{Q}{2m}$$

$$M = \frac{QL}{2m}$$

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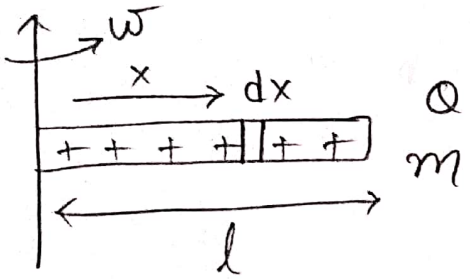
$$= \frac{Q}{2m}(I\omega)$$

$$= \frac{Q}{2m}\left(\frac{mR^2}{2}\right)\omega$$

$$\boxed{M = \frac{QR^2\omega}{4}}$$

Ex:-



Sol:-

M-I

$$M = \frac{QL}{2m} = \frac{Q}{2m}\left(\frac{ml^2}{3}\omega\right)$$

$$= \frac{Ql^2\omega}{6}$$

$$\boxed{M = \frac{Ql^2\omega}{6}}$$

M-II

$$di = dq \cdot \frac{\omega}{2\pi} = \left(\frac{Q}{L}\right)\cdot\frac{\omega}{2\pi}\,dx$$

$$dM = \int_0^l di\,A = \int_0^l \frac{Q}{L}\frac{\omega}{2\pi}(\pi x^2)\,dx = \frac{Ql^2\omega}{6} = \frac{Ql^2\omega}{6}$$

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Ex:- A non-conducting spherical shell of radius R and Uniform charge of surface charge density σ rotates around the axis passing through its centre at an angular velocity ω. Find Magnetic Moment

Sol:-

$$\frac{Q}{2m} = \frac{M}{L}$$

$$M = \frac{Q}{2m}(I\omega)$$

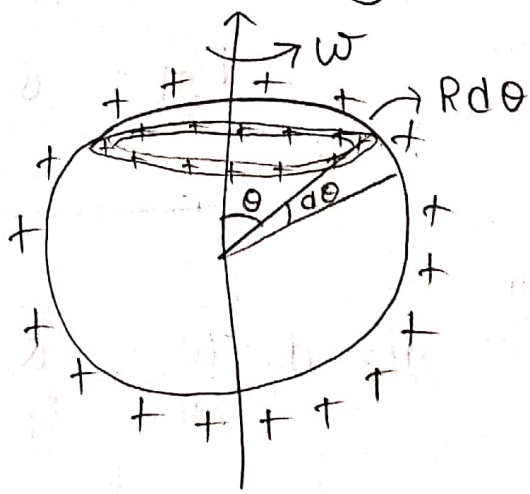
$$= \frac{Q}{2m}\left(\frac{2}{3}MR^2\omega\right)$$

$$= \frac{QR^2\omega}{3}$$

$$= \frac{4\pi R^4\omega\sigma}{3}$$

$$\left[\sigma = \frac{Q}{4\pi R^2}\right]$$

M-II



From fig,

$$\sin\theta = \frac{r}{R}$$

$$r = R\sin\theta \rightarrow (1)$$

$$dq = \sigma 2\pi r dr$$

$$= \sigma 2\pi (R\sin\theta) R d\theta$$

$$dq = \sigma 2\pi R^2 \sin\theta d\theta$$

$$di = dq\frac{\omega}{2\pi}$$

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$$dM = (di)A = di(\pi r^2)$$

$$M = \int_0^{\pi} dM = \int_0^{\pi} \sigma 2\pi R^2 \sin\theta \, d\theta \cdot \left(\frac{\omega}{2\pi}\right)(\pi r^2)$$

$$= \int_0^{\pi} \sigma \pi R^4 \sin^3\theta \, d\theta \, (\omega)$$

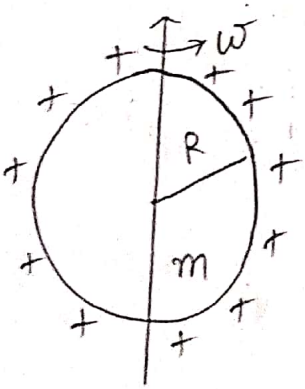
$$= \sigma \pi R^4 \omega \int_0^{\pi} \sin^3\theta \, d\theta$$

$$= 2\sigma \pi R^4 \omega \int_0^{\pi/2} \sin^3\theta \, d\theta$$

$$= 2\sigma \pi R^4 \omega \, \frac{(3-1)!!}{3}$$

$$= \frac{4\sigma \pi R^4 \omega}{3}$$

Ex:- JEE 2025



A Conducting Solid sphere carries charge 'Q' & mass 'm' rotates about an axis passing through its centre.

Find ratio of magnetic moment & its Angular momentum.

Sol:- While rotating, it behaves as Hollow Spherical shell i.e, charge is on outer surface.

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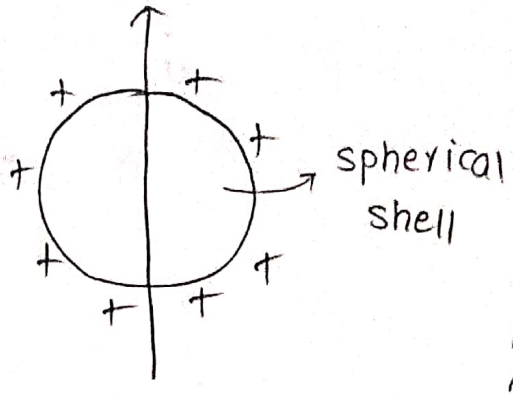
$$\frac{M}{L} = \frac{Q}{2m}$$

To find Magnetic Moment,



$$M = \frac{Q}{2m} L$$

$$= \frac{Q}{2m} (I\omega)$$

$$= \frac{Q}{2m} \left(\frac{2}{3} mR^2 \omega \right)$$

$$M = \frac{Q}{3} R^2 \omega$$

To find L :- we should take solid sphere

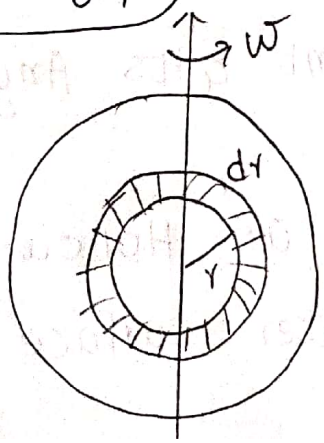
$$L = I\omega$$

$$L = \frac{2}{5} mR^2 \omega$$

$$\frac{M}{L} = \frac{\frac{Q}{3} R^2 \omega}{\frac{2}{5} mR^2 \omega}$$

$$\boxed{\frac{M}{L} = \frac{5Q}{6m}}$$

Ex:-



~~Between r~~

$+\rho_0$ b/w $r=0$ & $r=\frac{R}{2}$

$-\rho_0$ b/w $r=\frac{R}{2}$ & $r=R$

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Sol:- Small element is the spherical shell of radius 'r' & thickness dr.

Charge on the small element,

$$dq = \rho_0 4\pi r^2 dr$$

we have,

$$dM = \frac{(dq) r^2 \omega}{3}$$

$$dM = \frac{1}{3}\left(\rho_0 4\pi r^2 dr\right) r^2 \omega$$

$$\overline{M}_1 = \int_0^{R/2} dM = \frac{4}{3}\rho_0 \pi \omega \int_0^{R/2} r^4$$

$$= \frac{4}{15}\rho_0 \pi \omega \left[\frac{R^5}{32}\right]$$

$$\overline{M}_2 = \int_{R/2}^{R} dM = \frac{4}{3}(-\rho_0)\pi\omega\left[\frac{R^5}{5} - \frac{R^5}{5\times 32}\right]$$

$$= \frac{4}{15}\rho_0 \pi\omega\left[R^5 - \frac{R^5}{32}\right]$$

$$\overline{M} = \overline{M}_1 + \overline{M}_2 = \frac{-4}{15}\rho_0 \pi\omega\left[R^5 - \frac{R^5}{16}\right]$$

$$= -\frac{\pi\omega\rho_0 R^5}{4}$$

Net charge,

$$dq = \rho_0 4\pi r^2 dr$$

$$q_1 = \int_0^{R/2} dq = 4\pi\rho_0\left[\frac{r^3}{3}\right]_0^{R/2}$$

$$dM = \frac{dq}{2m}\left(\frac{2}{3}mr^2\omega\right)$$

$$dM = \frac{dq\, r^2\omega}{3}$$

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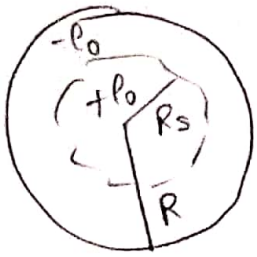
$$= \frac{\pi \rho_0 R^3}{6}$$

$$q_2 = \int_{\frac{R}{2}}^{R} (-\rho_0)\, 4\pi r^2 dr$$

$$= -\frac{4}{3}\pi \rho_0 \left[R^3 - \frac{R^3}{8} \right]$$

$$= -\frac{7\pi \rho_0 R^3}{6}$$

Net charge, $Q = q_1 + q_2 = -\rho_0 \pi R^3$



Given,

$$\bar{m} = \bar{m}_1 + \bar{m}_2 = 0$$

$$\Rightarrow \bar{m}_1 = -\bar{m}_2$$

$$\Rightarrow \int_0^{R_s} \frac{1}{3}\rho_0\, 4\pi r^4 \omega\, dr = -\int_{R_s}^{R} \frac{1}{3}(-\rho_0)\, 4\pi r^4 \omega\, dr$$

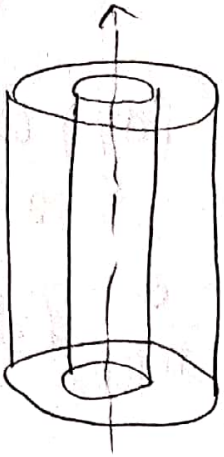
$$\frac{R_s^5}{5} = \frac{R^5 - R_s^5}{5}$$

$$\boxed{R_s = R \cdot 2^{-1/5}}$$

Ex:- A hollow cylinder has a length l & inner radius and outer radius are R_1 & R_2. The cylinder carries

a uniform charge density σ. Derive the equation for magnetic moment interms of ω

Sol:-



$I = mr^2$

$dq = \sigma\, dA$

$dA = 2\pi r l\, dr$

$$dM = \frac{dq}{2m}(I\omega)$$

$$dM = \frac{\omega \cdot dq \cdot r^2}{2}$$

$$M = \int_{R_1}^{R_2} dM$$

$$= \int_{R_1}^{R_2} \frac{\omega(2\pi r dr) l r^2 \sigma}{2}$$

$$= \frac{\omega \pi l \sigma}{4}\left(R_2^4 - R_1^4\right)$$

Ex:- A Solid Cylinder R, l carries a uniform charge density $+\rho$ from 0 to R_s & $-\rho$ from R_s to R Find R_s so that magnetic moment = 0.

Sol:- From above question,

$$\left(\frac{\pi\omega l\sigma}{4}\right)\left(R_s^4 - 0\right) = \left(\frac{\pi\omega l\sigma}{4}\right)\left(R^4 - R_s^4\right)$$

$$2R_s^4 = R^4$$

$$\boxed{R_s = R \cdot 2^{-1/4}}$$

Ex:- A charge q is uniformly distributed over the same surface of right circular cone of semi verte

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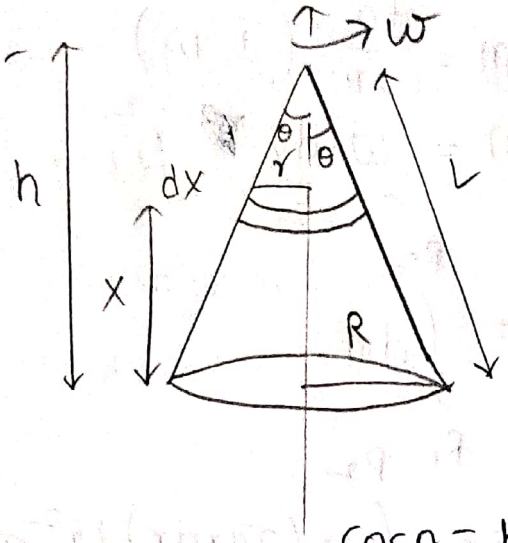
angle θ and height h. The cone is rotated about its axis with ω. Calculate magnetic moment

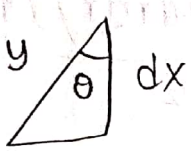
Sol:-



$$dM = (dq)\frac{\omega}{2\pi} A \rightarrow (1)$$

$$dM = \frac{Q}{\pi R \ell}(2\pi r y)\pi r^2 \left(\frac{\omega}{2\pi}\right)$$

$$dM = \frac{Q}{\pi R \ell}\left(2\pi r \cdot \frac{dx}{\cos\theta}\right)\pi r^2 \left(\frac{\omega}{2\pi}\right)$$



$$\cos\theta = \frac{h}{L}$$

$$\sin\theta = \frac{R}{L}$$

$$\cos\theta = \frac{dx}{y}$$

$$\Rightarrow y = \frac{dx}{\cos\theta}$$

$$dM = \frac{Q r^3 \omega}{R \ell \cos\theta} \cdot dx$$

From fig,

$$\frac{h}{R} = \frac{h-x}{r} \Rightarrow r = \frac{(h-x)R}{h}$$

$$r = (h-x)\tan\theta$$

$$M = \int_{x=0}^{x=h} dM = \frac{Q\omega}{R\ell\cos\theta} \int_0^h (h-x)^3 \tan^3\theta \cdot dx$$

$$= \frac{Q\omega}{R\ell\cos\theta} \cdot \tan^3\theta \left[-\frac{(h-x)^4}{4}\right]_0^h$$

$$= \frac{Q\omega}{R\ell\cos\theta} \cdot \tan^3\theta \left(\frac{h^4}{4}\right)$$

$$= \frac{Q\omega}{Rh}\left(\frac{R^3}{h^3}\right)\left(\frac{h^4}{4}\right)$$

$$= \frac{Q\omega}{4} R^2 = \boxed{\frac{Q\omega}{4} h^2 \tan^2\theta}$$

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M-II

$I \rightarrow$ hollow cone $= \frac{mR^2}{2}$

$I \rightarrow$ Solid Cone $= \frac{3mR^2}{10}$

$$\frac{M}{L} = \frac{q}{2m}$$

$$M = \frac{q}{2m}\left(\frac{mR^2}{2}\omega\right) = \frac{qR^2}{4}\omega = \boxed{\frac{Q\omega}{4}h^2\tan^2\theta}$$

Ex:- JEE 2025 Model

A Conducting Solid Cone carries charge Q & mass 'm' rotates about an axis passing through its centre. Find ratio $\frac{M}{L}$

Sol:- While rotating, it behaves as Hollow Cone ie, Charge is on Outer surface.

$$\frac{M}{L} = \frac{q}{2m}$$

$$M = \frac{q}{2m}\left(\frac{mR^2}{2}\omega\right) \qquad [L = I\omega]$$

$$M = \frac{qR^2\omega}{4}$$

$$L = I\omega = \frac{3mR^2}{10}\omega$$

$$\frac{M}{L} = \frac{1}{4} \times \frac{10}{3} \times \frac{q}{m} = \frac{5q}{6m}$$

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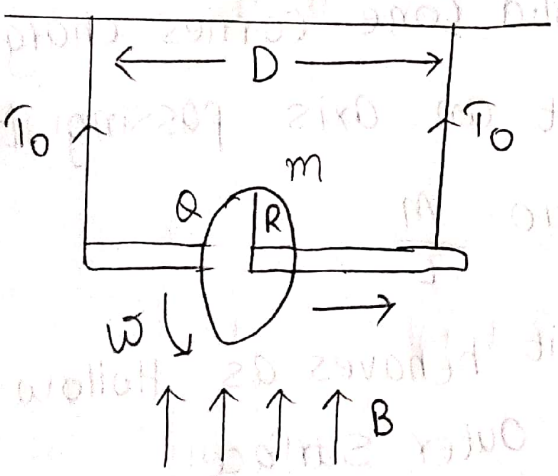
Ex:- A ring of radius R has uniformly distributed charge q mounted on a rod, suspended by two identical strings. The Tension in string in equilibrium is T_0. Now a vertical magnetic field is switched on and the ring is rotated at a constant angular velocity ω. Find max ω with which the ring can be rotated if the string can withstand a max Tension of $\frac{3T_0}{2}$

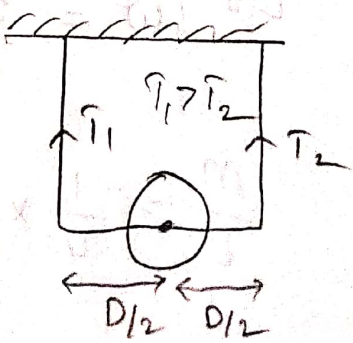


Sol:- $2T_0 = mg \Rightarrow T_0 = \frac{mg}{2} \longrightarrow (1)$

$$M = iA = \frac{Q\omega}{2\pi} \pi R^2$$

$$|\bar{\tau}| = |\bar{M} \times \bar{B}| = MB\sin 90^\circ$$

$$\boxed{\tau = \frac{Q\omega R^2 B}{2}}$$



$$T_1 + T_2 = mg \longrightarrow (2)$$

$$T_1\left(\frac{D}{2}\right) - T_2\left(\frac{D}{2}\right) = \frac{Q\omega R^2 B}{2}$$

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$$T_1 - T_2 = \frac{Q\omega R^2 B}{D}$$

$$2T_1 = mg + \frac{Q\omega R^2 B}{D}$$

$$2\left(\frac{3T_0}{2}\right) = 2T_0 + \frac{Q\omega R^2 B}{D} \qquad [\text{From } (1)]$$

$$T_0 = \frac{Q\omega R^2 B}{D}$$

$$\boxed{\omega = \frac{D T_0}{B Q R^2}}$$

Ex:- A rectangular loop PQRS has length a, breadth b and mass m. It is free to rotate about arm PQ which remains hinged on a horizontal line taken as the Y-axis. A Uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})\, B_0$ exists in the region. The loop is now released and is found to stay in horizontal position in equillibrium

i) What is the direction of current i in PQ

ii) Find magnetic force on arm RS.

iii) Find expression for i interms of B_0, A, m

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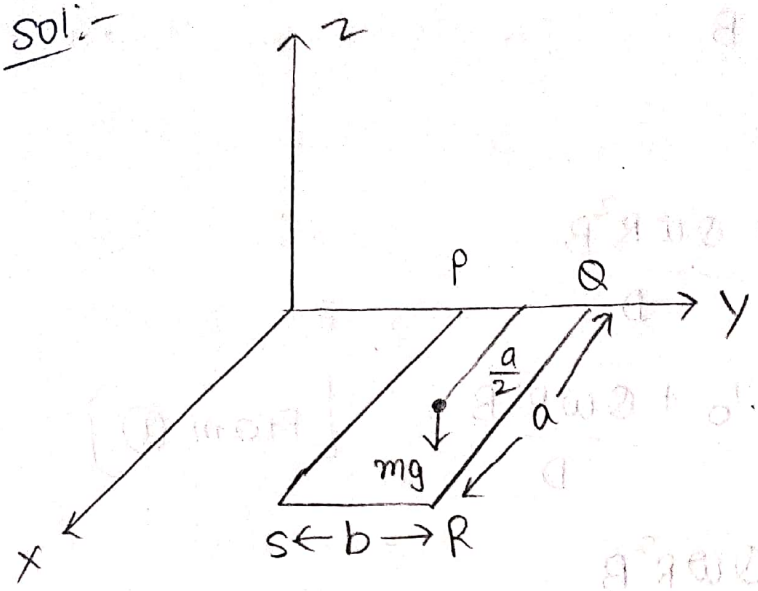
Sol:-



$$\tau_g = mg\frac{a}{2}\hat{j}$$

τ_{Fm} must be along $-\hat{j}$

Let $\bar{m} = m\hat{k}$

Now, $\bar{\tau}_B = \bar{m} \times \bar{B}$

$$= m\hat{k} \times B_0(3\hat{i} + 4\hat{k})$$

$$= 3mB_0\hat{j}$$

$\therefore \bar{m} = m_0(-\hat{k})$

Current flows in the order $P \to Q \to R \to S \to P$

Force on RS, $\bar{b} = b(-\hat{j})$

$$\bar{F} = i(\bar{b} \times \bar{B}) = i\left(b(-\hat{j}) \times B_0(3\hat{i} + 4\hat{k})\right)$$

$$= i\left(-4bB_0\hat{i} + 3bB_0\hat{k}\right)$$

$$= ibB_0(-4\hat{i} + 3\hat{k})$$

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$$\tau_g = \tau_B$$

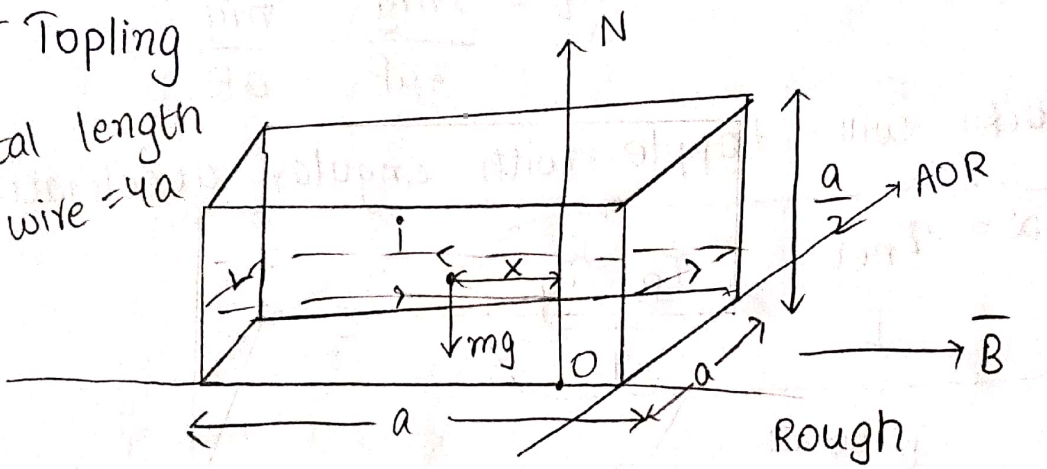
$$mg\frac{a}{2} = 3MB_0$$

$$mg\frac{a}{2} = 3iabB_0$$

$$\boxed{i = \frac{mg}{6bB_0}}$$

Ex:- Topling

Total length of wire = 4a



Torque due to magnetic field,

$$\tau_B = MB = ia^2B \longrightarrow (1) \text{ (Clockwise)}$$

Torque due to gravity about point O.

$$\tau_g = mgx \longrightarrow (2)$$

For no toppling, $\tau_g = \tau_B$

$$\Rightarrow mgx = ia^2B$$

$$\boxed{x = \frac{ia^2B}{mg}}$$

Here 'i' is variable,

For just toppling, $x = \frac{a}{2}$

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$$\frac{a}{2} = \frac{i_{max}\, a^2 B}{mg}$$

$$i_{max} = \frac{mg}{2aB}$$

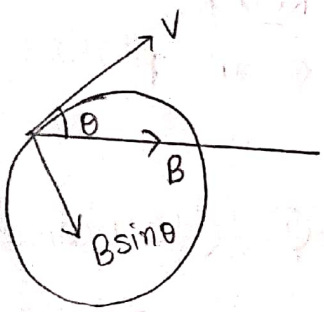
When $i > \frac{mg}{2aB}$ i.e, For ex,

$$\tau = \frac{2mg}{2aB} = \frac{mg}{aB}$$

then Block will topple with angular acceleration

$$\alpha = \frac{\tau_{net}}{I} = \frac{\tau_B - \tau_g}{I}$$

Ex:-



$$R = \frac{V^2}{a_\perp} = \frac{V^2}{\frac{BqV\sin\theta}{m}} = \frac{mV}{Bq\sin\theta}$$

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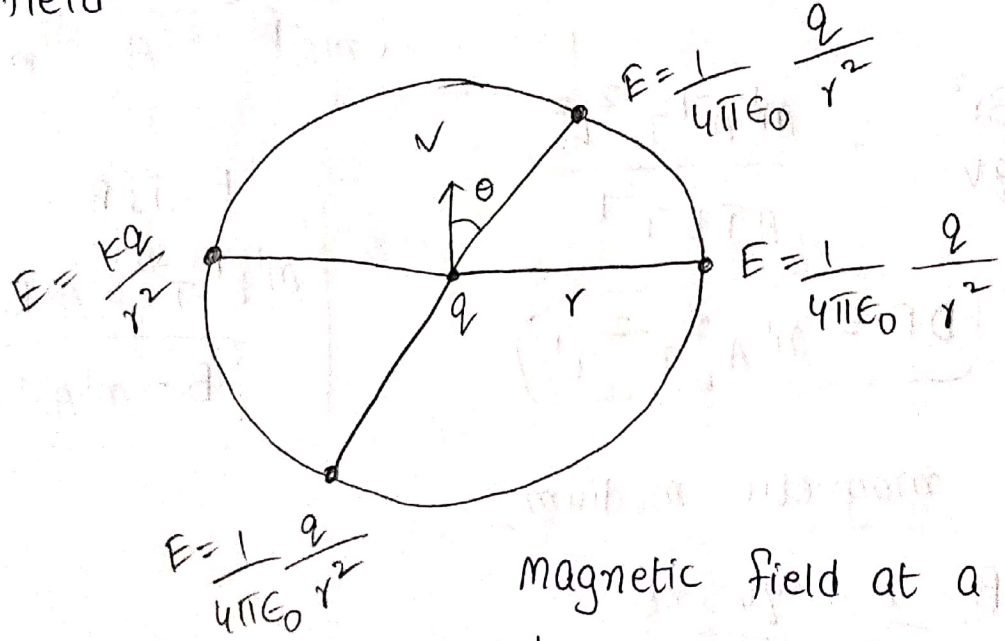
Calculation of B:-

* A Stationary charge can produce constant electric field known as Electrostatic field.
* Whereas, a moving charge w.r.t observer can produce both electric and magnetic field known as Electromagnetic field. [Refer H.C.Verma]
* A Variable Electric field can produce Magnetic field.



Magnetic field at a point due to moving charge is,

(1) $B \propto q$ (magnitude)

(2) $B \propto v$ (speed)

(3) $B \propto \sin\theta$ (θ is angle b/w $\bar{v}$ & $\bar{r}$)

(4) $B \propto \frac{1}{r^2}$ (distance)

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By combining,

$$B \propto \frac{qv\sin\theta}{r^2}$$

$$B = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$$

μ_0 = Permeability of free space

$= 4\pi \times 10^{-7}$ henry/metre (or) $\frac{Tm^2}{Cms^{-1}} = \frac{T \cdot m}{A} = \frac{Wb}{m-A}$

$$D.F = \frac{Br^2}{qv} = \frac{m^1 A^{-1} T^{-2} L^2}{ATLT^{-1}}$$

$$D.F = m^1 A^{-2} T^{-2} L^1$$

$F = ilB$

$m^1 L^1 T^{-2} = ALB$

$B = m^1 A^{-1} T^{-2}$

In case of magnetic medium,

$$B = \frac{\mu}{4\pi}\frac{qv\sin\theta}{r^2}$$

where, $\mu = \mu_0 \mu_r$

$\mu_r = \frac{\mu}{\mu_0}$ ← relative permeability.

i) $B = 0$ when $v = 0$ (or) $\theta = 0^\circ, \theta = 180^\circ$

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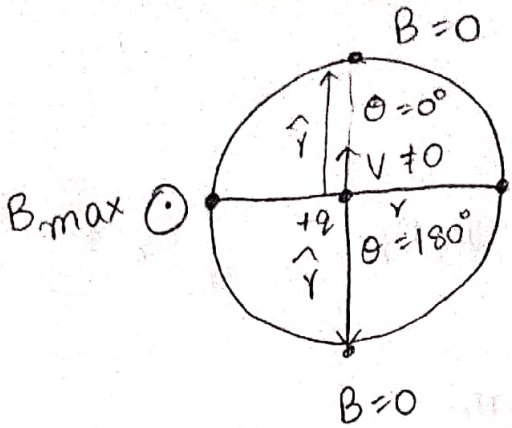

B_{max} ⊙ ⊗ $B_{max} = \frac{\mu_0 qV}{4\pi r^2}$

② In vector form,

$$\bar{B} = \frac{\mu_0 q}{4\pi} \frac{(\bar{v} \times \hat{r})}{r^2} \rightarrow ①$$

③ Electric field at that point is,

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \rightarrow ②$$

Eq ① becomes,

$$\bar{B} = \frac{\mu_0}{4\pi} \left(\bar{v} \times \frac{q\hat{r}}{r^2} \right)$$

$$\bar{B} = \frac{\mu_0}{4\pi} \left(\bar{v} \times 4\pi\epsilon_0 \bar{E} \right)$$

$$\bar{B} = \frac{\mu_0 4\pi\epsilon_0}{4\pi} \left(\bar{v} \times \bar{E} \right)$$

$$\bar{B} = \mu_0 \epsilon_0 \left(\bar{v} \times \bar{E} \right)$$

We have, $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ $\quad v = \sqrt{\frac{1}{\mu\epsilon}}$

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Refractive Index,

$$n = \frac{c}{v} = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}} = \sqrt{\mu_r \epsilon_r} \qquad [\epsilon_r = k]$$

For non-conducting medium,

$$\boxed{\mu_r = 1} \Rightarrow \boxed{n = \sqrt{\epsilon_r}}$$

$$\therefore \boxed{\bar{B} = \frac{\bar{V} \times \bar{E}}{c^2}}$$

and, $\bar{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$; $\bar{B} = \frac{\mu_0}{4\pi} \cdot \frac{qv\sin\theta}{r^2}$

$$\frac{E}{B} = \frac{1}{\mu_0 \epsilon_0} \cdot \frac{1}{v\sin\theta}$$

$$\boxed{\frac{E}{B} = \frac{c^2}{v\sin\theta}}$$

Clearly, $\boxed{E > B}$ at all points.

To find direction of magnetic field, we use, ① Maxwell's Cork Screw rule

② Ampere's swimming rule

③ Right hand Thumb rule.

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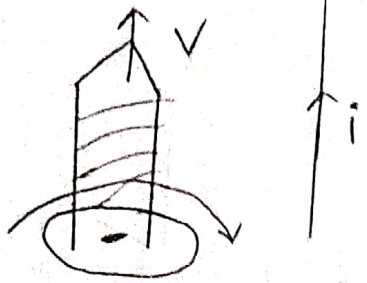
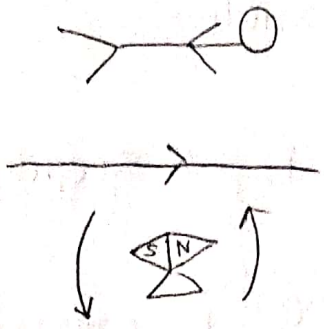
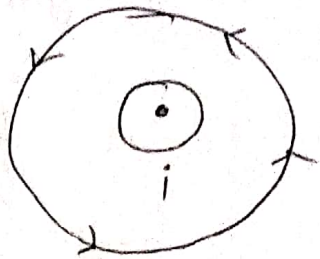
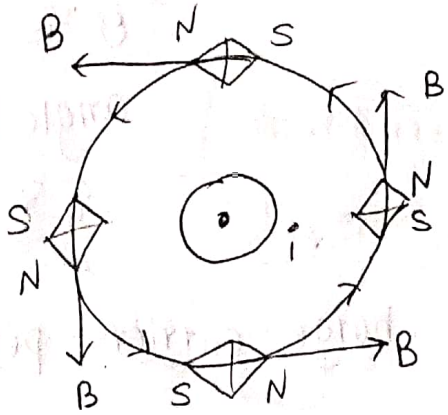


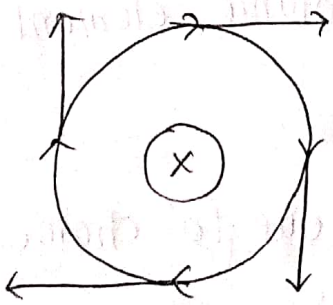




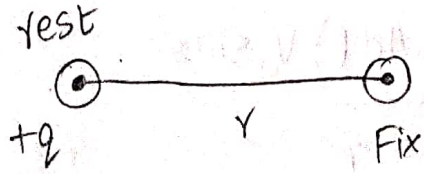
* From all these rules, we can conclude that the plane of the magnetic field is always $\perp^{lar}$ to the current carrying conductor



Direction of B is always tangent to circle and $\perp$ to the conductor



Biot - Savart's Law:-



$\bar{E}$ = constant = Electro static field.

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} = \text{constant}$$

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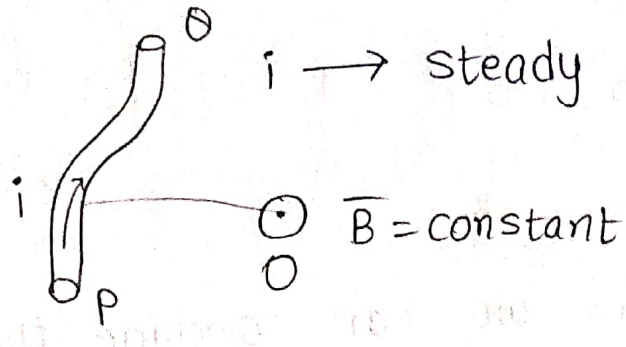
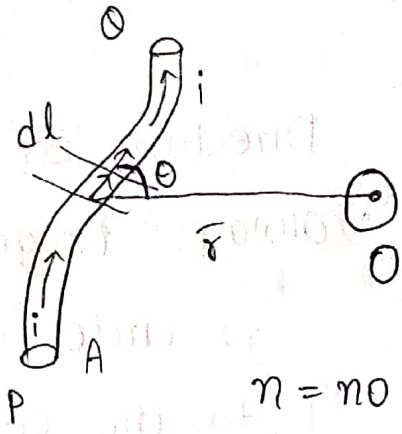
In Magnetic field,



$i \longrightarrow$ steady current

$\bar{B}$ = constant

* There should be no charge accumulation in case of Steady Current



$\bar{B}$ = constant

'θ' is the angle blw $d\bar{l}$ & $\bar{r}$

n = no. of charge carriers per unit Volume

Charge contained in the volume element,

$$dq = neA\,dl$$

Magnetic field at point 'P' due to charge flowing in small element,

$$dB = \frac{\mu_0}{4\pi} \frac{(dq)\, V_d \sin\theta}{r^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{(neA\,dl)\, V_d \sin\theta}{r^2}$$

$[i = neAV_d]$

$$\boxed{dB = \frac{\mu_0}{4\pi} \frac{i\,dl \sin\theta}{r^2}}$$

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Magnetic field at 'O' due to current in entire conductor is,

$$B = \int_P^Q dB$$

Points:-

① $dB = \dfrac{\mu_0 i\, d\ell \sin\theta}{4\pi r^2}$

$$d\bar{B} = \frac{\mu_0}{4\pi} \frac{i\,(d\bar{\ell} \times \hat{r})}{r^2}$$

(or) $$d\bar{B} = \frac{\mu_0}{4\pi} \frac{(i d\bar{\ell} \times \hat{r})}{r^2}$$

$i d\bar{\ell}$ = Current element

(or) $$d\bar{B} = \frac{\mu_0}{4\pi} \frac{(i d\bar{\ell} \times \bar{r})}{r^3}$$

Clearly, $d\bar{B} \perp i d\bar{\ell}$

$d\bar{B} \perp \bar{r}$

$d\bar{B} \parallel i(d\bar{\ell} \times \bar{r})$

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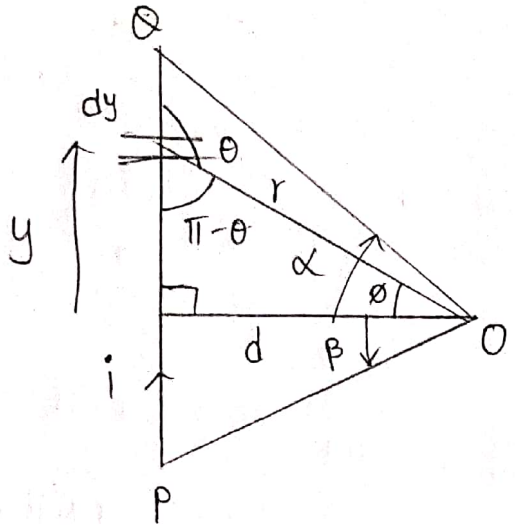
Applications :-

① Magnetic field at a point due to current carrying straight conductor :-



d → ⊥ lar distance

$r = \sqrt{y^2 + d^2}$

$\theta = \frac{\pi}{2} + \phi$

$$dB = \frac{\mu_0 i dy \sin\theta}{4\pi \left(\sqrt{d^2+y^2}\right)^2}$$

From fig, $\tan\phi = \frac{y}{d}$

$$y = d\tan\phi$$

$$dy = d\sec^2\phi\, d\phi$$

$$dB = \frac{\mu_0 i}{4\pi}\left(\frac{d\sec^2\phi\, d\phi}{d^2 + d^2\tan^2\phi}\right)\sin\left(\frac{\pi}{2}+\phi\right)$$

$$dB = \frac{\mu_0 i}{4\pi d}\cos\phi\, d\phi$$

Total magnetic field, $B = \int_{-\beta}^{\alpha} dB = \int_{-\beta}^{\alpha} \frac{\mu_0 i}{4\pi d}\cos\phi\, d\phi$

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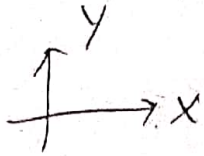
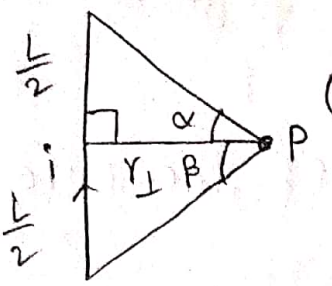
$$B = \frac{\mu_0 i}{4\pi d}(\sin\alpha + \sin\beta)$$

Points:-



① 

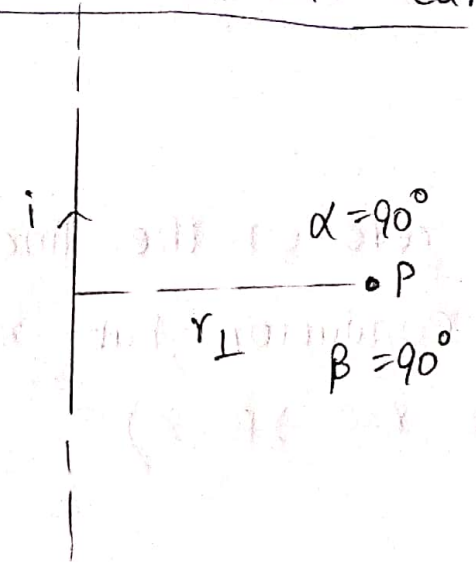
⊗ B

$\alpha = \beta$

$$B = \frac{\mu_0 i}{4\pi r_\perp}(2\sin\alpha)(d\hat{\ell} \times \hat{r})$$

$$B = \frac{\mu_0 i}{2\pi r_\perp} \times \frac{L/2}{\sqrt{\left(\frac{L}{2}\right)^2 + r_\perp^2}}$$

② Infinite Current carrying wire:-



$$B = \frac{\mu_0 i}{2\pi r_\perp}$$

i.e $B \propto \frac{1}{r}$ ⇒ Rectangular Hyperbola

③ At one end of semi-infinite long current carrying wire:-

$\alpha = 90°$, $\beta = 0°$

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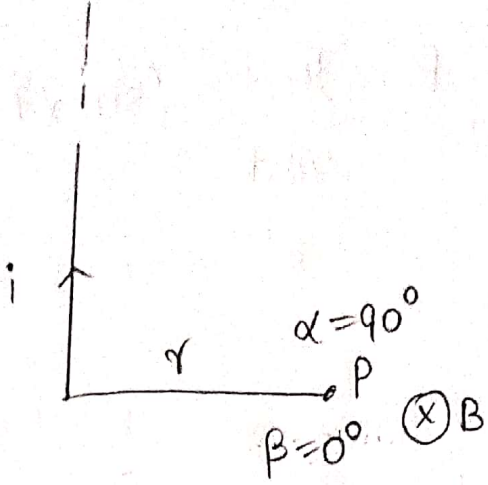
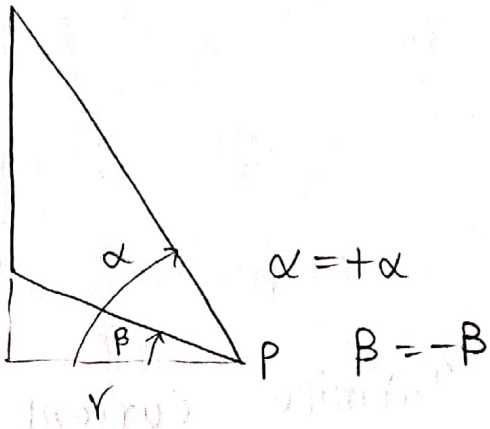
$$B = \frac{\mu_0 i}{4\pi r}$$



(4) If Point is not in front of the wire :-

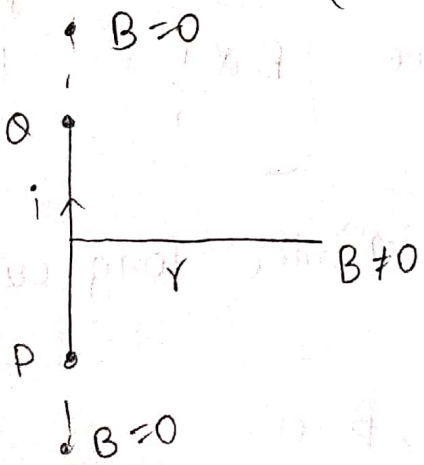
$$B = \frac{\mu_0 i}{4\pi r}\left[\sin\alpha + \sin(-\beta)\right]$$

$$B = \frac{\mu_0 i}{4\pi r}\left[\sin\alpha - \sin\beta\right]$$



(5) $d\vec{B} \parallel i d\vec{l} \times \vec{r}$

↳ magnetic field is zero on the line of the current carrying conductor, but not on the wire (on wire, $r = 0 \Rightarrow B = \infty$)



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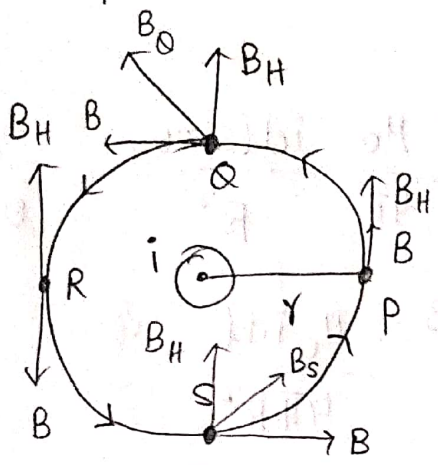
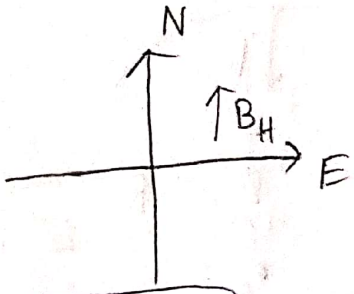
Note:- Current Carrying wire is kept vertically in Earth's magnetic field & current is flowing upwards.





Here $B = \frac{\mu_0 i}{2\pi Y}$

Net magnetic field

(a) at P is $B_P = B + B_H$

(b) at Q is $B_Q = \sqrt{B^2 + B_H^2}$

(c) at R is $B_R = B - B_H$

(d) at S is $B_S = \sqrt{B^2 + B_H^2}$

At R, if $B_{net} = 0$ (null point)

$\Rightarrow B = B_H$

$\Rightarrow B_H = \frac{\mu_0 i}{2\pi r}$

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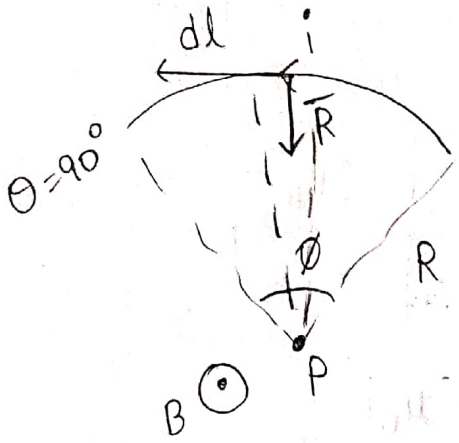
Magnetic field at the centre of current carrying circular arc shaped wire:-



$$dB = \frac{\mu_0}{4\pi}\frac{i\,dl\sin\theta}{R^2} \qquad [\theta = 90^\circ]$$

$$\Rightarrow dB = \frac{\mu_0 i\,dl}{4\pi R^2}$$

Total magnetic field at the centre,

$$B = \int dB = \int_{\phi} \frac{\mu_0 i\,dl}{4\pi R^2}$$

$$B = \int_0 \frac{\mu_0 i\,R\,d\phi}{4\pi R^2}$$

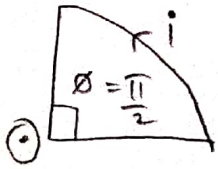
$$B = \frac{\mu_0 i \phi}{2\pi(2R)}$$

$$\boxed{B = \frac{\mu_0 i}{2R}\left(\frac{\phi}{2\pi}\right)} \quad \odot \quad \phi \to \text{rad}$$

Points:-

① For quarter shaped circular arc,

$$B = \frac{\mu_0 i}{2R}\left(\frac{\pi/2}{2\pi}\right) = \frac{\mu_0 i}{8R} \odot$$



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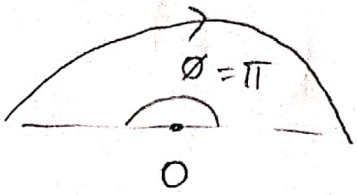
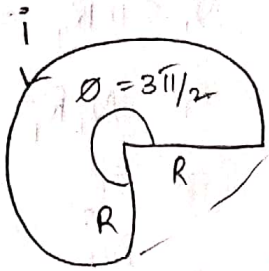
(2) Semi-circular arc



$$B = \frac{\mu_0 i}{2R}\left(\frac{\pi}{2\pi}\right) = \frac{\mu_0 i}{4R} \quad \otimes$$

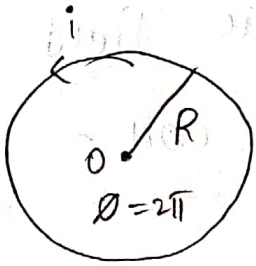
(3) Three fourt of circle



$$B = \frac{\mu_0 i}{2R}\left(\frac{3\pi/2}{2\pi}\right)$$

$$B = \frac{3\mu_0 i}{8R} \quad \odot$$

(4) For Complete Circle,



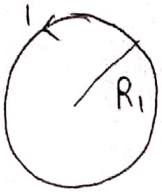
$$B_c = \frac{\mu_0 i}{2R}\left(\frac{2\pi}{2\pi}\right)$$

$$\boxed{B_c = \frac{\mu_0 i}{2R}} \quad \odot$$

If the coil contains N-turns then,

$$\boxed{B_c = \frac{\mu_0 N i}{2R}}$$

* If l is the length of wire used,



$$B_1 = \frac{\mu_0 N_1 i}{2R_1}$$

Coil is now unwounded and then rewounded,

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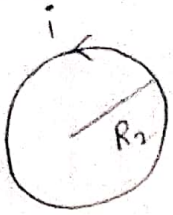


$$B_2 = \frac{\mu_0 N_2 i}{2R_2}$$

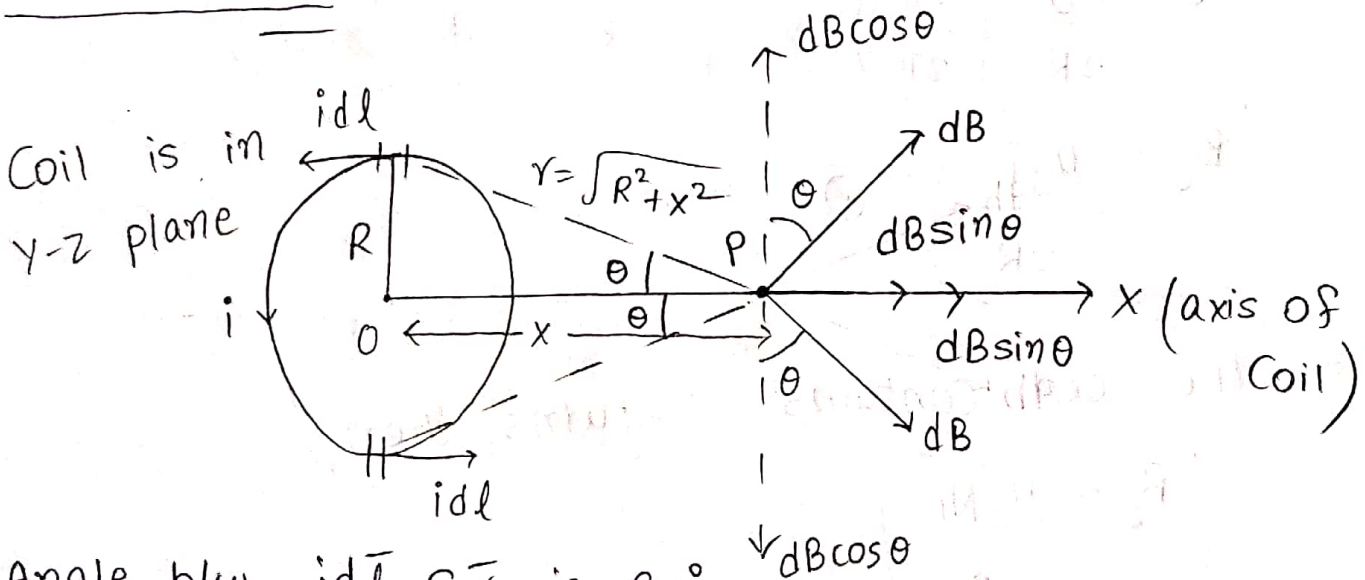
Now, $$\frac{B_1}{B_2} = \frac{N_1}{N_2}\left(\frac{R_2}{R_1}\right)$$

We have $l = N_1 2\pi R_1 = N_2 2\pi R_2$

$$\Rightarrow N_1 R_1 = N_2 R_2$$

$$\Rightarrow \boxed{\frac{B_1}{B_2} = \frac{N_1^2}{N_2^2} = \frac{R_2^2}{R_1^2}}$$

Magnetic field at a point on the axis of a circular coil :-



Angle b/w $id\bar{l}$ & $\bar{r}$ is 90°

Magnetic field at point 'P' due to current element,

$$dB = \frac{\mu_0 i dl \sin 90^\circ}{4\pi r^2}$$

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Net field at 'P' is,

$$B_{axis} = \int dB \sin\theta = \int \frac{\mu_0 i\, dl}{4\pi(R^2+x^2)} \frac{R}{\sqrt{R^2+x^2}}$$

$$B_a = \frac{\mu_0 iR}{4\pi(R^2+x^2)^{3/2}} \int dl$$

$$B_a = \frac{\mu_0 iR}{4\pi(R^2+x^2)^{3/2}} (2\pi R)$$

$$\boxed{B_a = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}}}$$

If the coil contains N-turns,

$$\boxed{B_a = \frac{\mu_0 N i R^2}{2(R^2+x^2)^{3/2}}}$$

$$E_a = \frac{kQx}{(x^2+R^2)^{3/2}}$$

$x=0$ (or) $x=\infty$

$\Rightarrow E_a = 0$ | max $\rightarrow \frac{R}{\sqrt{2}}$

Points:-

① $$B_a = \frac{\mu_0 N i R^2}{2(R^2+x^2)^{3/2}} = \frac{\mu_0 N i R^2}{2R^3\left(1+\frac{x^2}{R^2}\right)^{3/2}}$$

$$= \frac{\mu_0 N i}{2R\left(1+\frac{x^2}{R^2}\right)^{3/2}}$$

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$$B_a = \frac{B_c}{\left(1+\frac{x^2}{R^2}\right)^{3/2}}$$

If $x <<< R$, then

$$B_a = B_c \left(1+\frac{x^2}{R^2}\right)^{-3/2}$$

$$B_a = B_c \left(1-\frac{3x^2}{2R^2}\right)$$

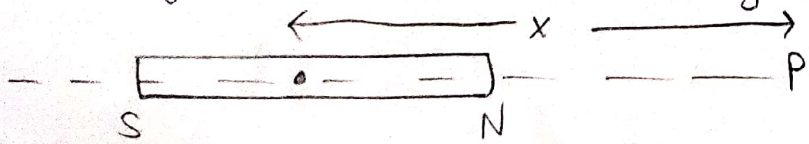
(2) $$B_a = \frac{\mu_0 N i \pi R^2}{2\pi (R^2+x^2)^{3/2}}$$

If $x >>> R$

then $$B_a = \frac{\mu_0 N i \pi R^2}{2\pi x^3}$$

$$B_a = \frac{\mu_0 \, 2(N i \pi R^2)}{4\pi \, x^3}$$

Comparing with short bar magnet,



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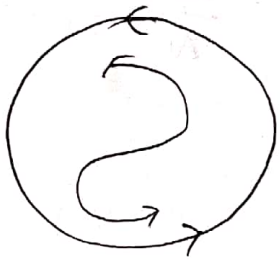
$$B_a = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$$

$$B_e = \frac{\mu_0}{4\pi}\frac{M}{x^3}$$

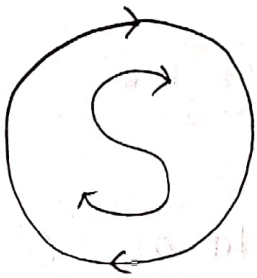
$$M = Ni\pi R^2 = NiA$$

i.e, $$B_a = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$$, for any current carrying loop

(It is applicable only for small sized coils)



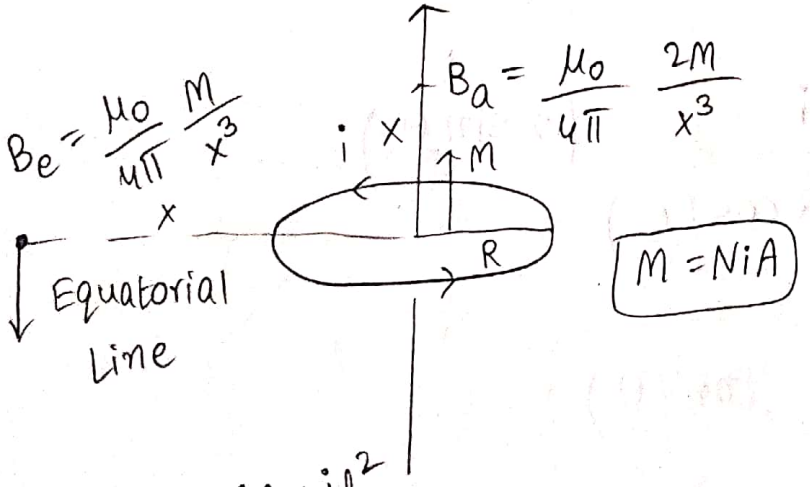
N-Pole



S-Pole

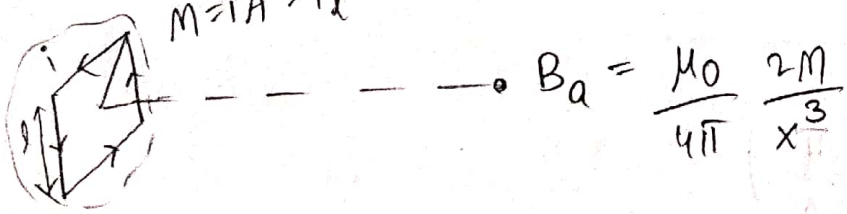
[Clock face Rule]

*



$M = NiA$

*



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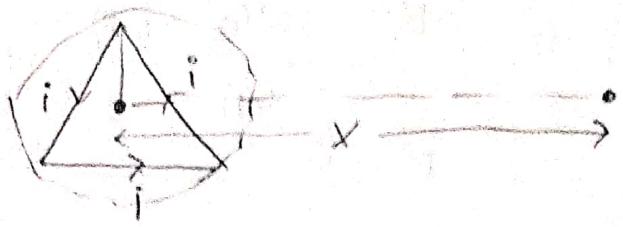
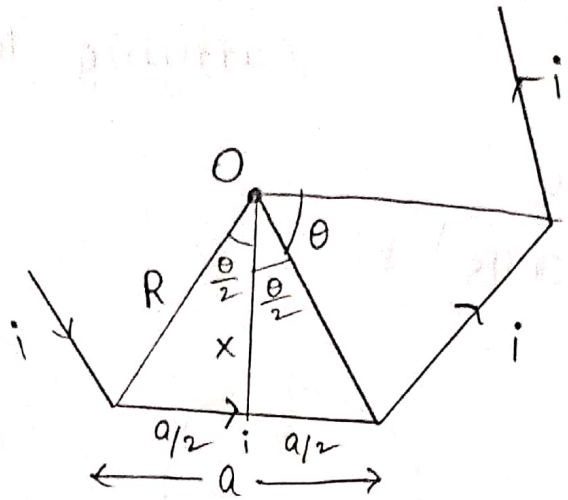


$$B_a = \frac{\mu_0}{4\pi}\frac{2M}{x^3}$$

$$M = iA = i\frac{\sqrt{3}}{4}a^2$$

Magnetic field at the centre of N-sided Regular Polygon:-



$$N\theta = 2\pi$$

$$\theta = \frac{2\pi}{N}$$

Magnetic field at 'O' due to one side.

$$B_1 = \frac{\mu_0 i}{4\pi x}(\sin\alpha + \sin\beta)$$

$$= \frac{\mu_0 i}{4\pi R\cos\left(\frac{\theta}{2}\right)}\left(2\sin\frac{\theta}{2}\right)$$

$$\boxed{B_1 = \frac{\mu_0 i}{2\pi R}\tan\left(\frac{\theta}{2}\right)}$$

$$B_1 = \frac{\mu_0 i}{2\pi R}\tan\left(\frac{\pi}{N}\right)$$

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due to all N-sides,

$$B = NB_1$$

$$B = \frac{\mu_0 Ni}{2\pi R} \tan\left(\frac{\pi}{N}\right)$$

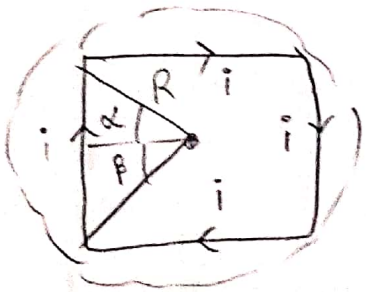
$$B = \frac{\mu_0 i}{2R} \frac{\tan\left(\frac{\pi}{N}\right)}{\left(\frac{\pi}{N}\right)}$$

If $N \to \infty$; Polygon becomes circle,

$$B = \frac{\mu_0 i}{2R} \lim_{N\to\infty} \frac{\tan\left(\frac{\pi}{N}\right)}{\frac{\pi}{N}}$$

$$B = \frac{\mu_0 i}{2R}$$

Ex:-



$$B_1 = \frac{\mu_0 i}{4\pi \frac{l}{2}} (2\sin 45^\circ)$$

$$B_1 = \frac{\mu_0 i}{2\pi l} \sqrt{2}$$

$$B_{net} = 4B_1 = \frac{2\sqrt{2}\,\mu_0 i}{\pi l}$$

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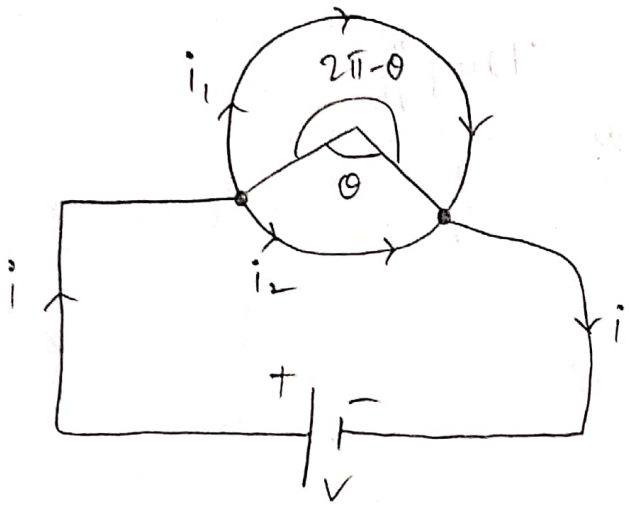
$$B = \frac{\mu_0 i}{2R} \frac{\tan\left(\frac{\pi}{N}\right)}{\pi/N}$$

$$B = \frac{\mu_0 i}{2\left(\frac{l}{\sqrt{2}}\right)} \frac{\tan(\pi/4)}{\pi/4}$$

$$\boxed{B = 2\sqrt{2}\,\frac{\mu_0 i}{\pi l}}$$

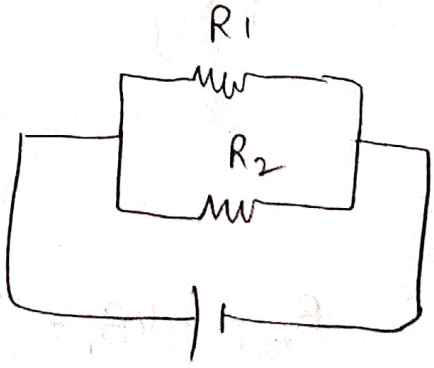
Battery is connected b/w any two points on coil:-



$$B_c = B_1 - B_2$$

where $B_1 = \frac{\mu_0 i_1}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)$

& $B_2 = \frac{\mu_0 i_2}{2R}\left(\frac{\theta}{2\pi}\right)$



Since Both plates are in parallel,

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$$i_1 R_1 = i_2 R_2$$

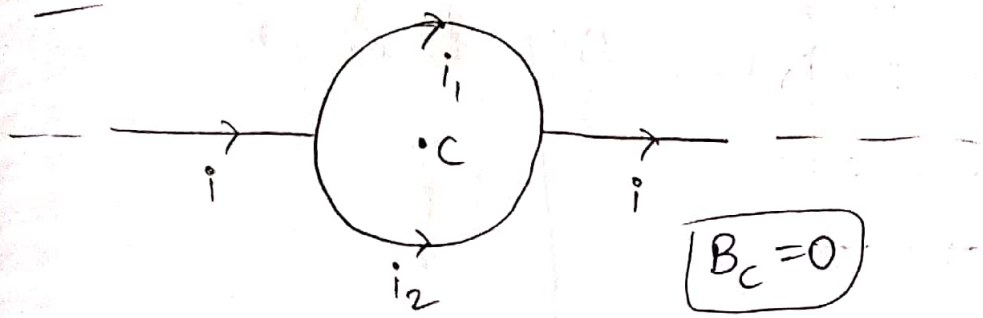
$$i_1 \lambda R (2\pi - \theta) = i_2 \lambda R (\theta)$$

$$i_1 (2\pi - \theta) = i_2 \theta$$

$$B_c = \frac{\mu_0}{4\pi R} \left(i_1 (2\pi - \theta) - i_2 \theta \right)$$

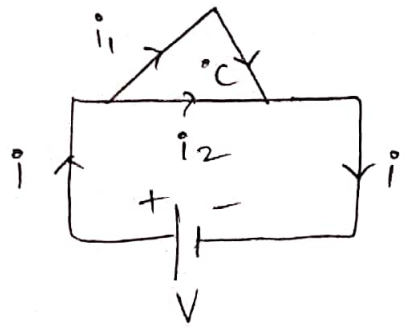
$\Rightarrow$ $\boxed{B_c = 0}$ ie, Independent of angle 'θ' & shape of coil

Ex:-



$B_c = 0$

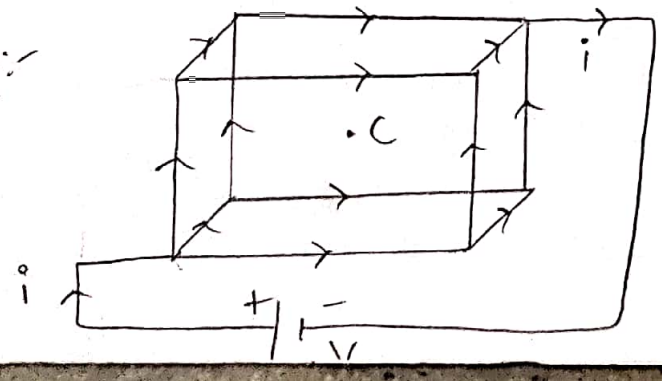
Ex:-



$B_c = 0$

wire with uniform thickness & same material

Ex:-



$B_c = 0$

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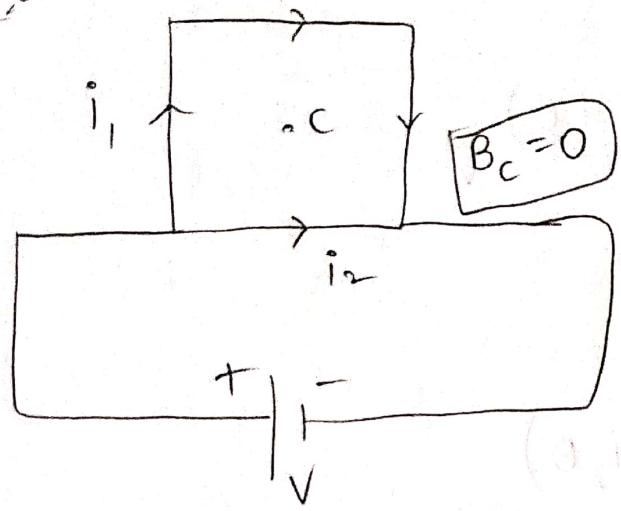
Ex:-



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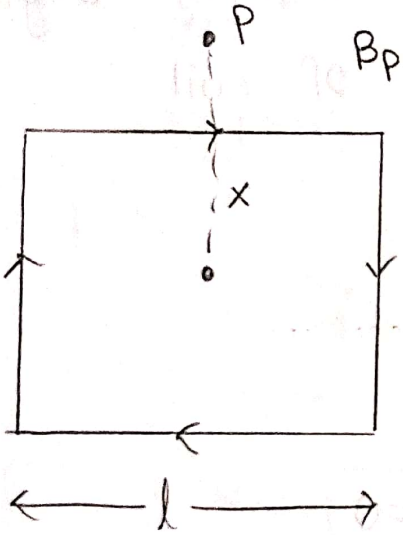
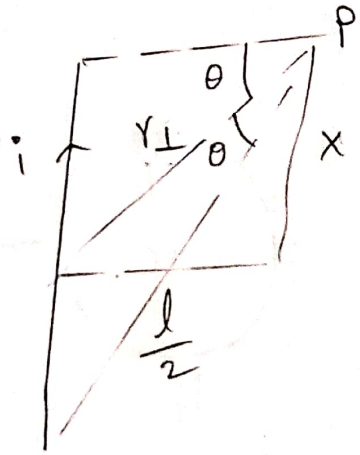
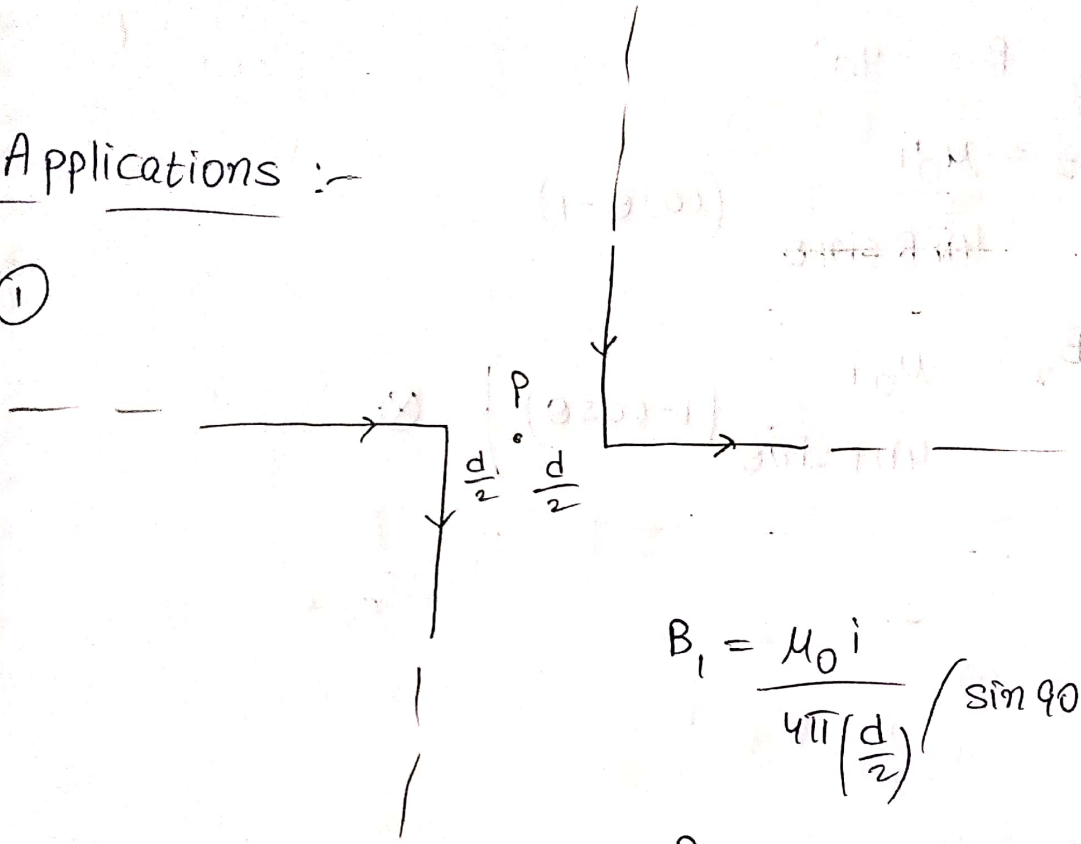




$$\frac{4\mu_0 i l^2}{\pi(4x^2+l^2)\sqrt{4x^2+2l^2}}$$

Applications :-

(1)



$$B_1 = \frac{\mu_0 i}{4\pi\left(\frac{d}{2}\right)}\left(\sin 90^\circ + \sin 0^\circ\right) \odot$$

$$B_1 = \frac{\mu_0 i}{2\pi d} \odot$$

$$B_2 = \frac{\mu_0 i}{2\pi d} \otimes$$

$$B_p = 0$$

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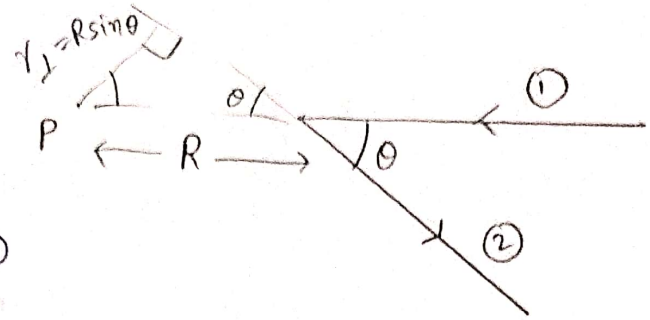
(2)

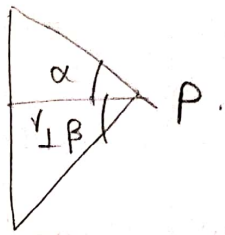


$B_1 = 0$

$$B_2 = \frac{\mu_0 i}{4\pi r_\perp}(\sin\alpha + \sin\beta)$$

$\alpha = 90 - \theta$

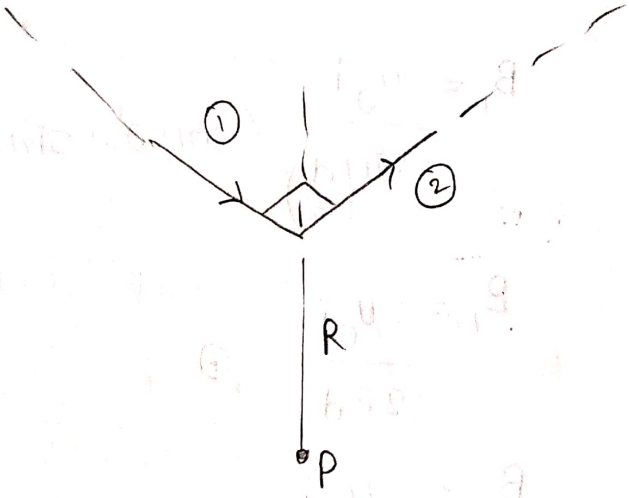
$\beta = -90°$



$$B_2 = \frac{\mu_0 i}{4\pi R\sin\theta}(\cos\theta - 1)$$

$$\boxed{B_2 = \frac{\mu_0 i}{4\pi R\sin\theta}(1-\cos\theta)} \otimes$$

Ex:-



Sol:-

$B = 2B_1 \quad [B_1 = B_2]$

$$= 2\left[\frac{\mu_0 i}{4\pi r_\perp}(\sin\alpha + \sin\beta)\right]$$

$$= 2\left[\frac{\mu_0 i}{4\pi\left(\frac{R}{\sqrt{2}}\right)}(\sin 90 + \sin(-45°))\right]$$

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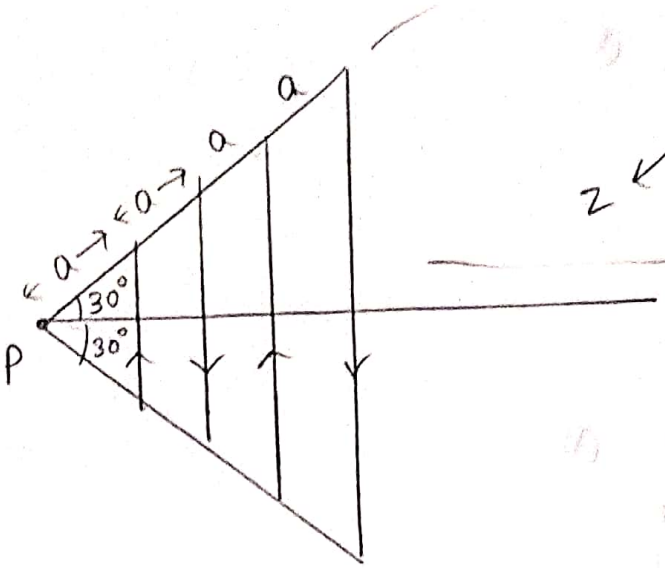
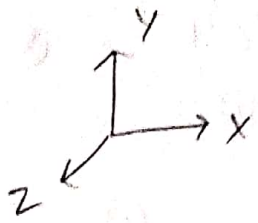
$$= \frac{\mu_0 i}{\sqrt{2}\pi R}\left(1-\frac{1}{\sqrt{2}}\right)$$

Ex:-





to ∞

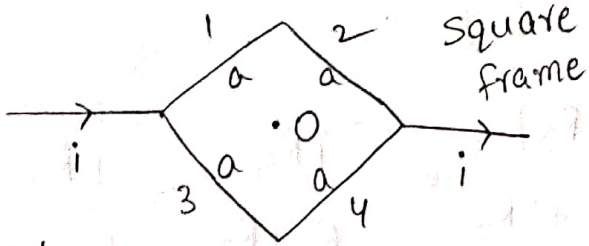
Sol:- $B = B_1 - B_2 + B_3 - \cdots$

$$\sin\alpha + \sin\beta = \sin 30^0 + \sin 30^0 = \frac{1}{2}+\frac{1}{2} = 1$$

$$B = \frac{\mu_0 i}{4\pi\left(\frac{\sqrt{3}a}{2}\right)}(\sin\alpha+\sin\beta) - \frac{\mu_0 i}{4\pi\left(\frac{2a\sqrt{3}}{2}\right)}(\sin\alpha+\sin\beta) - \cdots$$

$$B = \frac{\mu_0 i\,(1)}{2\pi a\sqrt{3}}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{3}-\cdots\infty\right) = \frac{\mu_0 i}{2\pi a\sqrt{3}}(\ln 2)$$

Ex:-

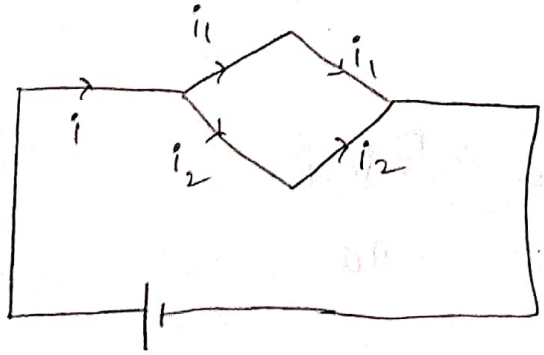


square frame

Find net magnetic field

A square frame is made of four metal rod of same material but of rod 1 & rod 2 of thickness 2d and 3 & 4 of thickness d

Sol:-



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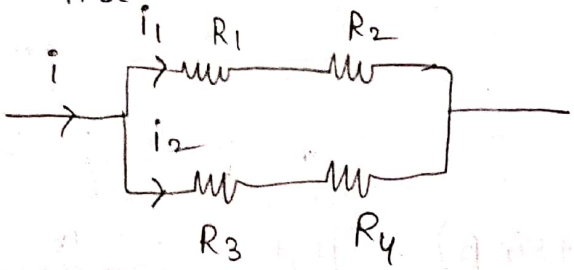
$$B' = 2\,\frac{\mu_0 i_1}{4\pi\left(\frac{a}{2}\right)}\,(2\sin 45^\circ)$$

$$B' = \frac{\sqrt{2}\mu_0 i_1}{\pi a} \quad \otimes$$

& $$B'' = \frac{\sqrt{2}\mu_0 i_2}{\pi a} \quad \odot$$

$$B = \frac{\sqrt{2}\mu_0}{\pi a}(i_1 - i_2) \quad \otimes$$



$$R_1 = R_2 = \frac{\rho a}{\pi (2d)^2} \qquad \& \quad R_3 = R_4 = \frac{\rho a}{\pi d^2}$$

$$\boxed{R_1 = \frac{R_3}{4}}$$

$$i_1 = \frac{i(R_3 + R_4)}{R_1 + R_2 + R_3 + R_4} = \frac{i(2R_3)}{2R_1 + 2R_3} = \frac{iR_3}{R_1 + R_3} = \frac{iR_3}{\frac{R_3}{4} + R_3}$$

$$\boxed{i_1 = \frac{4i}{5}} \qquad \& \qquad \boxed{i_2 = \frac{i}{5}}$$

$$B = \frac{\sqrt{2}\mu_0}{\pi a}\left(\frac{4i}{5} - \frac{i}{5}\right) = \frac{3\sqrt{2}\mu_0 i}{5\pi a}$$

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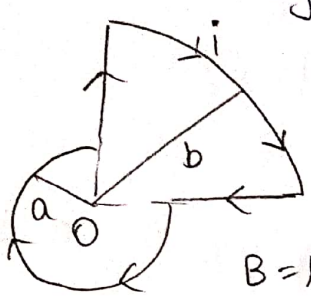
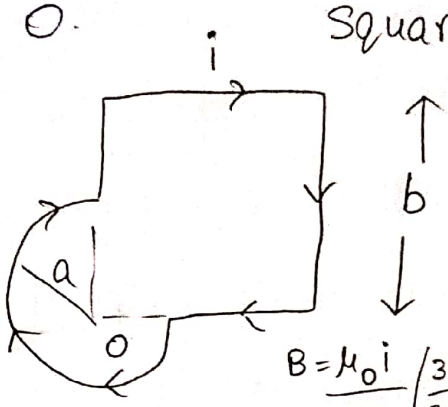
Ex:- Find magnetic field at O.

a.)



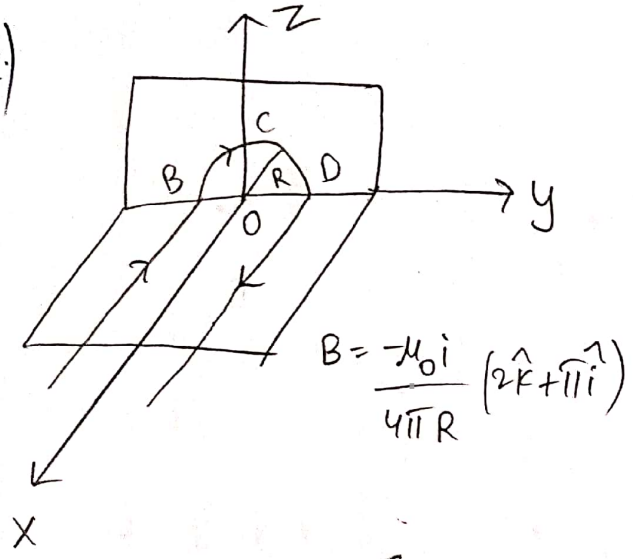
$$B = \frac{\mu_0 i}{8}\left(\frac{1}{b}+\frac{3}{a}\right)(-\hat{k})$$

b.) Square loop



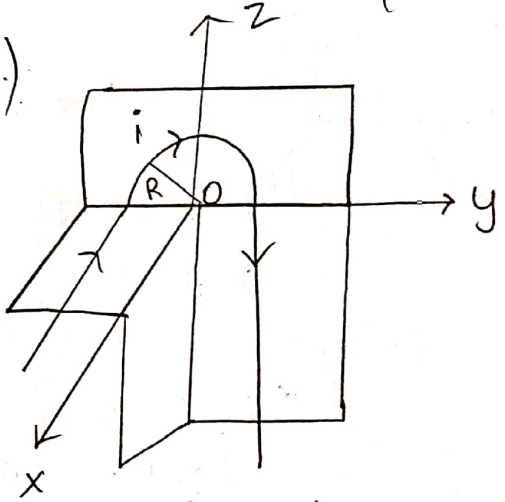
$$B = \frac{\mu_0 i}{4\pi}\left(\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right)(-\hat{k})$$

c.)



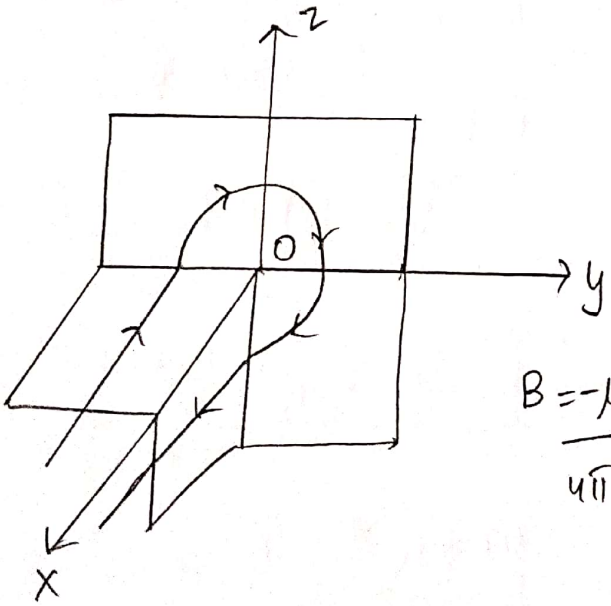
$$B = \frac{-\mu_0 i}{4\pi R}\left(2\hat{k}+\pi\vec{i}\right)$$

d.)



$$B = \frac{-\mu_0 i}{4\pi R}\left(\hat{k}+(\pi+1)\vec{i}\right)$$

e.)



$$B = \frac{-\mu_0 i}{4\pi R}\left(\hat{k}+\hat{j}+\frac{3\pi}{2}\vec{i}\right)$$

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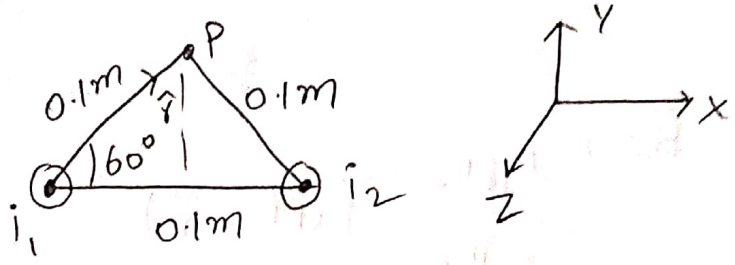
Ex:- Two straight infinitely long and thin parallel wires are 0.1m apart and carry a current of 10A each Find the magnetic field at a point distance 0.1m from both the wires in two cases when currents are in the a) same & b) opposite directions.



Sol:- $i_1 = i_2 = 10A$

a)

$$\bar{B}_1 = \frac{\mu_0 i_1}{2\pi r}(d\bar{\ell} \times \hat{r}) \qquad \hat{r} = 1\cos 60^\circ \hat{i} + 1\sin 60^\circ \hat{j}$$

$$\bar{B}_1 = \frac{\mu_0 i_1}{2\pi r}\left[\hat{k} \times \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)\right] = \frac{\mu_0 i_1}{2\pi r}\left[\frac{1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$\bar{B}_2 = \frac{\mu_0 i_2}{2\pi r}(d\bar{\ell} \times \hat{r})$$

$$= \frac{\mu_0 i_2}{2\pi r}\left[\hat{k} \times \left(\frac{-1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)\right] = \frac{\mu_0 i_2}{2\pi r}\left[\frac{-1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$\bar{B} = \bar{B}_1 + \bar{B}_2 = \frac{\mu_0 i_1}{2\pi r}\left[\frac{1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right] + \frac{\mu_0 i_2}{2\pi r}\left[\frac{-1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$= \frac{\mu_0 i}{2\pi r}\left[-\sqrt{3}\hat{i}\right] = \frac{4\pi \times 10^{-7}(10)}{2\pi(10^{-1})}(-\sqrt{3}\hat{i})$$

$$= \boxed{2\sqrt{3} \times 10^{-5}(-\hat{i})}$$

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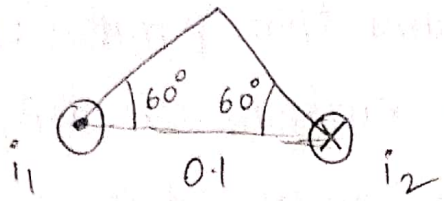
b)



$$\bar{B}_1 = \frac{\mu_0 i_1}{2\pi r}(d\bar{l} \times \hat{r})$$

$$\bar{B}_1 = \frac{\mu_0 i_1}{2\pi r}\left[\hat{k} \times \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)\right]$$

$$\bar{B}_1 = \frac{\mu_0 i_1}{2\pi r}\left[\frac{1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$\bar{B}_2 = \frac{\mu_0 i_2}{2\pi r}(d\bar{l} \times \hat{r})$$

$$= \frac{\mu_0 i_2}{2\pi r}\left[-\hat{k} \times \left(\frac{-1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)\right]$$

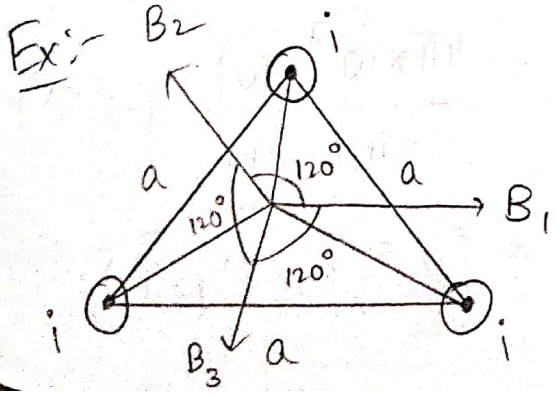
$$= \frac{\mu_0 i_2}{2\pi r}\left[\frac{1}{2}\hat{j} + \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$\bar{B} = \bar{B}_1 + \bar{B}_2 = \frac{\mu_0 i_1}{2\pi r}\left[\frac{1}{2}\hat{j} - \frac{\sqrt{3}}{2}\hat{i}\right] + \frac{\mu_0 i_2}{2\pi r}\left[\frac{1}{2}\hat{j} + \frac{\sqrt{3}}{2}\hat{i}\right]$$

$$= \frac{\mu_0 i_1}{2\pi r} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 10^{-1}}$$

$$= 2 \times 10^{-5}$$

Ex:-



Magnetic field at Centroid O = 0

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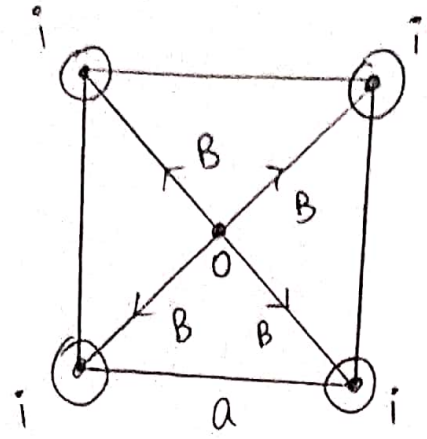
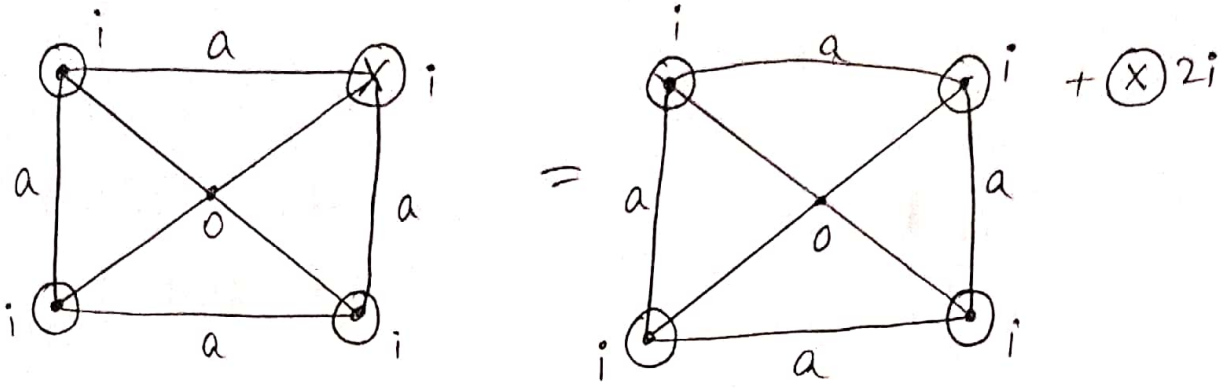
Ex:-



At O,

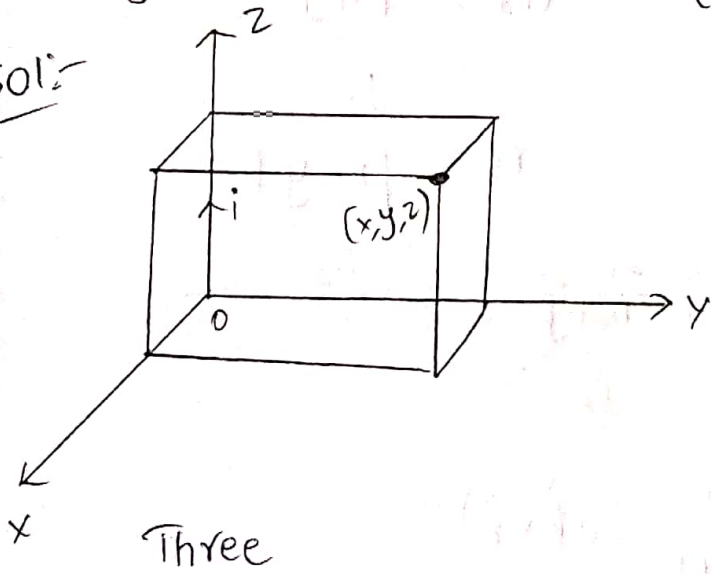
$$\boxed{B = 0}$$

Ex:-



Ex:- A long wire carries current 'i' along z-axis, Current is flowing along +ve z-direction. Find Magnetic induction at (x,y,z) in space

Sol:-



$$B = \frac{\mu_0 i}{2\pi r_\perp} (d\hat{l} \times \hat{r})$$

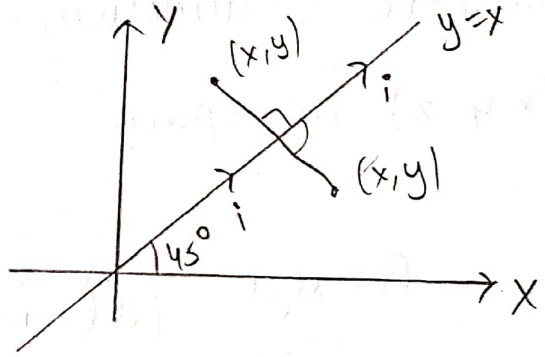
$$= \frac{\mu_0 i}{2\pi \sqrt{x^2+y^2}} \left[\hat{k} \times \frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}} \right]$$

Three

Ex:- long conducting wires each carrying current i along +ve x & y & z directions then magnetic induction at (3,4,5) =

Ex:- A wire carries current i along a direction making equal angles with +x & y-axis. Find B at (x,y)

Sol:-



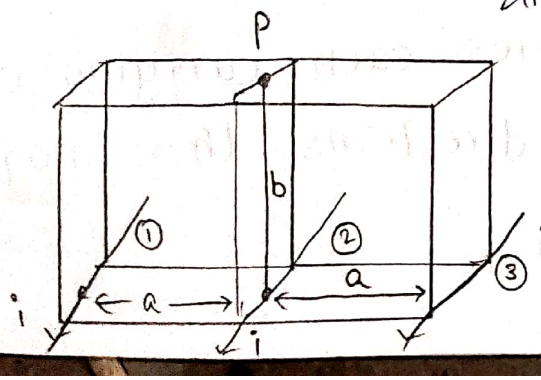
$$d\hat{\ell} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

$$r_\perp = \frac{|x-y|}{\sqrt{2}}$$

$$\hat{r} = \frac{\hat{i}-\hat{j}}{\sqrt{2}} \quad \text{(or)} \quad \frac{-\hat{i}+\hat{j}}{\sqrt{2}}$$

$$B = \frac{\mu_0 i}{2\pi r_\perp}\left(d\hat{\ell} \times \hat{r}\right)$$

Ex:-



Find magnetic field at P.

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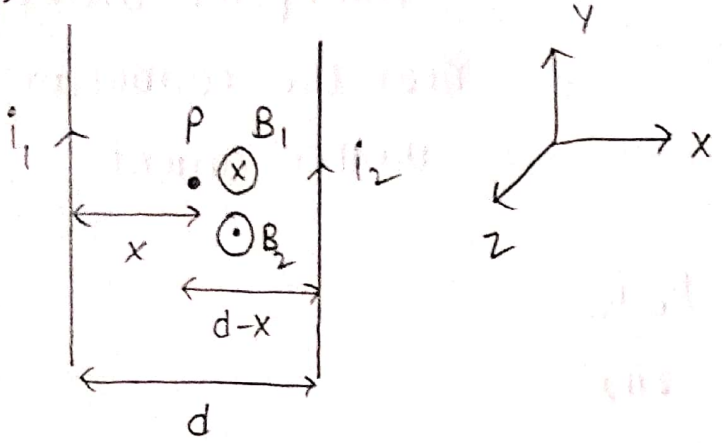
Sol:- $\vec{B_1} = \frac{\mu_0 i}{2\pi\sqrt{a^2+b^2}} \left(\hat{k} \times \frac{a\hat{i}+b\hat{j}}{\sqrt{a^2+b^2}}\right)$

$\vec{B_2} = \frac{\mu_0 i}{2\pi b} (\hat{k} \times \hat{j})$ & $\vec{B_3} = \frac{\mu_0 i}{2\pi\sqrt{a^2+b^2}} \left(\hat{k} \times \frac{-a\hat{i}+b\hat{j}}{\sqrt{a^2+b^2}}\right)$

Null Points:—

Case 1:-



At null points,

$\vec{B_1} + \vec{B_2} = 0$

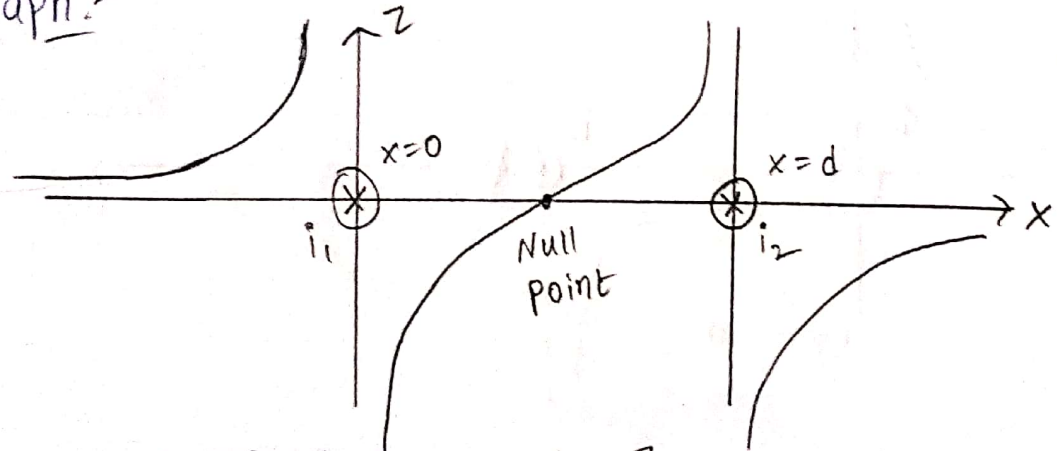
$\vec{B_1} = -\vec{B_2}$

$|\vec{B_1}| = |\vec{B_2}|$

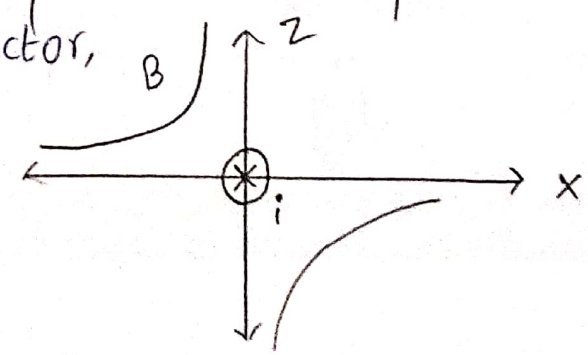
$\frac{\mu_0 i_1}{2\pi x} = \frac{\mu_0 i_2}{2\pi(d-x)}$

$\Rightarrow \boxed{\frac{i_1}{x} = \frac{i_2}{d-x}}$

Graph:-



For Single Conductor,



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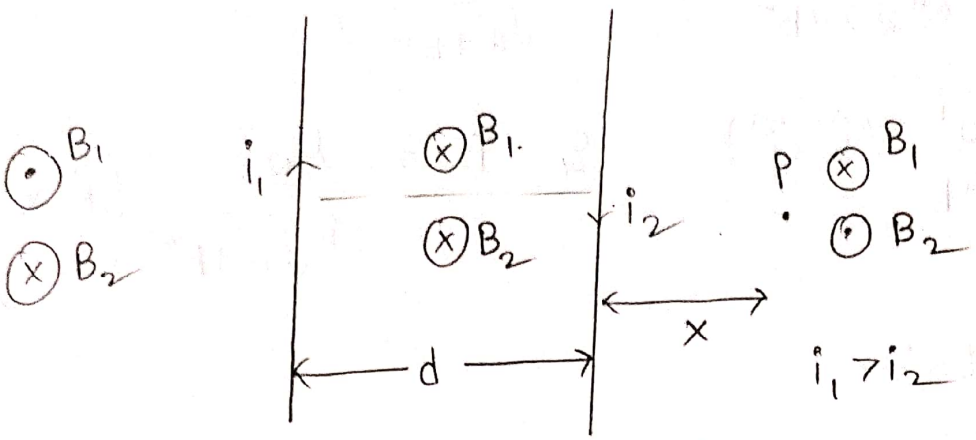
Case 2:-



Null point always forms near the conductor carrying weaker current.

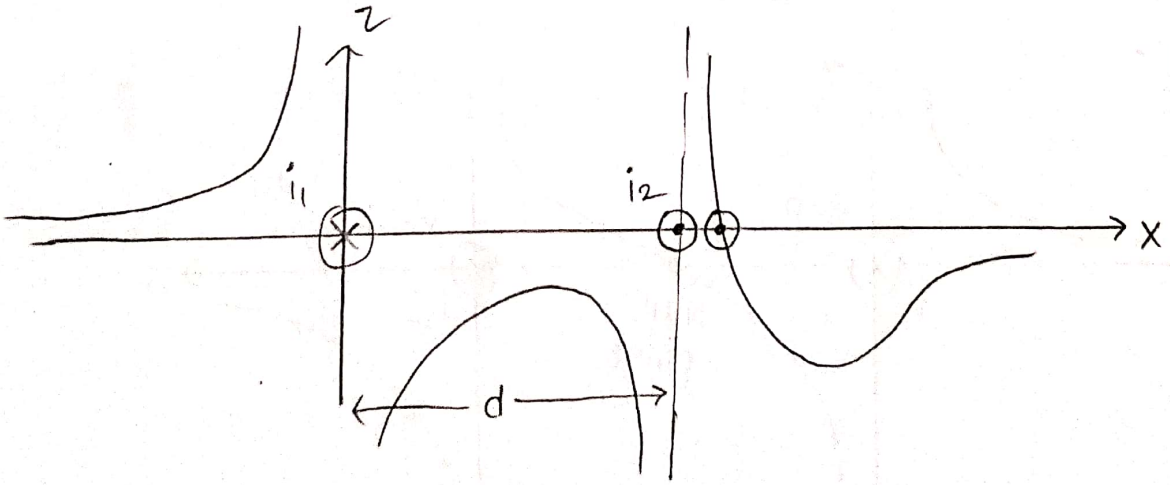
At null point,

$$B_1 = B_2$$

$$\frac{\mu_0 i_1}{2\pi (x+d)} = \frac{\mu_0 i_2}{2\pi x}$$

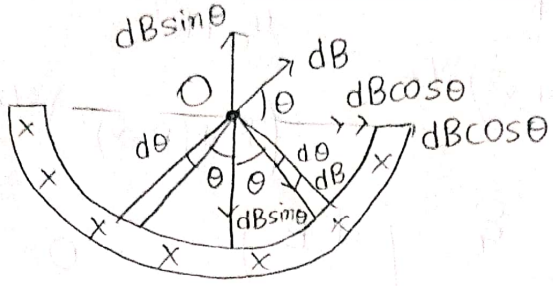
$$\frac{i_1}{x+d} = \frac{i_2}{x}$$

Solve for x



ELECTRO MAGNETISM

Magnetic field due to a long straight line with cross section in the form of thin half ring :-



i → distributed Current

$$di = \frac{i}{\pi a}(a\,d\theta)$$

$$\boxed{di = \frac{i}{\pi}\,d\theta}$$

Net field at O, $B = \int_{-\frac{\pi}{2}}^{\pi/2} dB\cos\theta = \int_{-\frac{\pi}{2}}^{\pi/2} \frac{\mu_0\,di}{2\pi a}\cos\theta$

$$= \int_{-\frac{\pi}{2}}^{\pi/2} \frac{\mu_0 i}{2\pi^2 a}\cos\theta\,d\theta$$

$$\boxed{B = \frac{\mu_0 i}{\pi^2 a}}$$, along Horizontal

Helmholtz Coils :-

* Two identicular circular coils carrying same current in same direction and are seperated by a distance of radius R of each coil.

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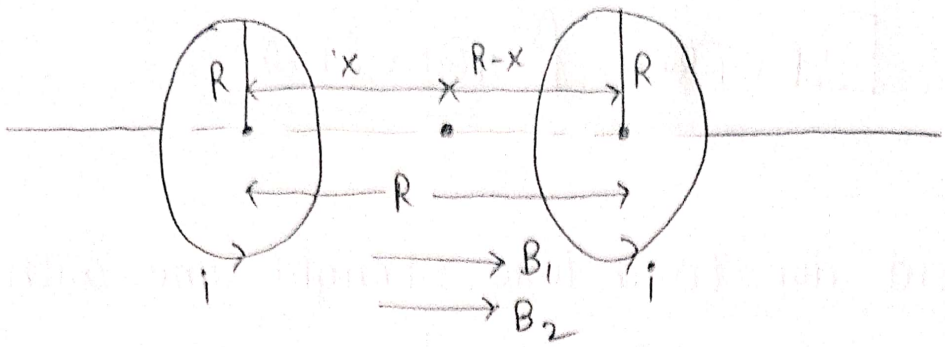
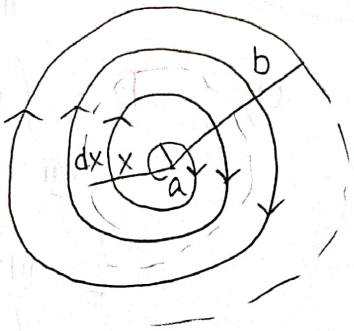


$$B = B_1 + B_2 = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}} + \frac{\mu_0 i R^2}{2\left(R^2+(R-x)^2\right)^{3/2}}$$

$$\left(\frac{dB}{dx}\right)_{\text{at } x=\frac{R}{2}} = 0 \quad \& \quad \left(\frac{d^2B}{dx^2}\right)_{x=\frac{R}{2}} = 0.$$

$\therefore$ B is Uniform at $x = \frac{R}{2}$

Magnetic field at the centre of current carrying Spiral coil :-



Number of turns per unit length along radial direction $= \frac{N}{b-a}$

Magnetic field at Centre due to elementary circular coil is, $dB = \frac{\mu_0 (dn) i}{2x} = \frac{\mu_0}{2x}\left(\frac{N}{b-a}\right) i \cdot dx$

Total Magnetic field at Centre is

$$B = \int_a^b dB = \frac{\mu_0 N i}{2(b-a)} \int_a^b \frac{dx}{x}$$

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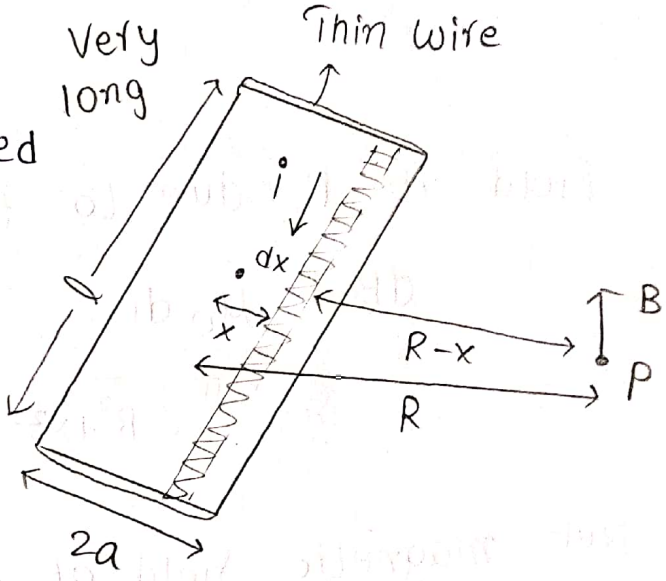
$$B = \frac{\mu_0 N i}{2(b-a)} \ln\left(\frac{b}{a}\right)$$

Magnetic field at a point due to flat strip

[Distributed Current] (Ribbon)

① At a point in the plane of strip :-



i = Total Current distributed in a width $2a$

Current per unit width

$$J = \frac{di}{dx} = \frac{i}{2a} = \text{constant}$$

Magnetic field at P due to thin wire,

$$dB = \frac{\mu_0 \, di}{2\pi (R-x)}$$

$$dB = \frac{\mu_0}{2\pi (R-x)} \left(\frac{i}{2a}\right) dx$$

$$dB = \frac{\mu_0 i}{4\pi a (R-x)} dx$$

Total Magnetic field, $B = \int_{x=-a}^{x=+a} dB = \int_{-a}^{a} \frac{\mu_0 i}{4\pi a} \cdot \frac{dx}{R-x}$

$$B = \frac{\mu_0 i}{4\pi a} \ln\left(\frac{R+a}{R-a}\right)$$ ↑

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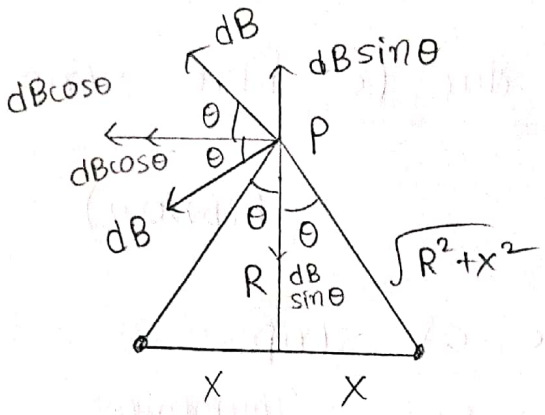
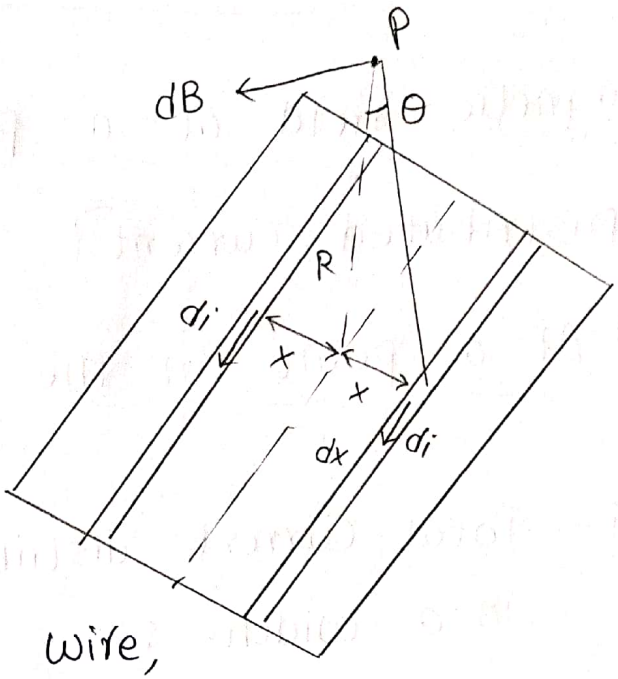
② At a distance 'R' from centre of strip and ⊥lar to its plane :-





Field at P due to thin wire,

$$dB = \frac{\mu_0 \, di}{2\pi \sqrt{R^2+x^2}}$$

Net magnetic field at 'P' is

$$B = \int dB\cos\theta$$

$$= \int \frac{\mu_0 \left(\frac{i}{2a}\right) dx}{2\pi \sqrt{R^2+x^2}} \cos\theta$$

From fig, $\tan\theta = \frac{x}{R}$

$$\Rightarrow x = R\tan\theta$$

$$dx = R\sec^2\theta \, d\theta$$

$$B = \int \frac{\mu_0 i}{4\pi a} \frac{R\sec^2\theta \cdot \cos\theta \cdot d\theta}{R\sec\theta \cdot}$$

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$$= \frac{\mu_0 i}{4\pi a} \int_{\theta=-\tan^{-1}\left(\frac{a}{R}\right)}^{\theta=\tan^{-1}\left(\frac{a}{R}\right)} d\theta$$

$$\boxed{B = \frac{\mu_0 i}{2\pi a} \tan^{-1}\left(\frac{a}{R}\right)} \text{ , to left}$$

If $a \longrightarrow$ infinity

$$\tan^{-1}(\infty) = \frac{\pi}{2}$$

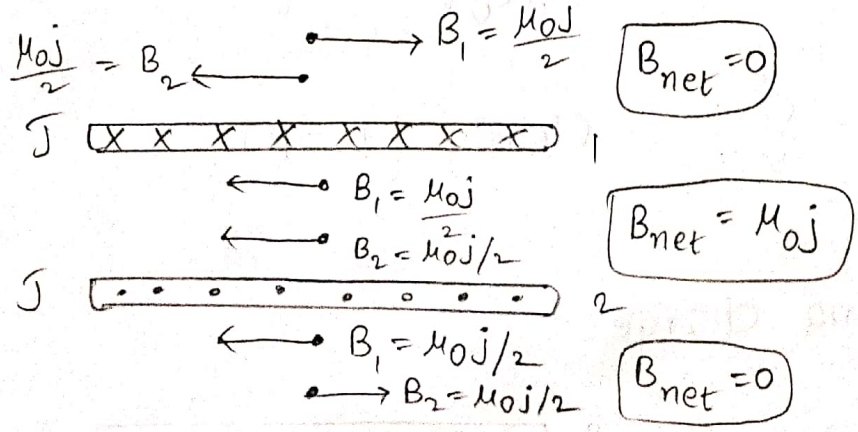
$$B = \frac{\mu_0 i}{2\pi a}\left(\frac{\pi}{2}\right) = \frac{\mu_0 i}{2(2a)} \quad \left[j = \frac{i}{2a} = \text{current density} \right]$$

$$\boxed{B = \frac{\mu_0 j}{2}}$$; which is independent of distance R

* In Infinite plane sheet, we are getting "Uniform Magnetic Field" in this case.

* Infinite Plane Sheet is the best way to produce Uniform Magnetic field.

Ex:



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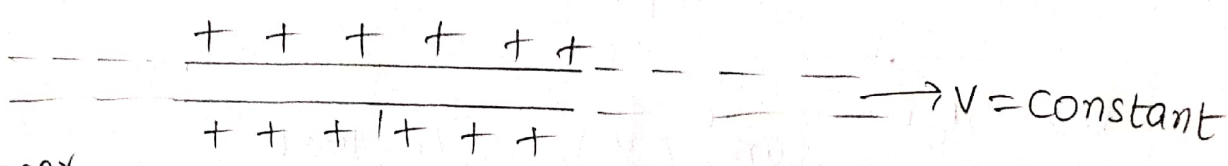
Magnetic field due to Translation & Revolution of Charge:-

① 

λ = Linear charge density

P ↓ E

Here $i = \frac{q}{t} = \frac{\lambda l}{t}$

$\boxed{i = \lambda V}$

Magnetic field at 'P' is,

$$B = \frac{\mu_0 i}{2\pi r_\perp} (d\hat{l} \times \hat{r})$$

$$\boxed{B = \frac{\mu_0 \lambda V}{2\pi r}} \otimes$$

Electric field at the same point,

$$E = \frac{2K\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r} \downarrow$$

$$\frac{E}{B} = \frac{\lambda}{2\pi\epsilon_0 r} \left(\frac{2\pi r}{\mu_0 \lambda V}\right) = \frac{1}{\mu_0 \epsilon_0 V} = \frac{c^2}{V}$$

$$\boxed{\frac{E}{B} = \frac{c^2}{V}}$$ Clearly $\boxed{E > B}$

② Due to revolving charge:-

$$i = fq = \frac{\omega}{2\pi} q$$

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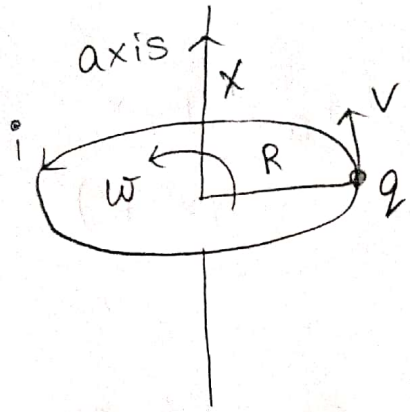


Magnetic field at Centre, & $E_c = 0$

$$B_c = \frac{\mu_0 i}{2R} = \frac{\mu_0}{2R}\left(\frac{\omega q}{2\pi}\right) = \boxed{\frac{\mu_0 \omega q}{4\pi R}}$$

Magnetic field at distance 'x' on the axis from centre is,

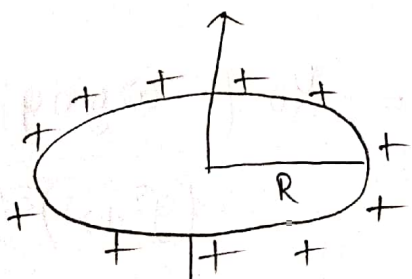
$$B_a = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}} = \boxed{\frac{\mu_0 \omega q R^2}{4\pi (R^2+x^2)^{3/2}}}$$

Electric field at a distance 'x' on the axis from centre is,

$$\boxed{E_a = \frac{kqx}{(x^2+R^2)^{3/2}}}$$

(3) Due to Spinning / revolution charged Ring :-

$\lambda = \frac{q}{2\pi R}$



$$i = fq = \frac{\omega q}{2\pi}$$

$$\boxed{B_c = \frac{\mu_0 i}{2R} = \frac{\mu_0 \omega q}{4\pi R}} \qquad \boxed{E_c = 0}$$

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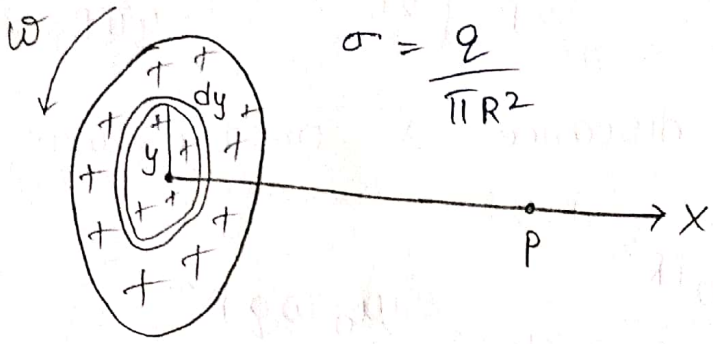
$$B_a = \frac{\mu_0 \omega q R^2}{4\pi (R^2+x^2)^{3/2}}$$

$$E_a = \frac{KqX}{(x^2+R^2)^{3/2}}$$

Magnetic field on axis of spinning non-conducting charged thin disc :-



$$\sigma = \frac{q}{\pi R^2}$$

Charge on elementary ring,

$$dq = \sigma dA$$
$$= \sigma 2\pi y \cdot dy$$

Now, $i = \frac{(dq)\omega}{2\pi} = (\sigma y \cdot dy)\omega$

Magnetic field at point P due to elementary ring,

$$dB = \frac{\mu_0 i y^2}{2(y^2+x^2)^{3/2}} = \frac{\mu_0 (\sigma \omega y dy) y^2}{2(y^2+x^2)^{3/2}}$$

Total magnetic field,

$$B = \int_{y=0}^{y=R} dB = \frac{\mu_0 \sigma \omega}{2} \int_{y=0}^{y=R} \frac{y^3 \cdot dy}{(y^2+x^2)^{3/2}}$$

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$y^2+x^2=t^2$

$\Rightarrow \quad y\,dy = t\,dt$

$$B = \frac{\mu_0 \sigma \omega}{2} \int_{t=}^{t=} \frac{(t^2-x^2)\,t\,dt}{t^3}$$

$$B = \frac{\mu_0 \sigma \omega}{2} \int_{t=}^{t=} \left(1 - \frac{x^2}{t^2}\right) dt$$

$$B = \frac{\mu_0 \sigma \omega}{2} \left[t + \frac{x^2}{t} \right]_{t=}^{t=}$$

$$B = \frac{\mu_0 \sigma \omega}{2} \left[\frac{t^2+x^2}{t} \right]_{t=}^{t=}$$

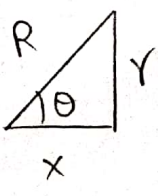
$$B = \frac{\mu_0 \sigma \omega}{2} \left[\frac{y^2+2x^2}{\sqrt{y^2+x^2}} \right]_0^R$$

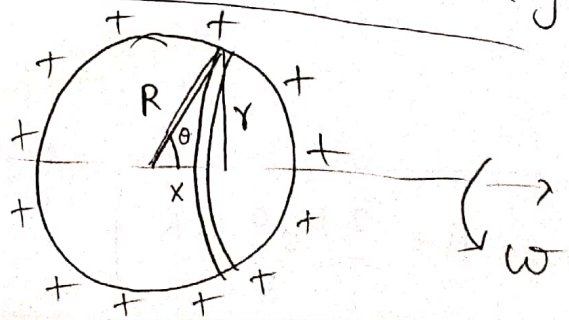
$$\therefore B = \frac{\mu_0 \sigma \omega}{2} \left[\frac{R^2+2x^2}{\sqrt{R^2+x^2}} - 2x \right]$$

At Centre, $x = 0$

$$\Rightarrow \quad B = \frac{\mu_0 \sigma \omega R}{2}$$

Magnetic Field at the Centre of Spinning Non-Conducting Sphere Charge on surface with charge density σ :-





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From fig, $\sin\theta = \frac{r}{R} \Rightarrow r = R\sin\theta$

$\cos\theta = \frac{x}{R} \Rightarrow x = R\cos\theta$

$dq = \sigma(2\pi r)\,dr = \sigma(2\pi R\sin\theta)\,R\,d\theta$

$dq = \sigma\, 2\pi R^2 \sin\theta \cdot d\theta$

$di = \frac{dq\,\omega}{2\pi} = \sigma\,\omega R^2 \sin\theta \cdot d\theta$

Magnetic field at Centre,

$$dB = \frac{\mu_0 (di) r^2}{2(r^2+x^2)^{3/2}} = \frac{\mu_0 (\sigma\,\omega R^2 \sin\theta\, d\theta)(R^2\sin^2\theta)}{2(R^2\sin^2\theta + R^2\cos^2\theta)^{3/2}}$$

$$dB = \frac{\mu_0 \sigma\,\omega\, R^4 \sin^3\theta \cdot d\theta}{2(R^3)} = \frac{\mu_0 \sigma\,\omega R \sin^3\theta \cdot d\theta}{2}$$

Net Magnetic field, $B = \int_{\theta=0}^{\theta=\pi} dB$

$$B = \int_0^{\pi} \frac{\mu_0 \sigma\,\omega R \sin^3\theta \cdot d\theta}{2} = 2\int_0^{\pi/2} \frac{\mu_0 \sigma\,\omega R \sin^3\theta}{2}\, d\theta$$

$$= \frac{\mu_0 \sigma\,\omega R}{1} \int_0^{\pi/2} \sin^3\theta \cdot d\theta$$

$$= \mu_0 \sigma\,\omega R \left(\frac{(2)}{3}\right)$$

$$\therefore B = \frac{2\mu_0 \sigma\,\omega R}{3}$$

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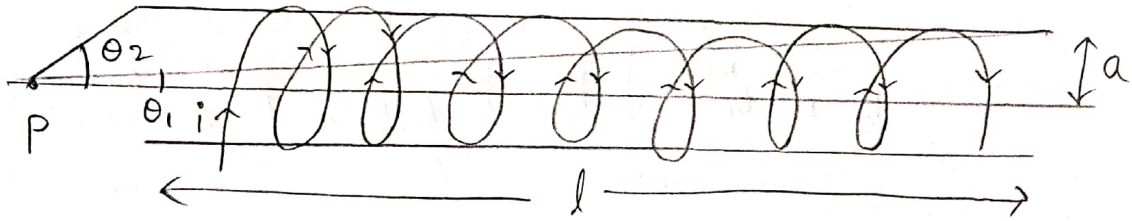
Magnetic Field due to Solenoid :-

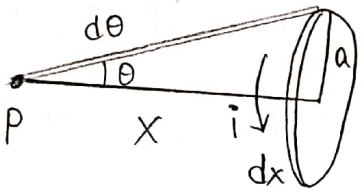
Let N = Total Number of turns in a total length 'l'

No. of turns per unit length, $n = \frac{N}{L} = \text{constant}$

Thightly wounded

Closely packed





$dN = n\,dx$

Magnetic field at P due to elementary coil is,

$$dB = \frac{\mu_0 (dn) i a^2}{2(a^2+x^2)^{3/2}}$$

$$dB = \frac{\mu_0 n i a^2 dx}{2(a^2+x^2)^{3/2}}$$

From fig,

$\tan\theta = \frac{a}{x}$

$x = a\cot\theta$

$dx = -a\,\text{cosec}^2\theta \cdot d\theta$

Net magnetic field at 'P' is,

$$B = \int_{\theta=\theta_2}^{\theta=\theta_1} dB = \int_{\theta_2}^{\theta_1} \frac{\mu_0 n i a^2 (-a\,\text{cosec}^2\theta\, d\theta)}{2(a^2 + a^2\cot^2\theta)^{3/2}}$$

$$B = \int_{\theta_1}^{\theta_2} \frac{\mu_0 n i}{2} \left(\frac{a^3\,\text{cosec}^2\theta \cdot d\theta}{a^3\,\text{cosec}^3\theta} \right)$$

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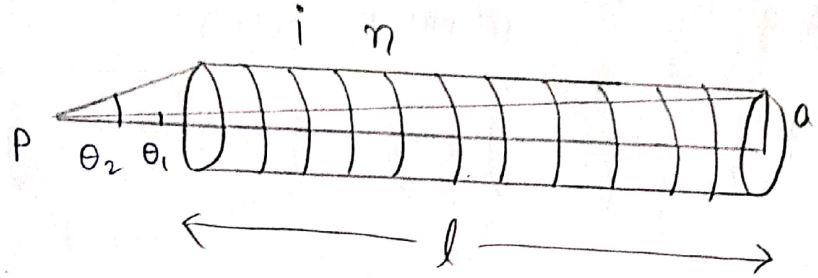
$$B = \int_{\theta_1}^{\theta_2} \frac{\mu_0 ni}{2} \sin\theta \cdot d\theta$$

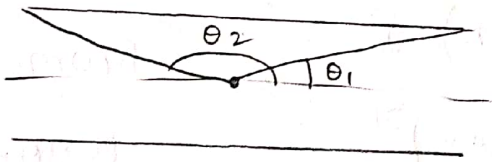
$$\boxed{B = \frac{\mu_0 ni}{2} (\cos\theta_1 - \cos\theta_2)}$$



$$B = \frac{\mu_0 ni}{2} (\cos\theta_1 - \cos\theta_2)$$

Points:-

① If Point 'P' is well inside the long solenoid,



Since Solenoid is too long,

$\theta_1 = 0°$

$\theta_2 = 180°$

$\boxed{B = \mu_0 ni}$ ↗ Field is Uniform

* If the medium inside the Solenoid is having "μ_r" then $\boxed{B = \mu_0 \mu_r ni}$

* Field is Uniform and is also same at "OFF the axis" (all points)

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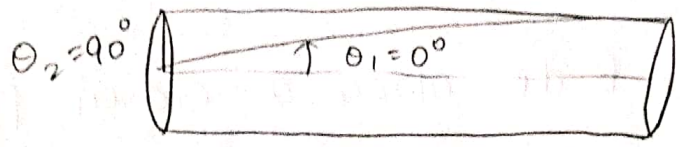
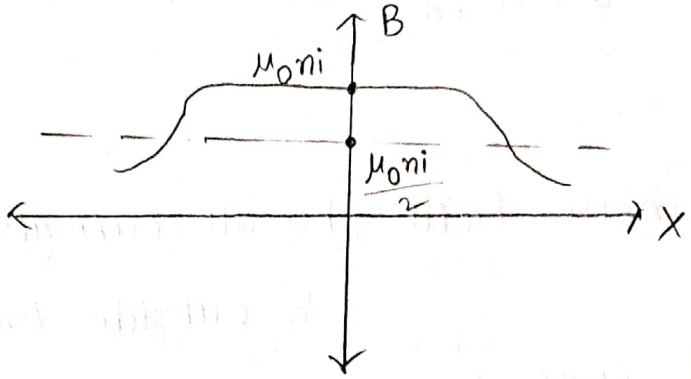
② At one end of Semi-infinite long solenoid:-

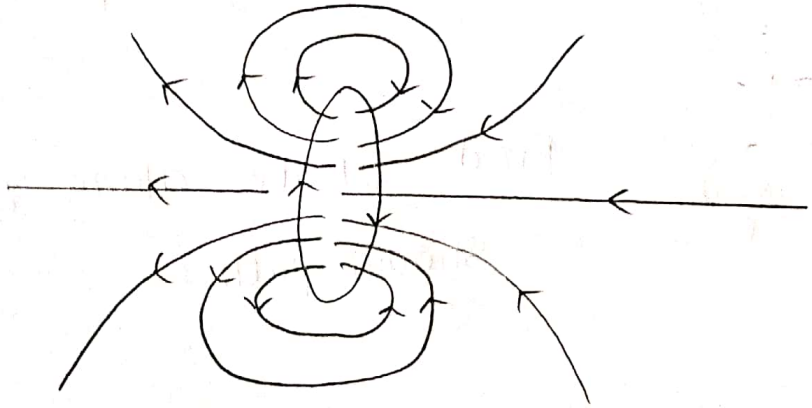


$$B = \frac{\mu_0 ni}{2}(\cos 0^\circ - \cos 90^\circ)$$

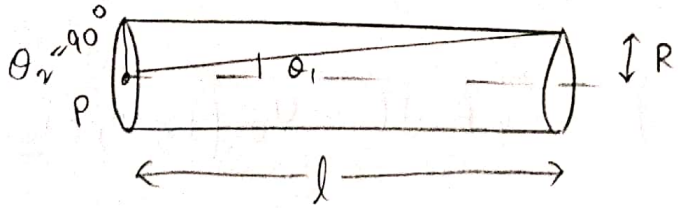
$$\boxed{B = \frac{\mu_0 ni}{2}}$$

Graph:-





③ For finite length rod:-



$$\cos\theta_1 = \frac{l}{\sqrt{R^2+l^2}}$$

$$\cos\theta_2 = 0$$

$$\boxed{B = \frac{\mu_0 ni}{2}\frac{l}{\sqrt{R^2+l^2}}}$$

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Ampere's Law :-

"

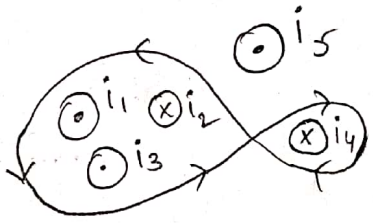
* Line Integral of $\bar{B} \cdot d\bar{l}$ along a closed path (loop) around a current carrying conductor is μ_0 times the net current enclosed by that loop."

i.e, $$\oint_l \bar{B} \cdot d\bar{l} = \mu_0 i_{enclosed}$$

Here $\bar{B}$ = Net Magnetic Field (by all currents inside & outside loop)

i_{enc} is net current enclosed by loop.

Ex:-

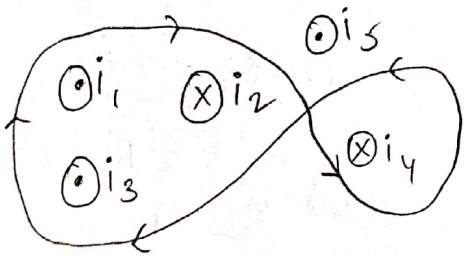


Find $\oint \bar{B} \cdot d\bar{l}$ along the shown path.

Sol:-

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 (i_1 - i_2 + i_3 + i_4)$$

Ex:-



$$\oint \bar{B} \cdot d\bar{l} = \mu_0 (i_2 - i_1 - i_3 - i_4)$$

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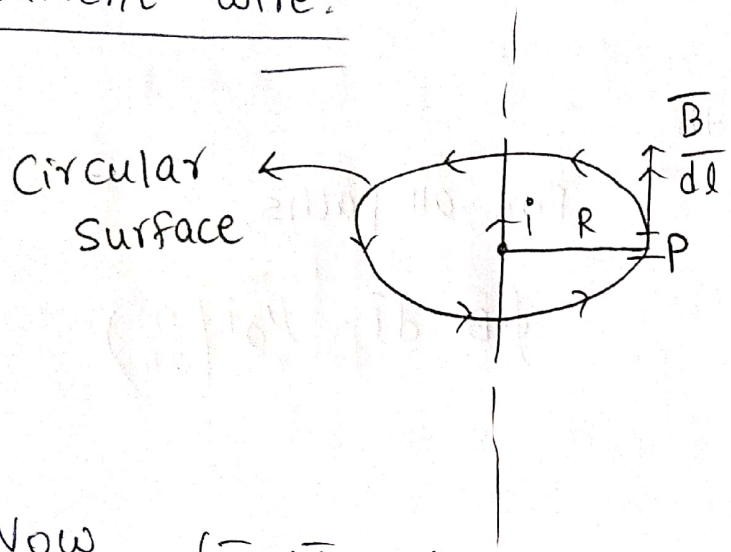
Magnetic field at a point due to infinite long Current wire:-



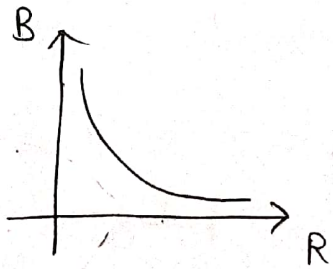
Now, $\int \bar{B} \cdot d\bar{l} = \int B dl \cos 0 = B \oint dl$

$= B(2\pi R)$

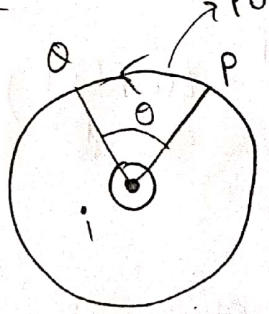
From Ampere's Law, $\int \bar{B} \cdot d\bar{l} = \mu_0 i$

$B(2\pi R) = \mu_0 i$

$$B = \frac{\mu_0 i}{2\pi R}$$



Note:-

① 

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i \left(\frac{\theta}{2\pi}\right)$$

for Complete loop,

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

2π rad $\longrightarrow \dfrac{\oint \bar{B} \cdot d\bar{l}}{\mu_0 i}$

1 rad $\longrightarrow \mu_0 i / 2\pi$

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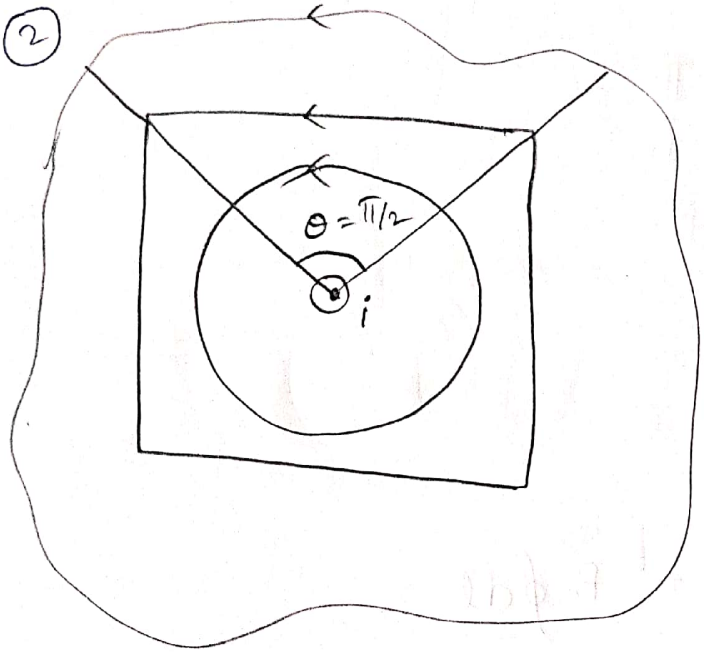
θ rad $\longrightarrow$ $\frac{\mu_0 i \theta}{2\pi}$

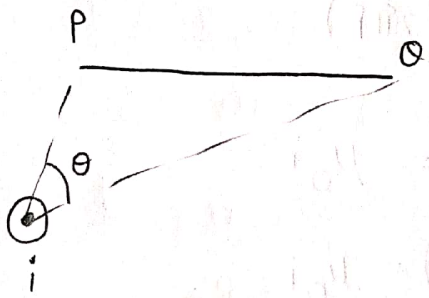
(2)



For all paths,

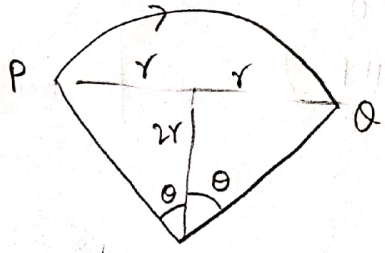
$$\oint \bar{B} \cdot d\bar{\ell} = \mu_0 i \left(\frac{\theta}{2\pi}\right)$$

(3)



$$\int_P^Q \bar{B} \cdot d\bar{\ell} = \mu_0 i \left(\frac{\theta}{2\pi}\right)$$

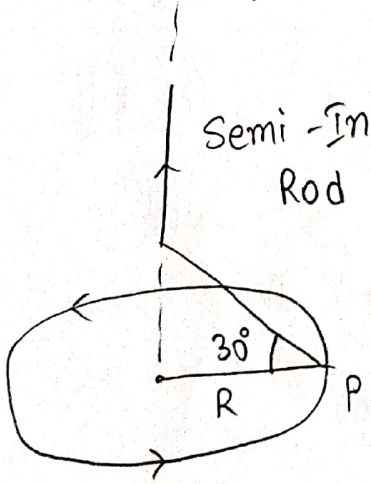
Ex:-



$$\int_P^Q \bar{B} \cdot d\bar{\ell} = \mu_0 i \left(\frac{2\theta}{2\pi}\right)$$

$$= \frac{\mu_0 i}{\pi} \tan^{-1}\left(\frac{1}{2}\right)$$

Ex:-



Find $\int \bar{B} \cdot d\bar{\ell}$ along closed circular path.

Sol:- $B = \frac{\mu_0 i}{4\pi R}(\sin\alpha + \sin\beta)$ $\alpha = 90^\circ$ & $\beta = -30^\circ$

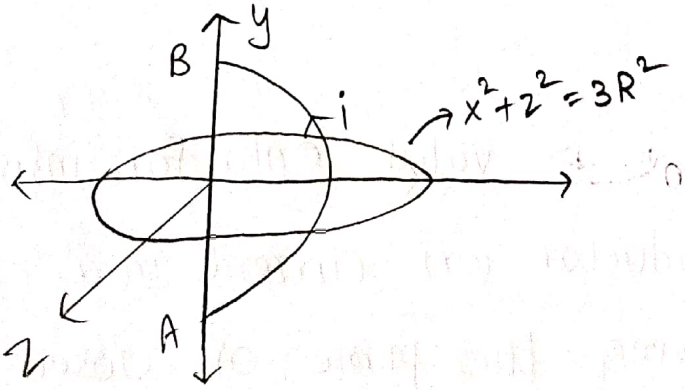
$$B = \frac{\mu_0 i}{4\pi R}\left(\frac{-1}{2}+1\right)$$

$$B = \frac{\mu_0 i}{8\pi R}$$

$$\oint \bar{B}\cdot d\bar{l} = \oint B\,dl\cos 0 = B\oint dl = B(2\pi R)$$

$$= \frac{\mu_0 i}{4}$$

Ex:- A Conductor carrying current 60A is in the form of semi-circle AB of radius R & lying in x-y plane of radius R with its centre at O. as shown. The magnitude of $\oint \bar{B}\cdot d\bar{l}$ for the circle $x^2+z^2=3R^2$ in x-z plane due to current in curve AB is $10n\mu_0$. Find value of n.



Sol:- We can take two wires that carry current in opposite directions along AB.

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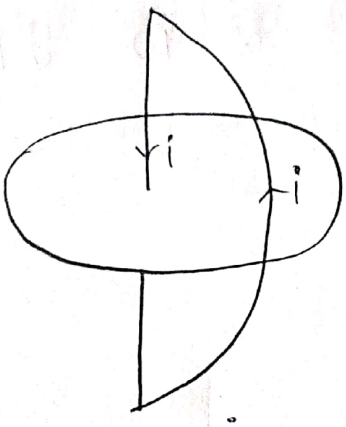
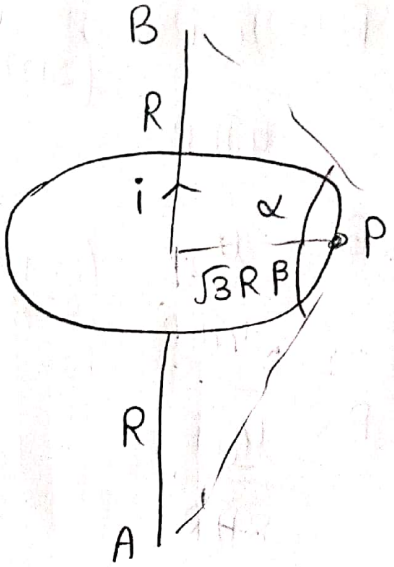
So,



$i_{enc} = 0$

$\oint \bar{B} \cdot d\bar{l} = 0$

+



$i_{enc} = i$

$$\alpha = \beta = \tan^{-1}\left(\frac{R}{\sqrt{3}R}\right) = 30^\circ$$

$$B = \frac{\mu_0 i}{4\pi(\sqrt{3}R)}(\sin\alpha + \sin\beta)$$

$$B = \frac{\mu_0 i}{4\pi(\sqrt{3}R)}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0 i}{4\sqrt{3}\pi R}$$

$$\oint \bar{B} \cdot d\bar{l} = \frac{\mu_0 i}{4\sqrt{3}\pi R}(2\pi\sqrt{3}R) = \frac{\mu_0 i}{2}$$

Note:- $\oint \bar{B} \cdot d\bar{l} = \mu_0 i$ is valid only for infinite long straight conductor (or) current in a closed circuit piercing the plane of closed loop.

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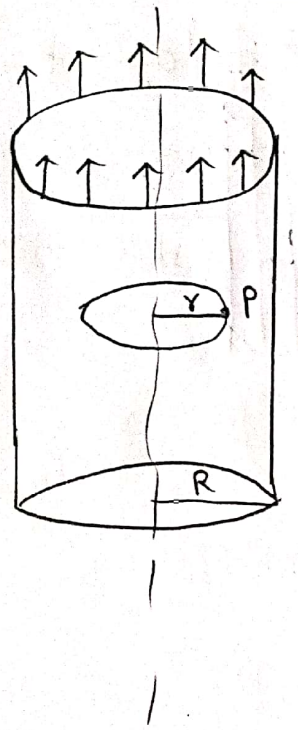
Magnetic Field at a point due to Current carrying hollow Cylindrical Pipe:-



① Point lies inside the Pipe.

$$\oint \bar{B} \cdot d\bar{l} = B(2\pi r)$$

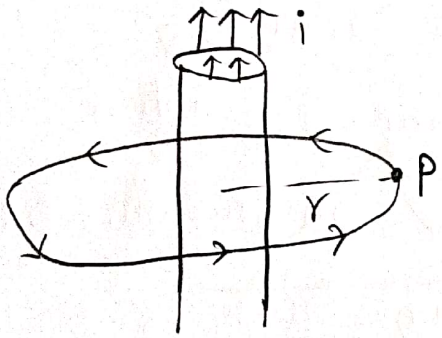
$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

Here $i = 0$

$$\Rightarrow B(2\pi r) = 0$$

$$\Rightarrow \boxed{B = 0}$$

② Point lies outside the pipe



$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

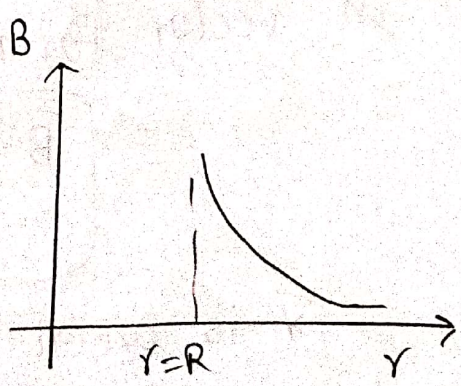
$$B(2\pi r) = \mu_0 i$$

$$\boxed{B = \frac{\mu_0 i}{2\pi r}}$$

$B \propto \frac{1}{r}$

③ On the Surface

$r = R \Rightarrow \boxed{B = \frac{\mu_0 i}{2\pi R} \ (max)}$



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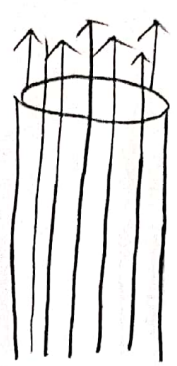
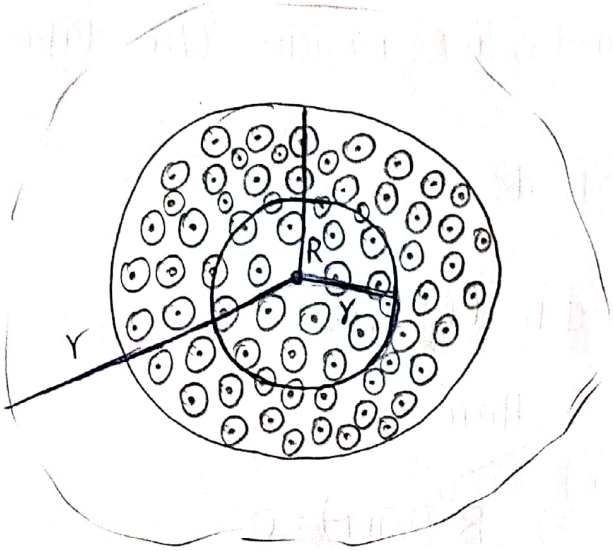
Magnetic field at a point due to current carrying Solid Cylindrical Conductor:-





Case i:- If Current Density $(J) = \frac{i_0}{\pi R^2}$ = constant

① **Point lies inside:-**

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i_{enc}$$

$$B(2\pi r) = \mu_0 j \pi r^2$$

$$\boxed{B = \frac{\mu_0 j r}{2}} \quad \text{i.e., } B \propto r$$

In Vector form,

$$\boxed{\bar{B} = \frac{\mu_0 (\bar{j} \times \bar{r})}{2}}$$

$\bar{r}$ is taken from centre

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② Point lies on the Conductor:-

$r = R$

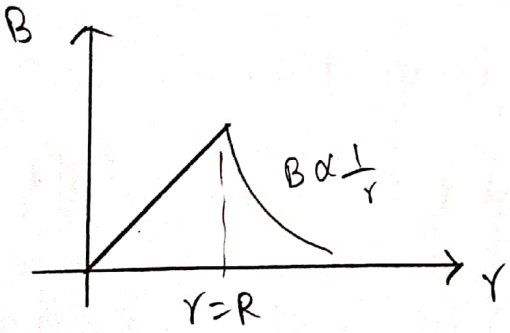
$$B = \frac{\mu_0 j R}{2}$$

③ Point lies outside the Conductor:-

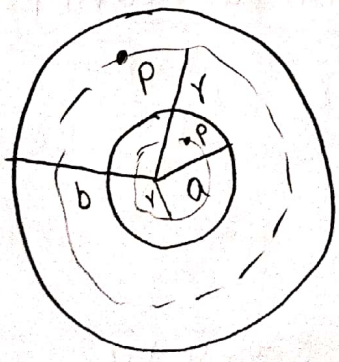
$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

$$B(2\pi r) = \mu_0 j \pi R^2$$

$$B = \frac{\mu_0 j R^2}{2r}$$ i.e, $$B \propto \frac{1}{r}$$



Hollow Cylindrical Pipe having inner radius 'a' & outer radius 'b' :-



i_0 = Total Current

$$j = \frac{i_0}{\pi(b^2 - a^2)} = \text{constant}$$

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① For $r<a$:-

$$\oint \bar{B}\cdot d\bar{l} = \mu_0(0)$$

$$\Rightarrow \boxed{B=0}$$

② For $a<r<b$:-

$$\oint \bar{B}\cdot d\bar{l} = \mu_0 i_{enc}$$

$$B(2\pi r) = \mu_0 j\pi (r^2-a^2)$$

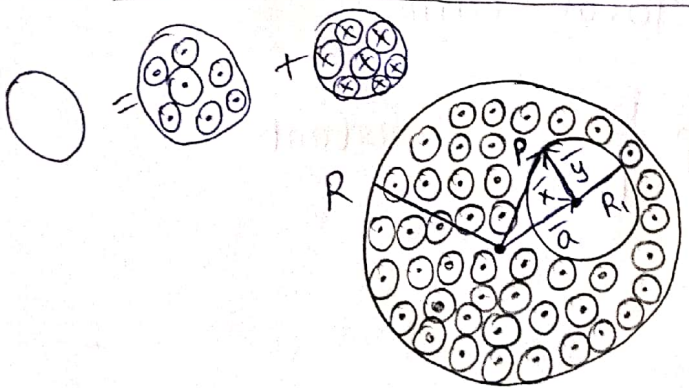
$$\boxed{B = \frac{\mu_0 i_0 (r^2-a^2)}{2\pi r (b^2-a^2)}}$$

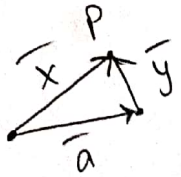
③ For $r>b$:-

$$\oint \bar{B}\cdot d\bar{l} = \mu_0 i_0$$

$$\boxed{B = \frac{\mu_0 i_0}{2\pi r}}$$

Magnetic field inside the cylindrical cavity :-





$$\bar{x} = \bar{y} + \bar{a}$$

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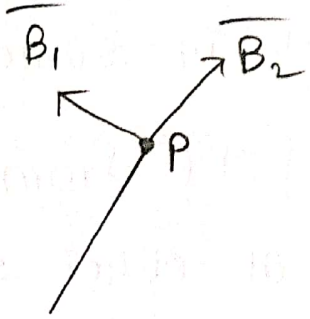
$$\bar{B}_p = \bar{B}_1 + \bar{B}_2$$

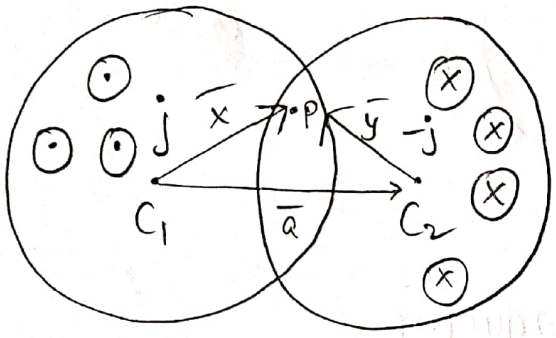
$$= \mu_0 \left(\frac{\bar{j} \times \bar{x}}{2} \right) + \mu_0 \left(\frac{-\bar{j} \times \bar{y}}{2} \right)$$

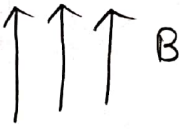
$$= \frac{\mu_0 \bar{j}}{2} \times (\bar{x} - \bar{y})$$



$$\boxed{\bar{B}_p = \mu_0 \left(\frac{\bar{j} \times \bar{a}}{2} \right)} \rightarrow \text{Uniform}$$

Ex:-





$$\bar{B}_p = \mu_0 \left(\frac{\bar{j} \times \bar{x}}{2} \right) + \mu_0 \left(\frac{-\bar{j} \times \bar{y}}{2} \right)$$

$$= \frac{\mu_0 \bar{j}}{2} \times (\bar{x} - \bar{y})$$

$$\boxed{\bar{B}_p = \frac{\mu_0}{2} (\bar{j} \times \bar{a})}$$

Ex:- A long conductor of circular cross section with radius R has current density $\bar{J}_r = \begin{cases} \bar{J}_0 \left(\frac{r^2}{R^2} \right) & \frac{R}{2} \le r \le R \\ 0 & r \le \frac{R}{2} \end{cases}$ (into the plane of paper)

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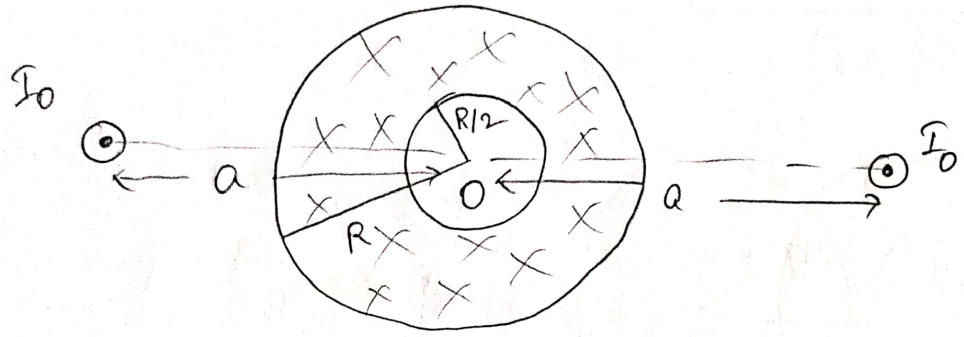
Two infinitely long thin conducting wire carry current i_0 in same direction are placed at a distance 'a' [>R] from O $\perp^{lar}$ to OP & $\parallel^{el}$ to the conductor at either side such that magnetic field at P is zero. Find current i_0 in the wires and the direction of current as compared with the direction of current in the conductor. Current density is into plane of paper.



Sol:- Current in the conductor,

$$i = \int \bar{J} \cdot d\bar{s}$$

$$i = \int_{\gamma=R/2}^{\gamma=R} J_0 \frac{r^2}{R^2} (2\pi r)\, dr$$

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 \oint \bar{J} \cdot d\bar{s}$$

$$i = \frac{J_0}{R^2} \cdot \left(\frac{2\pi}{4}\right) \left[R^4 - \frac{R^4}{16}\right]$$

$$= \frac{15 J_0 R^2 \pi}{32}$$

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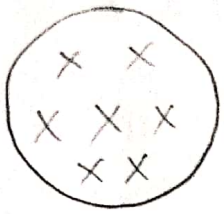
Ex:-



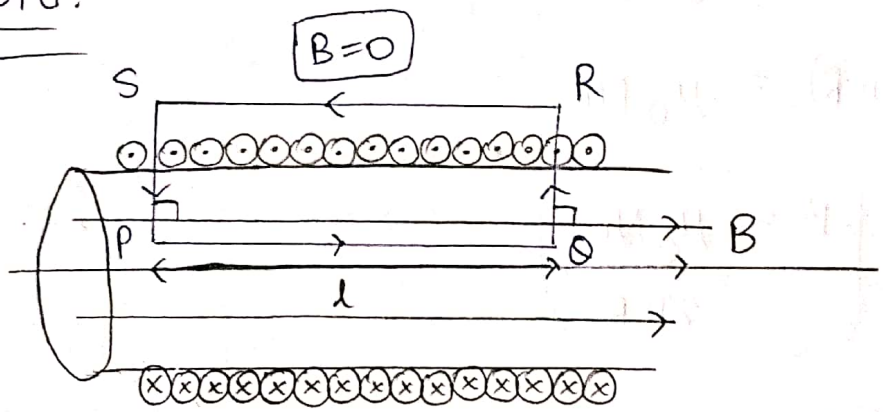
$j = j_0\left(1 - \frac{r}{R}\right)$ for $0 < r < R$

Find magnetic field at ① $r = \frac{R}{2}$

② $r = \frac{3R}{2}$

Sol:-

Solenoid:-



$$\oint \bar{B} \cdot d\bar{l} = PQ + QR + RS + SP$$

$$= B(l) + 0 + 0 + 0 \qquad [\cos 90^\circ = 0]$$

PQRS

$$\oint \bar{B} \cdot d\bar{l} = Bl$$

$$\Rightarrow \mu_0 n l i = Bl$$

$$\Rightarrow \boxed{B = \mu_0 n i}$$

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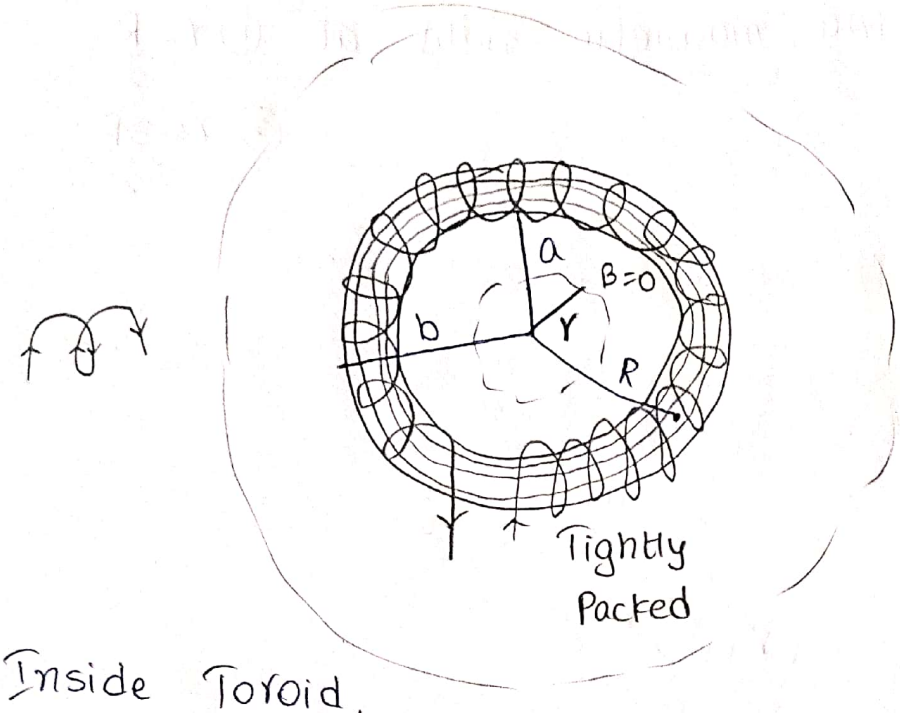
Toroid:- (End-less Solenoid)



$i_{en} = 0$

$B = 0$.

Inside Toroid,

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 i$$

$$B(2\pi R) = \mu_0 N i$$

$$\boxed{B = \frac{\mu_0 N i}{2\pi R}}$$

R = mean radius $= \frac{a+b}{2}$

$$\boxed{B = \mu_0 n i} \quad ; n = \frac{N}{l} = \frac{N}{2\pi R}$$

Force between moving Charges:-

$\bar{F}_{12}$ = Force acting on q_2 by q_1

$\bar{F}_{21}$ = Force acting on q_1 by q_2

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We have,

$$\bar{F} = q(\bar{v} \times \bar{B})$$

$$\bar{F}_{12} = q_2(\bar{v}_2 \times \bar{B}_1)$$

where $\bar{B}_1 = \dfrac{\mu_0 q_1 (\bar{v}_1 \times \hat{r}_{12})}{4\pi r_{12}^2}$

$\hat{r}_{12}$ = Unit vector from q_1 to q_2

$$\bar{F}_{12} = q_2 \left(\bar{v}_2 \times \frac{\mu_0}{4\pi} \frac{q_1 (\bar{v}_1 \times \hat{r}_{12})}{r_{12}^2} \right)$$

$$\boxed{\bar{F}_{12} = \frac{\mu_0}{4\pi} \frac{q_1 q_2}{r_{12}^2} \left(\bar{v}_2 \times (\bar{v}_1 \times \hat{r}_{12}) \right)}$$

& $\bar{F}_{21} = q_1 (\bar{v}_1 \times \bar{B}_2)$

$$= q_1 \left(\bar{v}_1 \times \frac{\mu_0}{4\pi} \frac{q_2 (\bar{v}_2 \times \hat{r}_{21})}{r_{21}^2} \right)$$

$$\boxed{\bar{F}_{21} = \frac{\mu_0}{4\pi} \frac{q_1 q_2}{r_{21}^2} \left(\bar{v}_1 \times (\bar{v}_2 \times \hat{r}_{21}) \right)}$$

Clearly, $\boxed{\bar{F}_{12} \neq -\bar{F}_{21}}$ (in most of cases)

i.e, Newton's III-law may not valid.

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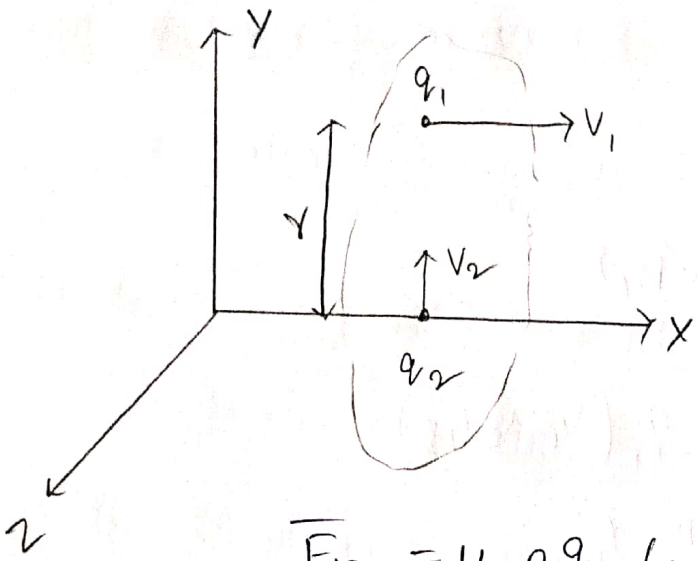
Ex:-



$$\bar{F}_{12} = \frac{\mu_0 q_1^2 q_2^2}{4\pi r^2}\left(V_2\hat{j} \times \left(V_1\hat{i} \times (-\hat{j})\right)\right)$$

$$= \frac{\mu_0 q_1 q_2}{4\pi r^2}\left(V_2\hat{j} \times V_1(-\hat{k})\right)$$

Force acting on q_2

$$\bar{F}_{12} = \frac{\mu_0 q_1 q_2}{4\pi r^2}\left(V_1 V_2 (-\hat{i})\right)$$

$$\bar{F}_{21} = \frac{\mu_0 q_1 q_2}{4\pi r^2}\left(V_1\hat{i} \times \left(V_2\hat{j} \times \hat{j}\right)\right)$$

Force acting on q_1

$$\bar{F}_{21} = 0$$

$$\bar{F}_{12} \neq -\bar{F}_{21}$$

Points:-

① Net Force on two particle system due to each other is to left

⇒ Acceleration is to left ⇒ Momentum is not conserved

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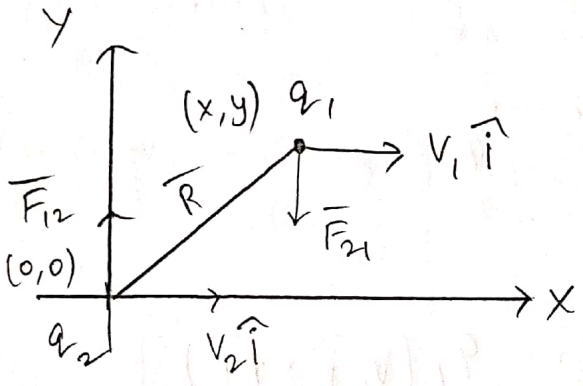
(2) Here we take that magnetic & electric fields that are produced due to moving charges, can carry both Energy & Momentum. This point helps the Law of Conservation of Energy & momentum.

(3) Net Momentum of System Containing both the charges and both the fields is always conserved.

Ex:-



$$\bar{R} = x\hat{i} + y\hat{j}$$

$$\bar{F}_{2,1} = \frac{\mu_0}{4\pi}\left(\frac{q_1 q_2}{R^3}\right)\left[V_1\hat{i} \times \left(V_2\hat{i} \times (x\hat{i} + y\hat{j})\right)\right]$$

$$\bar{F}_{21} = \frac{\mu_0 q_1 q_2}{4\pi R^3} V_1 V_2 y(-\hat{j})$$

& $$\bar{F}_{12} = \frac{\mu_0}{4\pi}\left(\frac{q_1 q_2}{R^3}\right)\left[V_2\hat{i} \times \left(V_1\hat{i} \times \left(x(-\hat{i}) + y(-\hat{j})\right)\right)\right]$$

$$\bar{F}_{12} = \frac{\mu_0 q_1 q_2}{4\pi R^3} V_1 V_2 y(\hat{j})$$

Here $\bar{F}_{12}$ & $\bar{F}_{21}$ are equal & opposite but non collinear

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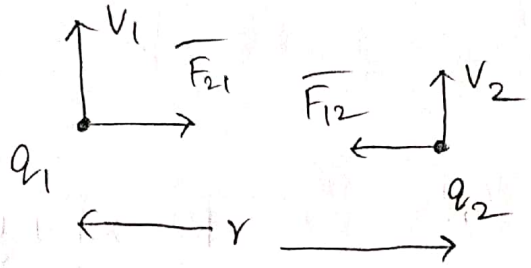
They constitute Torque

⇒ Angular Momentum is not Conserved ×

* To conserve Angular Momentum of two particle system, we can say that both the fields (E & B) can carry Angular Momentum in Opposite directions

Ex:-



$$\vec{F}_{12} = q_2 V_2 \hat{j} \times \left(\frac{\mu_0}{4\pi r^3} q_1 (V_1 \hat{j} \times r \hat{i}) \right]$$

$$\vec{F}_{12} = \frac{\mu_0 q_1 q_2 V_1 V_2}{4\pi r^2} (-\hat{i})$$

$$\vec{F}_{21} = q_1 V_1 \hat{j} \times \left(\frac{\mu_0}{4\pi r^3} q_2 (V_2 \hat{j} \times r(-\hat{i})) \right]$$

$$\vec{F}_{21} = \frac{\mu_0 q_1 q_2 V_1 V_2}{4\pi r^2} \hat{i}$$

F_m is attractive

Electric Force, $F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$, repulsive.

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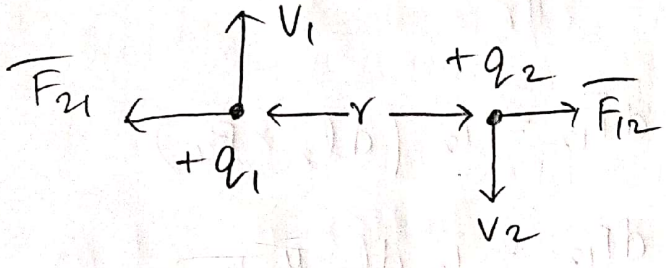
Now $\frac{F_e}{F_m} = \frac{c^2}{v_1 v_2} > 1$

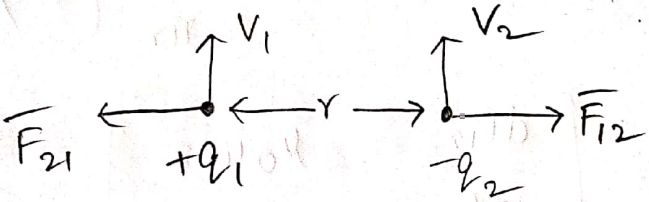
$\Rightarrow$ $F_e > F_m$ $\Rightarrow$ Net force is repulsive



F_m is repulsive

F_e is repulsive

$\therefore$ Net Force is repulsive.

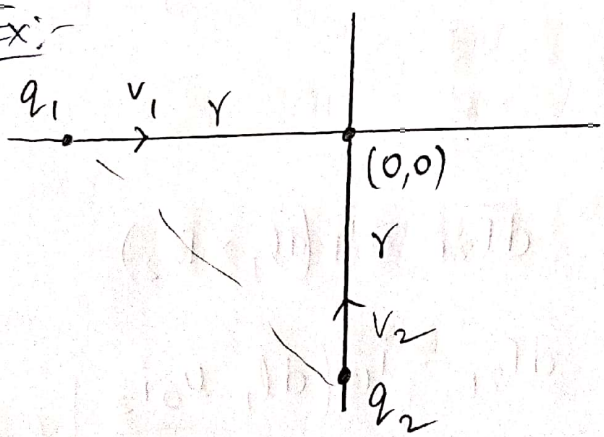


F_m is repulsive

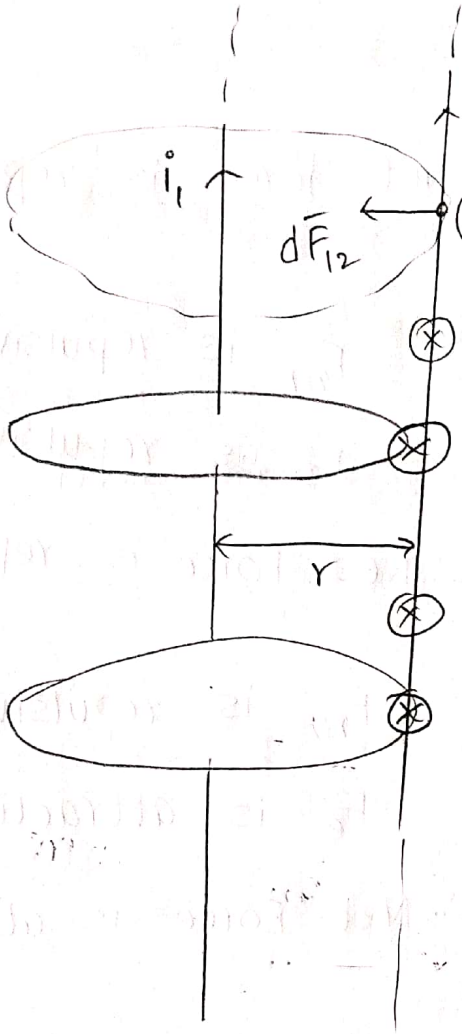
F_e is attractive

$\therefore$ Net Force is attractive

Ex:-



Force between two parallel Current Carrying Wires

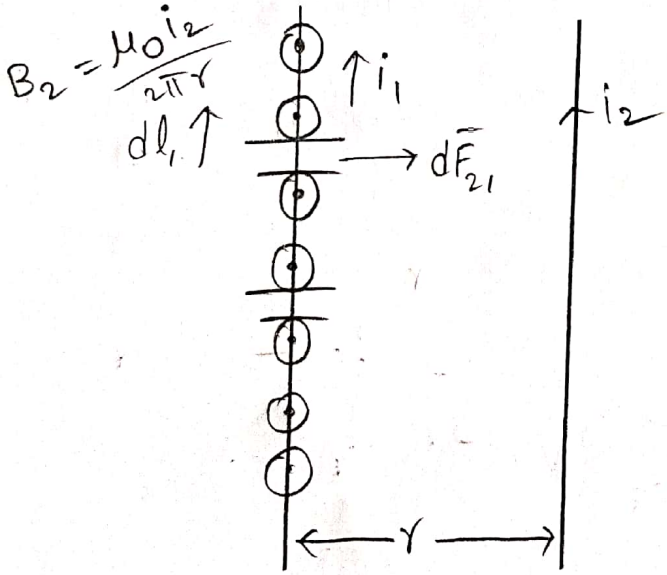


$$d\bar{F}_{12} = i_2 (d\bar{l}_2 \times \bar{B}_1)$$

$$dF_{12} = i_2 \, dl_2 \, \frac{\mu_0 i_1}{2\pi r} \; (\leftarrow)$$

$$\boxed{\frac{dF_{12}}{dl_2} = \frac{\mu_0 i_1 i_2}{2\pi r}}$$

Force per unit length.



$$d\bar{F}_{21} = i_1 (d\bar{l}_1 \times \bar{B}_2)$$

$$dF_{21} = i_1 \left(dl_1 \, \frac{\mu_0 i_2}{2\pi r} \right) (\rightarrow)$$

$$\boxed{\frac{dF_{21}}{dl_1} = \frac{\mu_0 i_1 i_2}{2\pi r}}$$

Force per unit length

Clearly, $\frac{dF}{dl} = \frac{\mu_0 i_1 i_2}{2\pi r}$ = constant

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$$F = \frac{\mu_0 i_1 i_2 l}{2\pi r}$$ → Force on finite length

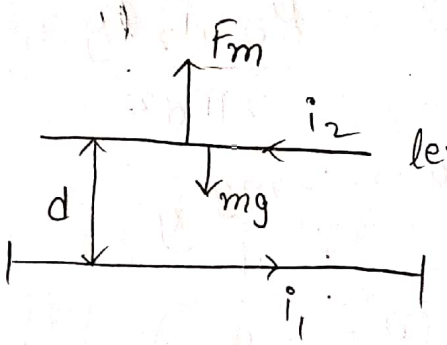
Points:-

① Force between the wires is attractive, if current flow is in same direction.

② Force between the wires is repulsive, if current flow is in opposite direction.

Applications:-

①

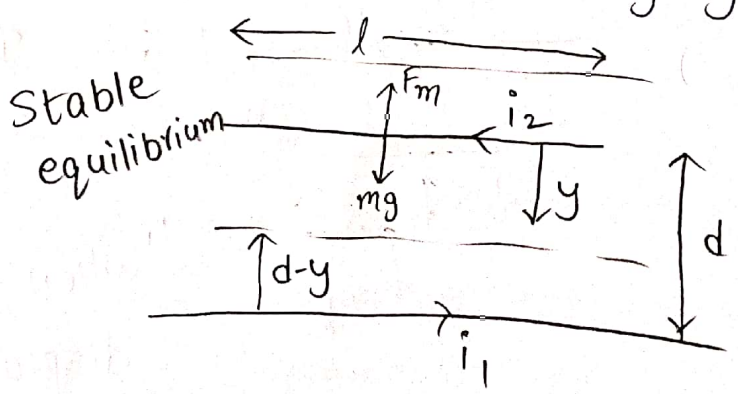


In equilibrium,

$F_m = F_g$

$$\frac{\mu_0 i_1 i_2 l}{2\pi d} = mg$$

Now, if the wire is displaced slightly down by 'y' and released,



$F_{net} = F_m' - F_g$

$$F = \frac{\mu_0 i_1 i_2 l}{2\pi (d-y)} - mg$$

$$F = \frac{\mu_0 i_1 i_2 l}{2\pi (d-y)} - \frac{\mu_0 i_1 i_2 l}{2\pi d}$$

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$$F = \frac{\mu_0 i_1 i_2 l}{2\pi}\left(\frac{1}{d-y} - \frac{1}{d}\right)$$

$$F = \frac{\mu_0 i_1 i_2 l}{2\pi}\left(\frac{y}{d(d-y)}\right) \quad \because y \ll\ll d$$

$$\boxed{F = -\frac{\mu_0 i_1 i_2 l y}{2\pi d^2}}$$

→ Motion is S.H.M

$a \propto -y$

$\Rightarrow a = -\omega^2 y$

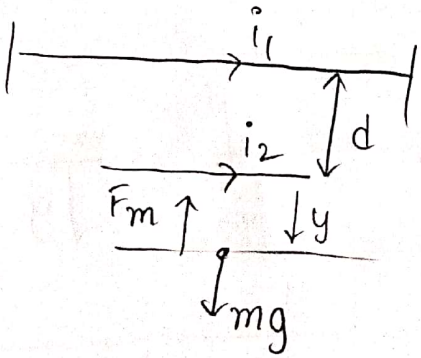
$$ma = -\frac{\mu_0 i_1 i_2 l y}{2\pi d^2}$$

$$ma = -\frac{mg}{d} y$$

$$\Rightarrow \boxed{a = -\frac{g}{d} y} \quad \Rightarrow \omega = \sqrt{\frac{g}{d}}$$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{d}{g}}}$$

(2)



In equilibrium,

$$\boxed{F_m = mg}$$

When the wire is slightly displaced down by 'y' and released

$F_m < F_g$

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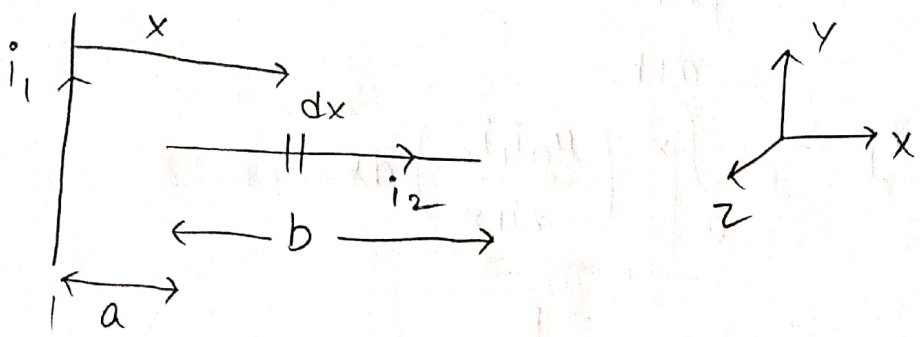
⇒ Wire falls down which is Unstable Equilibrium

Ex:-



$$dF = i_2 (d\bar{l} \times \bar{B}_1)$$

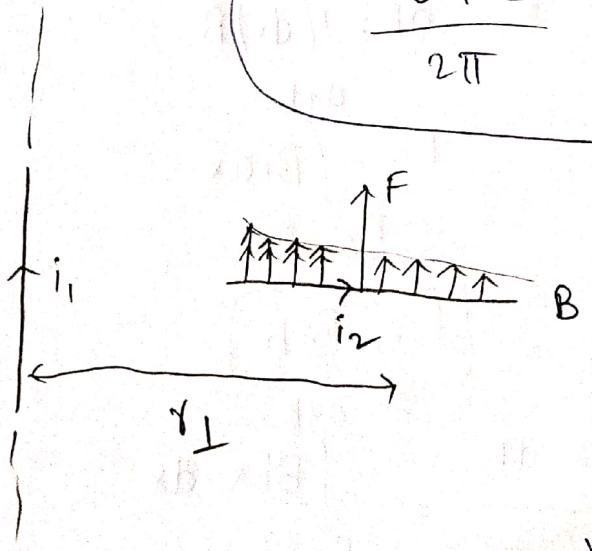
$$dF = i_2 \left(dx\,\hat{i} \times \frac{\mu_0 i_1}{2\pi x} (-\hat{k}) \right)$$

$$dF = \frac{\mu_0 i_1 i_2}{2\pi x} dx\,(\hat{j})$$

$$F = \int_a^{a+b} dF$$

$$\boxed{F = \frac{\mu_0 i_1 i_2}{2\pi} \ln\left(\frac{a+b}{a}\right) \hat{j}}$$ → B is non Uniform

*



To find line of action of Force on Short Conductor,

$$r_\perp = \frac{\int x\,dF}{\int dF}$$

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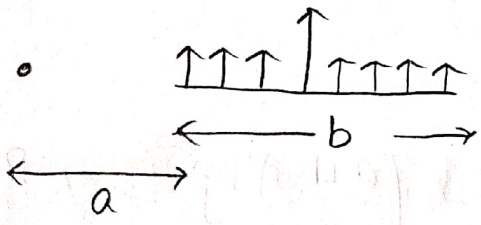
$$r_\perp = \frac{\int x \cdot dF}{\int dF}$$

$$r_\perp = \frac{\int_a^{a+b} x \left(\frac{\mu_0 i_1 i_2}{2\pi x}\right) dx}{F}$$

$$r_\perp = \frac{\frac{\mu_0 i_1 i_2}{2\pi}(b+a-a)}{\frac{\mu_0 i_1 i_2}{2\pi} \ln\left(\frac{a+b}{a}\right)}$$

$$\boxed{r_\perp = \frac{b}{\ln\left(\frac{a+b}{a}\right)}}$$

*



If field is Uniform

$$dF = i(dx)B$$

$$F = \int_a^{a+b} Bi\,dx$$

$$\boxed{F = Bib}$$

Now, $r_\perp = \frac{\int x \cdot dF}{\int dF} = \frac{\int_a^{a+b} Bix \cdot dx}{Bib}$

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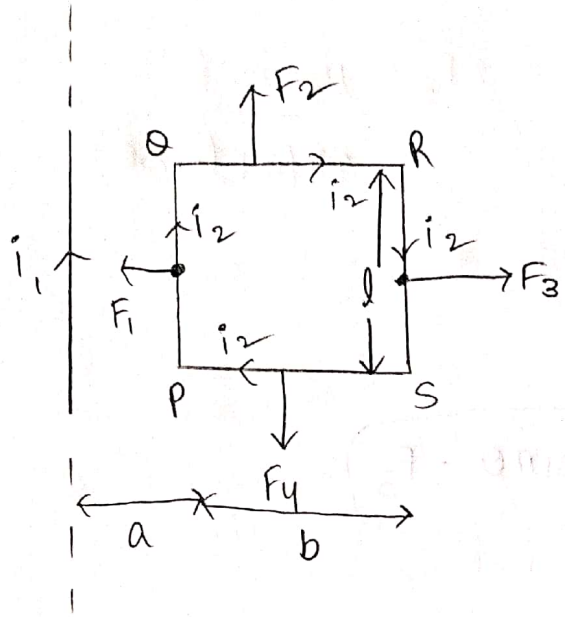
$$r_\perp = \frac{iB(b^2+2ab)}{2iBb} = \frac{b}{2}+a$$

Ex:-



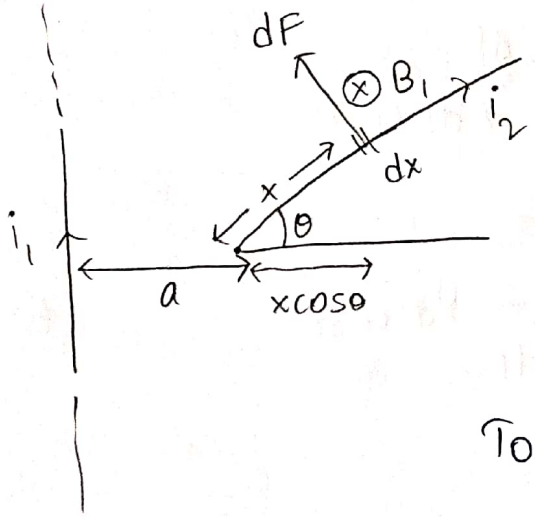
F_2 & F_4 cancel out

$F_{net} = F_1 - F_3$

$$F_{net} = \frac{\mu_0 i_1 i_2 l}{2\pi a} - \frac{\mu_0 i_1 i_2 l}{2\pi(a+b)}$$

, attractive.

Ex:-



$$dF = i_2 (dx \times B_1)$$

$$dF = i_2 (dx)\left(\frac{\mu_0 i_1}{2\pi(a+x\cos\theta)}\right)$$

Total Force,

$$F = \int_0^l dF = \frac{\mu_0 i_1 i_2}{2\pi}\int_0^l \frac{dx}{a+x\cos\theta}$$

$$F = \frac{\mu_0 i_1 i_2}{2\pi\cos\theta}\ln\left(\frac{a+l\cos\theta}{a}\right)$$

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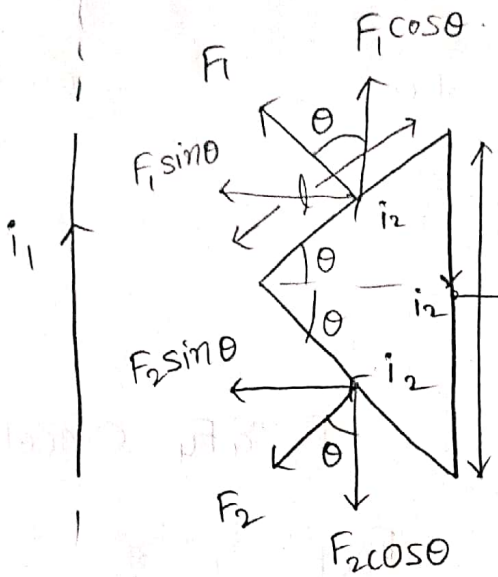
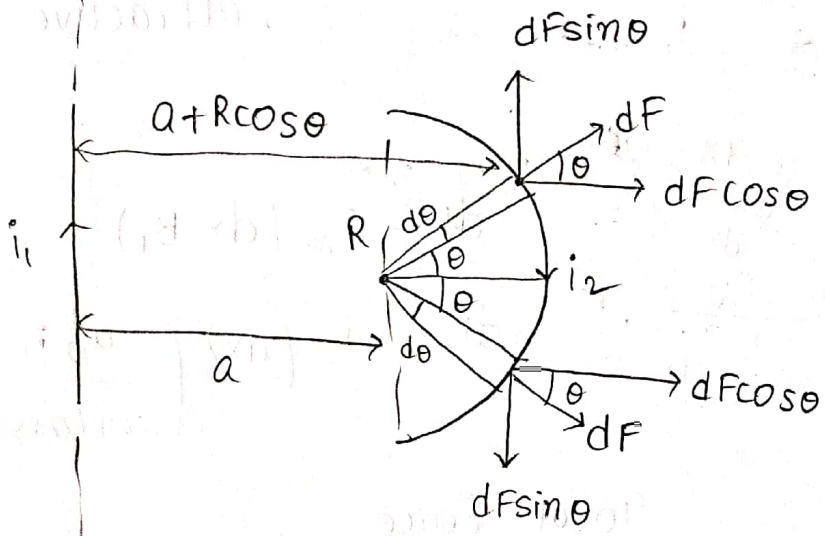
Ex:-



$$F_3 = \frac{\mu_0 i_1 i_2 l}{2\pi(a + l\cos\theta)}$$

$$F = 2F_1 \sin\theta - F_3$$

Ex:-



$$dF = i_2\, dl \times B_1$$

$$= i_2 (R\,d\theta) \frac{\mu_0 i_1}{2\pi(a + R\cos\theta)}$$

$$dF = \frac{\mu_0 i_1 i_2 R\,d\theta}{2\pi(a + R\cos\theta)}$$

Net force, $F = 2\int_0^{\pi/2} dF\cos\theta$, to Right

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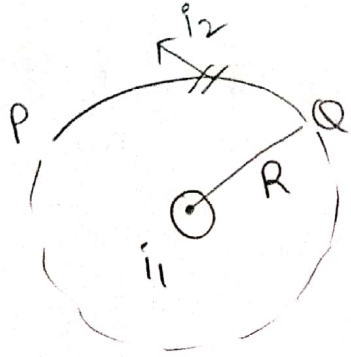


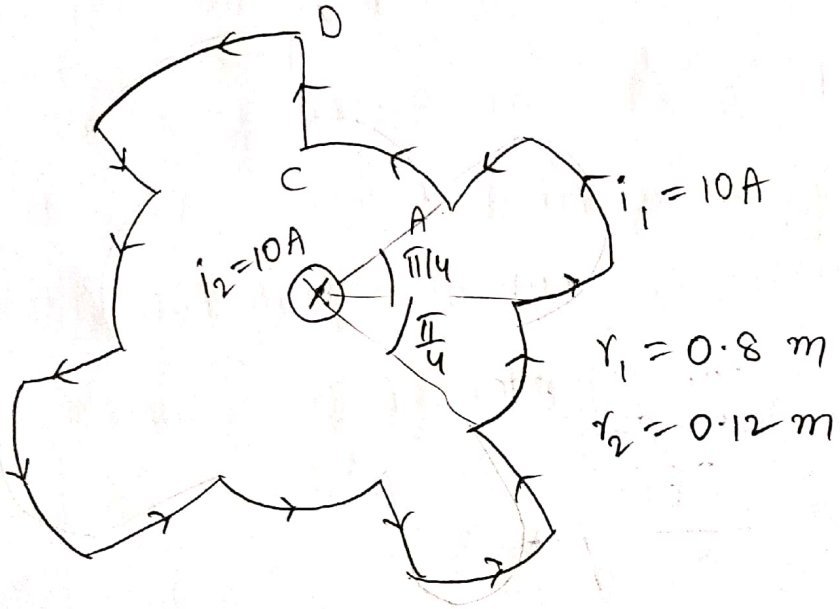
Force on PQ

$$dF = i_2 (d\bar{\ell}_2 \times \bar{B}_1) \quad [\theta = 0^\circ]$$

$$\boxed{dF = 0}$$

$$\Rightarrow \boxed{F = 0}$$

Ex:-



Find ① Magnetic field at centre

② An infinite long wire carrying current of 10A is passing through centre. What is the force acting on the wire at the centre due to current in the circuit? What is the force acting on arc AC & straight segment CD?

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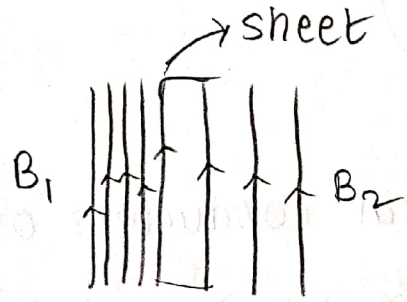
Magnetostatic Pressure:-

* Infinite plane sheet is placed in external magnetic field.



$B_1 > B_2$

$B_{ext} = B_0$

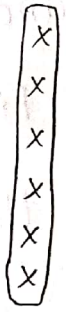
we have,

$B_1 = B_0 + B$

$B_2 = B_0 - B$

$2B_0 = B_1 + B_2$

$B_0 = \frac{B_1 + B_2}{2}$

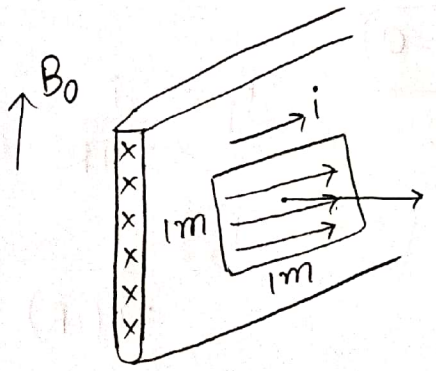


$2B = B_1 - B_2$

$B = \frac{B_1 - B_2}{2}$

$\frac{\mu_0 j}{2} = \frac{B_1 - B_2}{2}$

$$j = \frac{B_1 - B_2}{\mu_0}$$



$A = 1 \times 1 = 1m^2$

$dF = i l B_0$

$= j(1)(1) B_0$

$dF = jB_0 = \left(\frac{B_1 - B_2}{\mu_0}\right)\left(\frac{B_1 + B_2}{2}\right)$

$$dF = \frac{B_1^2 - B_2^2}{2\mu_0}$$, to Right

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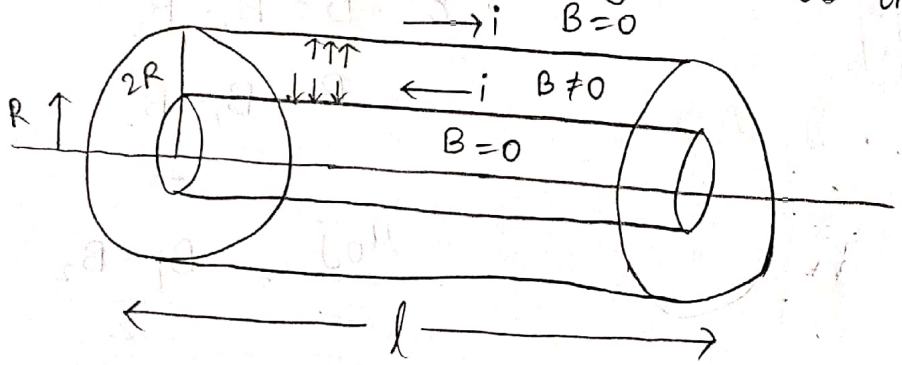
Pressure, $P = \frac{dF}{dA}$

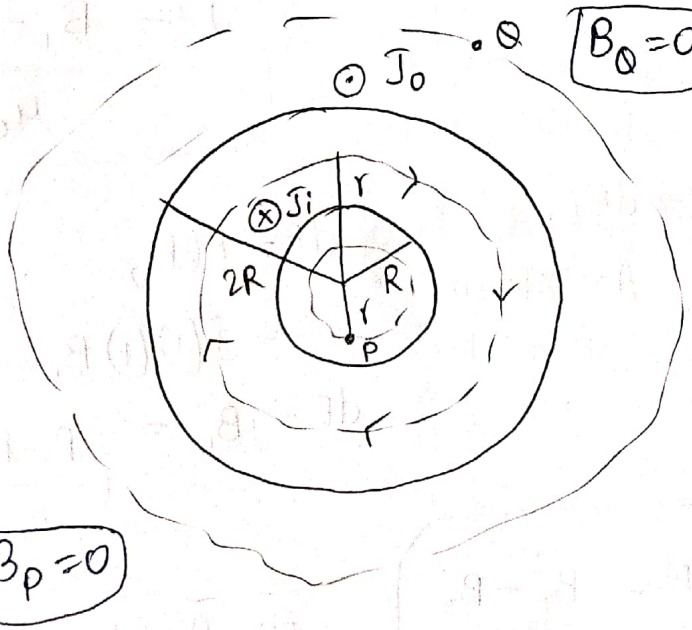
$$P = \frac{B_1^2 - B_2^2}{2\mu_0}$$

Ex:- Two thin long cylindrical conductors of same material (same Young's Modulus) having thickness 't' & raddi 'R' & '2R' are fitted co-axially as shown. Now equal current i and in opposite direction is flowing axially in cylinder. Find Ratio of Change in Radii of Outer Cylinder to that of inner Cylinder.



$R >>> t$

Sol:-



$$J_i = \frac{i}{2\pi R}$$

$$J_0 = \frac{i}{2\pi(2R)} = \frac{i}{4\pi R}$$

B/w $R \le r \le 2R$

$$\oint B\,dl = \mu_0 i$$

$$B(2\pi r) = \mu_0 i$$

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$$\Rightarrow \boxed{B = \frac{\mu_0 i}{2\pi r}}$$

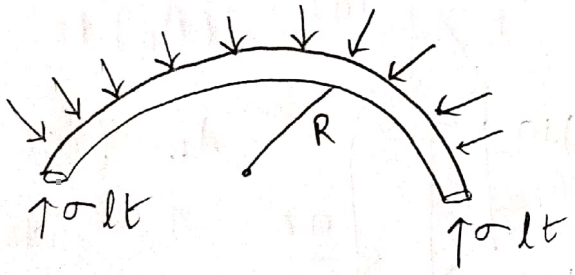
Pressure on the inner cylinder,

$$P_1 = \frac{B^2 - 0^2}{2\mu_0} = \frac{\mu_0^2 i^2}{2\mu_0 (4\pi^2 R^2)} = \frac{\mu_0 i^2}{8\pi^2 R^2}$$

Pressure on the outer cylinder,

$$P_2 = \frac{B^2 - 0^2}{2\mu_0} = \frac{\mu_0 i^2}{8\pi^2 (2R)^2}$$

Clearly $\boxed{P \propto \frac{1}{R^2}}$



Force due to Pressure
= Pressure x Projection Area
= P(2R l)

In equilibrium, $2\sigma lt = P(2Rl)$

$$\sigma = \frac{PR}{t}$$

$$\text{Strain} = \frac{\text{Stress}}{Y}$$

$$\frac{2\pi \Delta R}{2\pi R} = \frac{\sigma}{y} \Rightarrow \frac{\Delta R}{R} = \frac{PR}{Yt}$$

$$\boxed{\Delta R = \frac{PR^2}{Yt}}$$ we have, $P \propto \frac{1}{R^2}$

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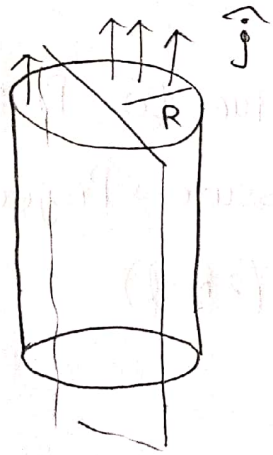
$\Rightarrow$ `$\Delta R$' is independent of `R' & `P'

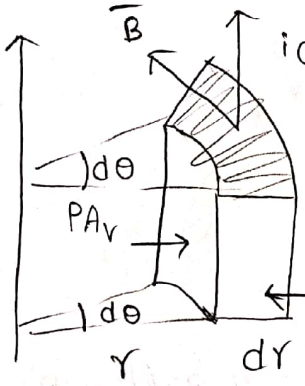
$\therefore$ ratio $= -1$ $\qquad \frac{\Delta R_i}{\Delta R_o} = -1$

$$\left|\frac{\Delta R_i}{\Delta R_o}\right| = 1$$

Ex:- Consider a long current carrying cylindrical conductor of radius R. Current density J inside the conductor is uniform over its cross section. Deduce suitable expression for force of interaction blw two halves that are obtained by dividing the conductor by the plane containing axes of conductor. per unit length

Sol:-





$A_H = (r d\theta)dr$

$A_r = (r d\theta)dz$

Since element is in equilibrium,

$$(P+dP)A_r + id\ell B = PA_r$$

$$(dP)(r d\theta)dz + j(r d\theta)dr\,(dz)\left(\frac{\mu_0 j r}{2}\right) = 0$$

$$\Rightarrow \quad dP = -\frac{\mu_0 j^2 r}{2}\,dr$$

$$\int_0^P dP = -\int_R^r \frac{\mu_0 j^2 r\,dr}{2}$$

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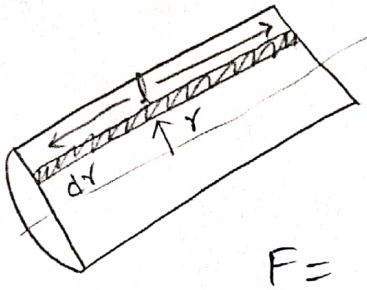
$$P = \frac{\mu_0 j^2}{4}(R^2 - r^2)$$

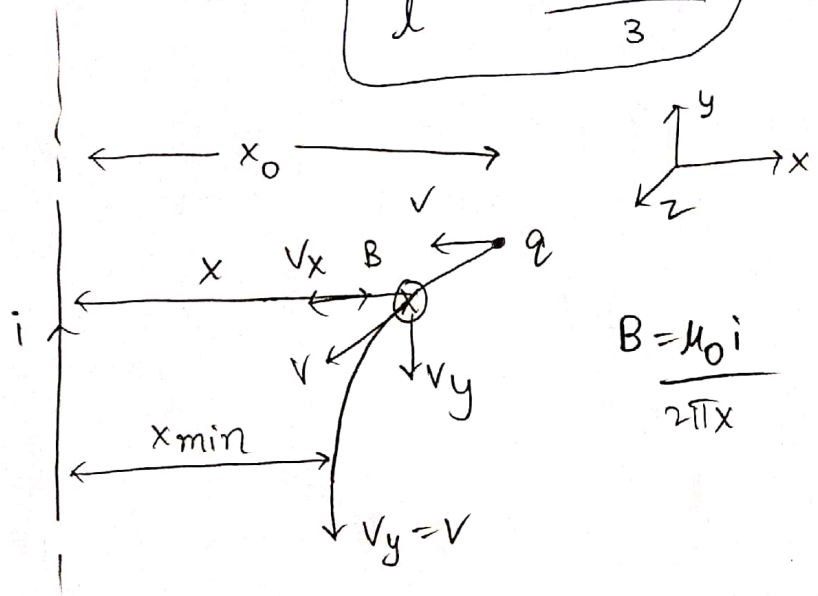


$$dF = PdA$$

$$F = \int dF = \int_{-R}^{R} \frac{\mu_0 j^2}{4}(R^2 - r^2)\, l\, dr$$

$$\frac{F}{l} = \frac{\mu_0 j^2 R^3}{3}$$

Ex:-



Find x_{min} as a function of x_0.

$$B = \frac{\mu_0 i}{2\pi x}$$

Sol:- At any instant, $\vec{F} = q(\vec{v} \times \vec{B})$

$$\vec{F} = q\left(v_x(-\hat{i}) + v_y(-\hat{j})\right) \times \frac{\mu_0 i}{2\pi x}(-\hat{k})$$

$$\vec{F} = \frac{q v_x \mu_0 i}{2\pi x}(-\hat{j}) + \frac{q v_y \mu_0 i}{2\pi x}(\hat{i})$$

$$a_y = \frac{-\mu_0 i q v_x}{2\pi m x} \quad ; v_x = \frac{-dx}{dt}$$

$$\frac{dv_y}{dt} = \frac{-\mu_0 i q}{2\pi m x}\left(\frac{-dx}{dt}\right) \Rightarrow \int_0^{v_y = -v} dv_y = \int_{x_0}^{x_{min}} \frac{\mu_0 i}{2\pi m x}\, dx$$

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$$\Rightarrow -V = \frac{\mu_0 i q}{2\pi m} \ln\left(\frac{x_{min}}{x_0}\right)$$

$$x_{min} = x_0 e^{-2\pi m V / \mu_0 i q}$$

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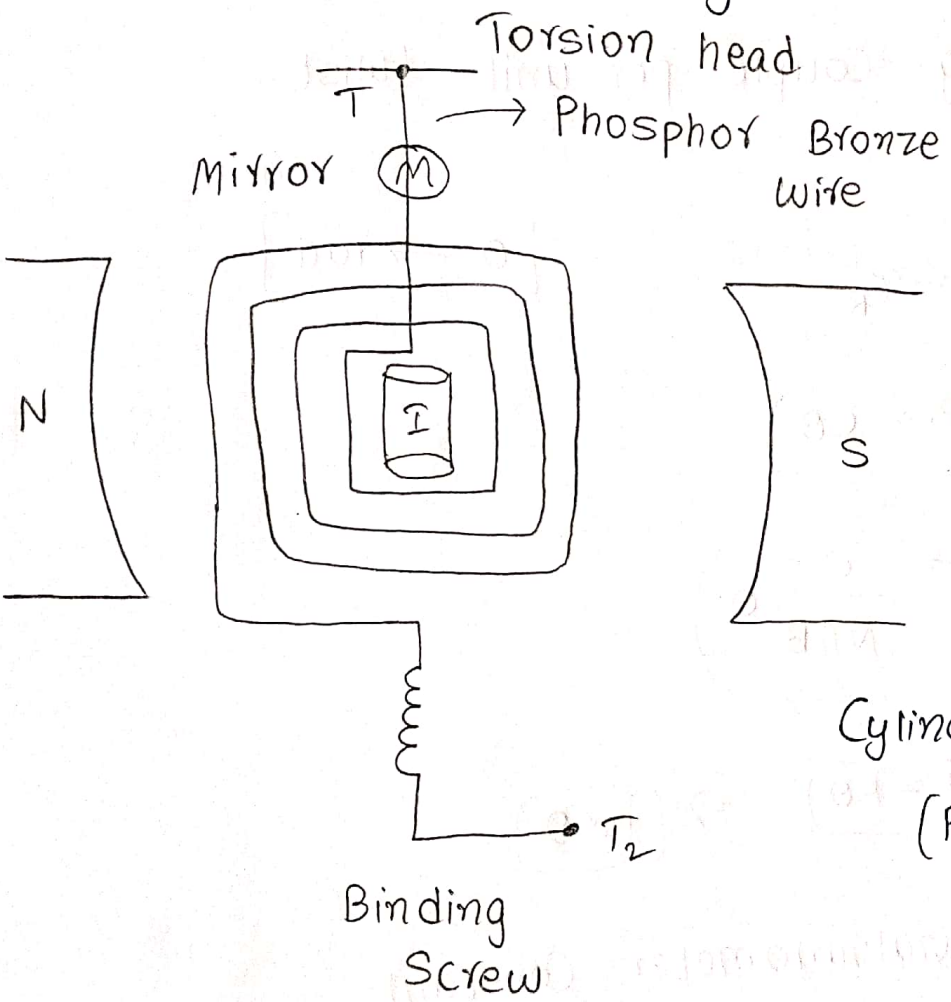
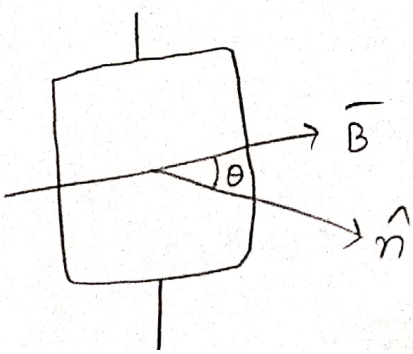
Moving Coil Galvanometer (MCG):-

* It is a device used to measure low currents of order 10^{-9} A.
* Principle:- Torque acting on a Rectangular Loop.



Cylinder is made of Iron (Ferromagnetic Substance)

* Strong magnetic field is present in b/w N-S poles.
* A Bar Magnet, Horshoe magnet cannot effect the deflection in Galvanometer.



$\tau = NiAB\sin\theta$

$\tau = NiAB\cos\alpha$

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Deflecting Torque acting on the coil,

$$\tau = NiAB \quad (max) \longrightarrow (1)$$

Restoring Couple on the coil

$$\tau_R = C\theta$$

C = Restoring Couple per unit twist.

In equilibrium,

$$\tau_D = \tau_R \qquad [\theta \rightarrow rad]$$

$$NiAB = C\theta$$

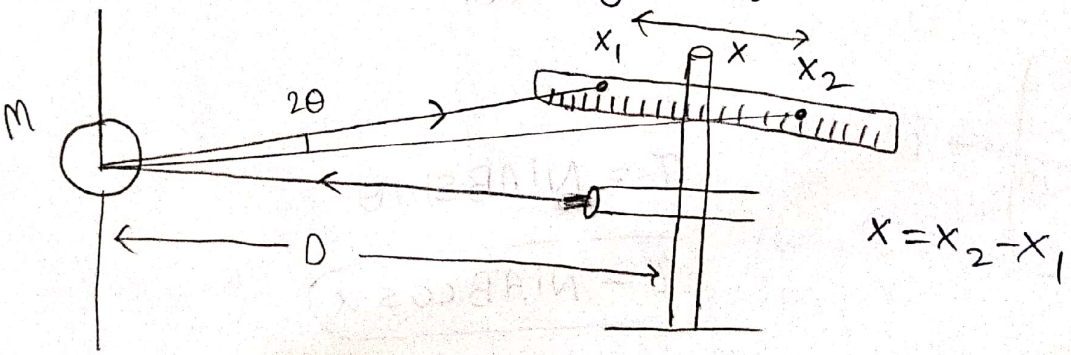
$$i = \frac{C}{NAB}\theta$$

$$i = k\theta \Rightarrow i \propto \theta$$

where, $k = \frac{C}{NAB}$ = Galvanometer Constant.

To find Angle of Deflection (θ):-

We use lamp & scale arrangement.



Since Mirror is rotated by θ when current is passed, reflected ray gets deflected by 2θ

From fig,

$$2\theta = \frac{x}{D}$$

$$x = |x_2 - x_1|$$

$$\boxed{\theta = \frac{x}{2D} \text{ rad}}$$

Sensitivity of Galvanometer :-

input $\longrightarrow i$

Output $\longrightarrow \theta$

① Current sensitivity (CS)

$$\boxed{C.S = \frac{\theta}{i} = \frac{NAB}{C}}$$

* Increasing the Magnetic field B is the effective way to increase C.S.
* Whereas, increasing N (or) A (or) decreasing C increases moment of Inertia which makes it difficult to rotate.

② Voltage Sensitivity (VS)

$$VS = \frac{\theta}{V} = \frac{\theta}{iG} = \frac{CS}{G} = \frac{NAB}{CG}$$

Points :- If A, B, C are constant and if only 'N' is increased $\Rightarrow$ Length of coil increases

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⇒ Galvanometer Resistance increases $\left[R = \frac{\rho l}{A}\right]$

⇒ $N \propto G$

⇒ $\frac{N}{G} = \text{constant}$

$\boxed{\text{V·S} = \text{constant}}$

Figure of Merit of Galvanometer:-

⇓

Current per unit deflection $= \frac{i}{\theta} = \frac{C\theta}{NAB\theta}$

$= \frac{e}{NAB}$

Ex:- Coil is initially in magnetic field. A sudden charge 'Q' flows through the coil in a short time 't'. Find max angle of deflection (θ_0) of the coil.

Sol:- Angular Impulse,

$$\int_0^t \tau\, dt = L_f - L_i^{\nearrow 0}$$

$$\int_0^t BiAN\, dt = I\omega_0$$

$$BiANt = I\omega_0$$

$$BQAN = I\omega_0$$

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$$\omega_0 = \frac{BQAN}{I}$$ ⇒ Max. Angular Velocity

$$W = \int_{\theta=0}^{\theta=\theta_0} C\theta \cdot d\theta = \frac{1}{2} C\theta_0^2$$

From L-C-E,

$$\frac{1}{2} I\omega_0^2 = \frac{1}{2} C\theta_0^2$$

$$\theta_0 = \omega_0 \sqrt{\frac{I}{C}}$$

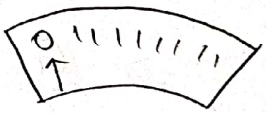
$$\theta_0 = \frac{BQAN}{I} \sqrt{\frac{I}{C}}$$

$$\theta_0 = \frac{BQAN}{\sqrt{CI}}$$

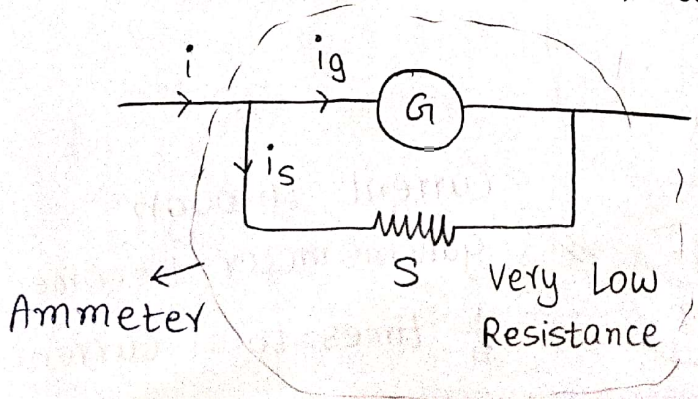
Conversion of Galvanometer into Ammeter:-

Max. deflection,

$$i_g = \frac{\text{Max. Voltage}}{\text{Resistance of Galvanometer}}$$



$0 \longrightarrow i_g$



* A Very Low Resistance Connected in Parallel to Galvanometer is called Shunt

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In parallel,

$$i = i_g + i_s \quad \rightarrow (1)$$

$$i_g G = i_s S \quad \rightarrow (2)$$

From eq (1) & (2),

$$i_g = \frac{iS}{G+S} \quad \rightarrow (3)$$

$$i_s = \frac{iG}{G+S} \quad \rightarrow (4)$$

Resistance of Ammeter,

$$R_A = \frac{GS}{G+S}$$

From eq (3),

$$\frac{i_g}{i} = \frac{S}{G+S} \Rightarrow \frac{i}{i_g} = \frac{G+S}{S}$$

$$\Rightarrow \frac{i}{i_g} = \frac{G}{S} + 1$$

$$\Rightarrow \frac{G}{S} = \frac{i}{i_g} - 1$$

$$\Rightarrow S = \frac{G}{\frac{i}{i_g} - 1}$$

Let $\frac{i}{i_g} = n \Rightarrow i_g = \frac{i}{n}$

Current through galvanometer becomes $\frac{1}{n}$ times total current

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For this to happen, $S = \frac{G}{n-1}$ This is called Principle of Shunt

$$\text{Sensitivity} \propto \frac{1}{\text{Range}}$$

100 div $\longrightarrow$ 100 mA (range)

Sensitivity $= \frac{\theta}{i}$ = 1 div/mA

= 1000 div/A

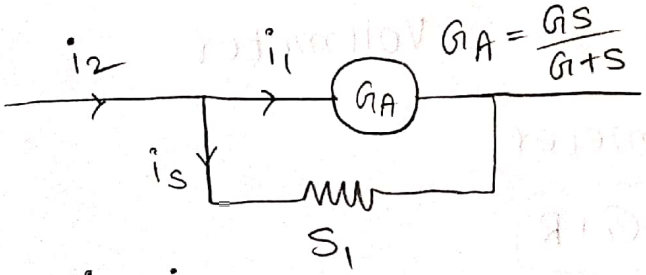
100 div $\longrightarrow$ 100 A (range)

Sensitivity = 1 div/A.

$\Rightarrow$ $$\text{Sensitivity} \propto \frac{1}{\text{Range}}$$

Conversion of Ammeter into Ammeter :-

$(0 \rightarrow i_1)$ $(0 \rightarrow i_2)$



$i_2 = i_1 + i_s$

$i_1 G_A = i_s S_1$

$i_1 G_A = (i_2 - i_1) S_1$

$$S_1 = \frac{G_A}{\frac{i_2}{i_1} - 1}$$

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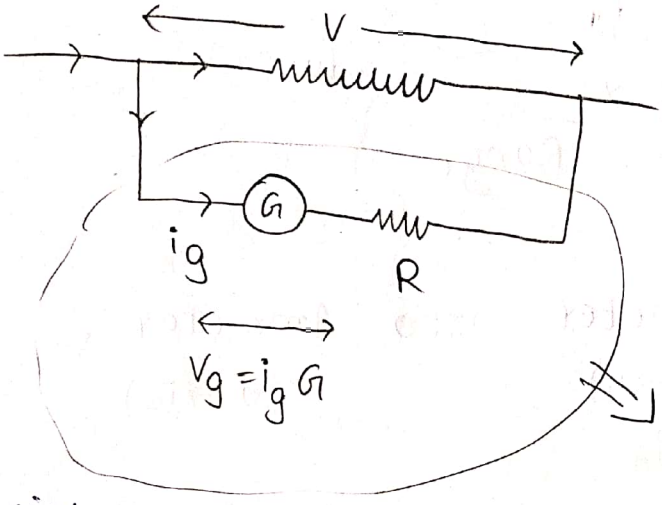
$$S_1 = \frac{G_A}{\frac{i_2}{i_1} - 1}$$

$$S_1 = \frac{G_A}{n-1}$$

$$n = \frac{i_2}{i_1} = \frac{\text{new current}}{\text{old current}} = \frac{\text{Old no. of divisions}}{\text{New no. of divisions}}$$

Conversion of Galvanometer into Voltmeter :-



A very large resistance R is connected in series to Galvanometer

Voltmeter

Resistance of Voltmeter,

$$G_V = G + R$$

$$V = i_g (G+R)$$

$$\frac{V}{i_g} = G+R$$

$$R = \frac{V}{i_g} - G$$

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$$R = \frac{V}{V_g} G - G$$

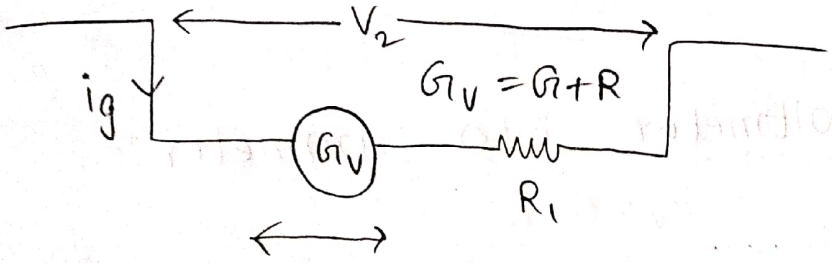
$$R = \left(\frac{V}{V_g} - 1\right) G$$

$$n = \frac{V}{V_g}$$

$$\boxed{R = (n-1)\,G}$$

Conversion of Voltmeter into Another Voltmeter :-

$(0 \rightarrow V_1)$ ⟵ $(0 \rightarrow V_2)$



$$V_1 = i_g G_V$$

$$R_1 = \frac{V_2}{i_g} - G_V$$

$$R_1 = \frac{V_2}{V_1} G_V - G_V$$

$$R_1 = \left(\frac{V_2}{V_1} - 1\right) G_V$$

$$\boxed{R_1 = (n-1)\,G_V}$$

$$n = \frac{V_2}{V_1} = \frac{\text{New Voltage}}{\text{Old Voltage}} = \frac{\text{Old no·of divisions}}{\text{New no·of divisions}}$$

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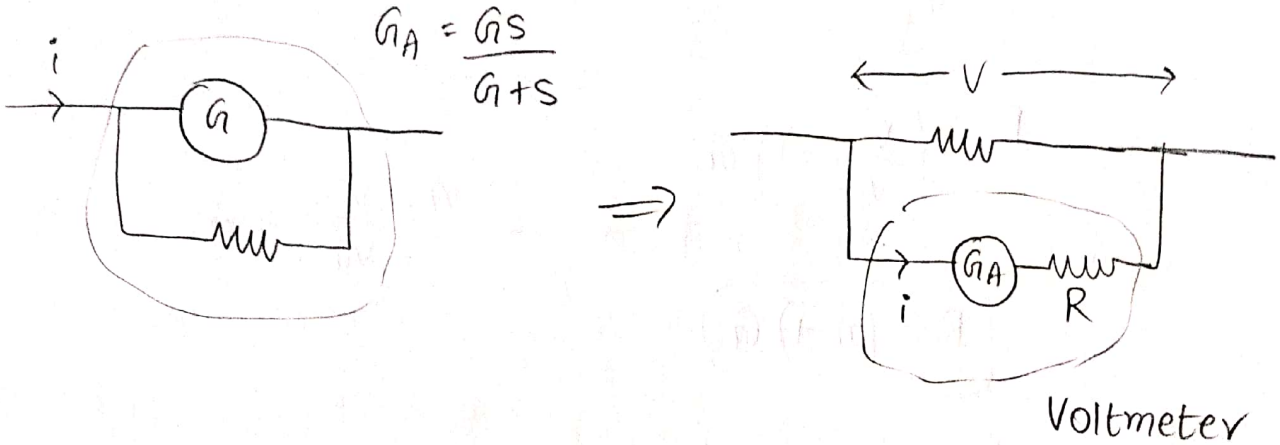
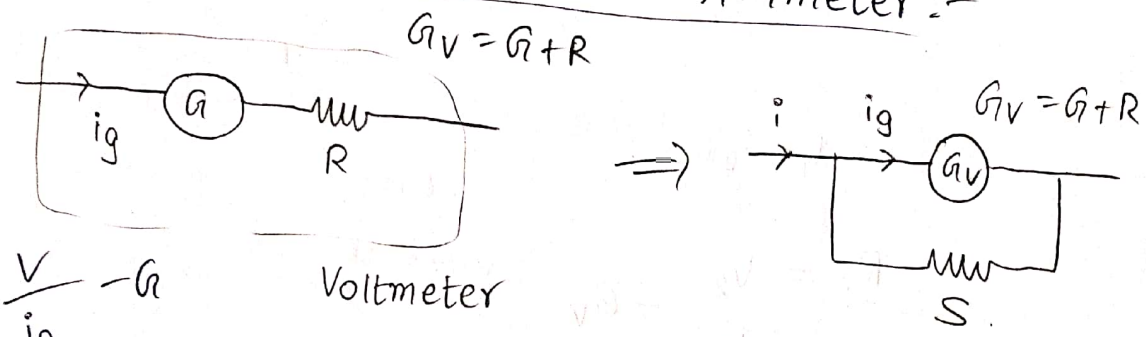
Conversion of Ammeter into Voltmeter :-



$$V = i(G_A + R)$$

$$R = \frac{V}{i} - G_A$$

Conversion of Voltmeter into Ammeter :-



$$R = \frac{V}{ig} - G$$

$$S = \frac{G_V}{\frac{i}{ig} - 1}$$

Ex:- A Galvanometer of Resistance $= 20\Omega$ is shunted by 2Ω resistance. What path of the main current flows through the galvanometer?

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Ex:- If $G = 98\ \Omega$. If 2% of the main current passes through the Galvanometer the Calculate Shunt Resistance ?

Ex:- A Galvanometer having 30 deflections has Sensitivity of 20 μA/division. It has a resistance of $2.5\ \Omega$. How will you convert it into

i) Ammeter measuring upto 1A.

ii) Voltmeter reading upto 1V.

Ex:- The deflection in a moving coil galvanometer falls from 50 divisions to 10 divisions. When a shunt resistance of $10\ \Omega$. What is the resistance of Galvanometer.

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Ex:- The scale of G is divided into 150 equal divisions. The Galvanometer has a current sensitivity of 10 divisions/mA. and Voltage sensitivity of 20 divisions/mV. How the galvanometer be designed to read i) 6A/division ii) 1V/division.

Sol:-

$1\ mA \longrightarrow 10$ div

$i_g \longleftarrow 150$ div

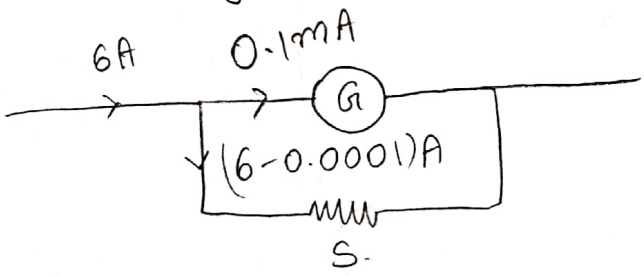
$\Rightarrow i_g = 15\ mA$

$1\ mV \longrightarrow 2$ div

$V_g \longleftarrow 150$ div

$V_g = 75\ mV$

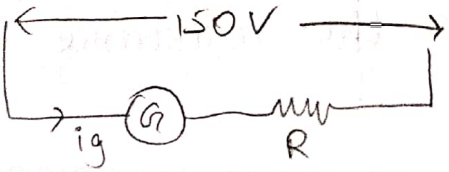
$$G = \frac{V_g}{i_g} = 5\Omega$$



$(0.0001)G = (5.9999)S$

$(0.0001)5 = (5.9999)S$

$\Rightarrow S =$



$V_g = 75mV$

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$V_g = i_g G = 75mV$

$V_R = i_g R = (150V - 75mV)$

$$\Rightarrow \frac{G}{R} = \frac{75 \times 10^{-3}}{150 - 75 \times 10^{-3}}$$

$$\Rightarrow \frac{S}{R} = \frac{75 \times 10^{-3}}{-75 \times 10^{-3} + 150}$$

Ex:- A Galvanometer has resistance 500Ω. It is shunted so that its sensitivity decreases by 100 times. Find the shunt Resistance.

Sol:- $n = 100$

$$S = \frac{G}{n-1} = \frac{500}{99}\ \Omega$$

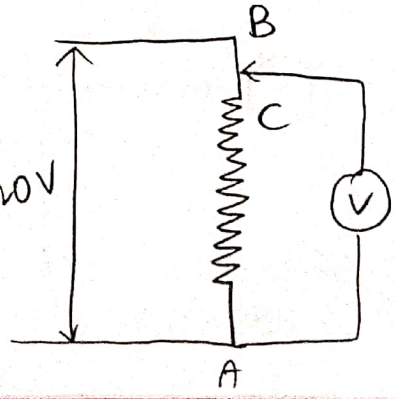
Ex:- A Potential difference of 220V is maintained across a 12000 Rheostat AB. The Voltmeter 'V' has a resistance of 6000Ω & point C is at $\frac{1}{4}$ of the distance from A to B. What is the reading of Voltmeter.

Sol:- $R_{AB} = 12000\ \Omega$

$$AC = \frac{AB}{4} \Rightarrow R_{AC} = \frac{R_{AB}}{4}$$

$$\Rightarrow R_{AC} = 3000\ \Omega$$

[Rdl]



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$R_1 = R_{BC} = 9000\ \Omega$

$\Rightarrow V_1 : V_2 = 9:2$

$\Rightarrow V_1 + V_2 = 220\ V$

$\Rightarrow \frac{11}{2} V_2 = 220.$

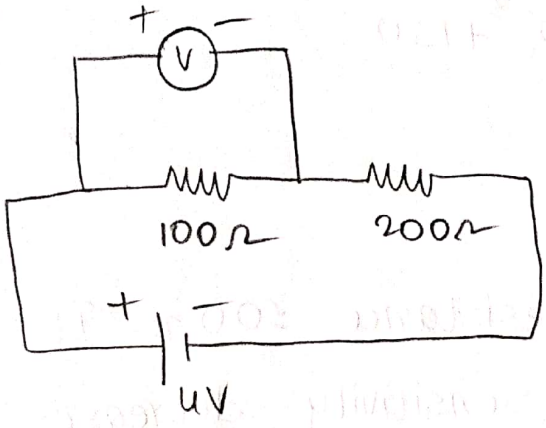
$\Rightarrow \boxed{V_2 = 40\ V}$

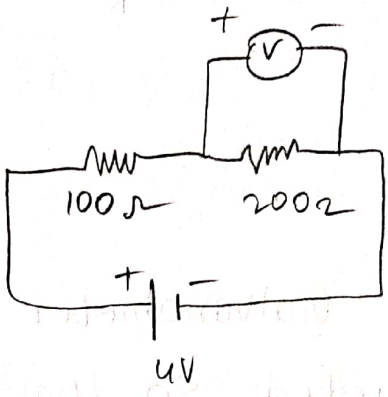
$\frac{1}{R_2} = \frac{1}{R_{AC}} + \frac{1}{R_V}$

$\frac{1}{R_2} = \frac{1}{3000} + \frac{1}{6000}$

$\boxed{R_2 = 2000\ \Omega}$

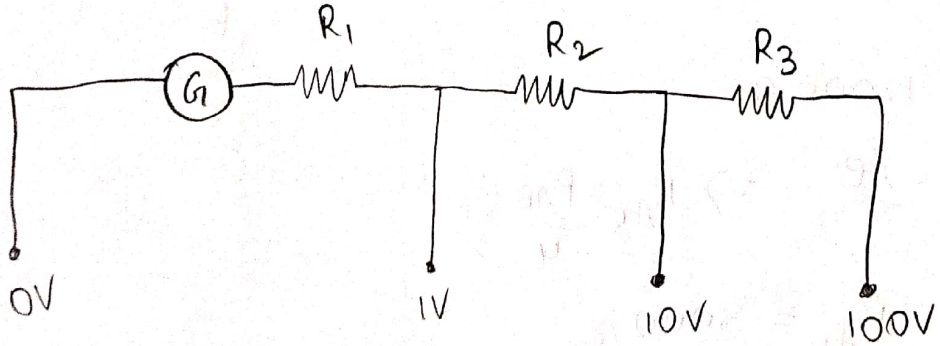
Ex:-





A Voltmeter of resistance $200\ \Omega$ is used to measure the voltage across the two resistances seperately. Calculate error in the Voltmeter Reading.

Ex:-



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A Galvanometer has internal resistance of 50Ω & Current required for full scale deflection is 1A. Find the series resistance required to use it as a Voltmeter with different ranges as shown.

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